

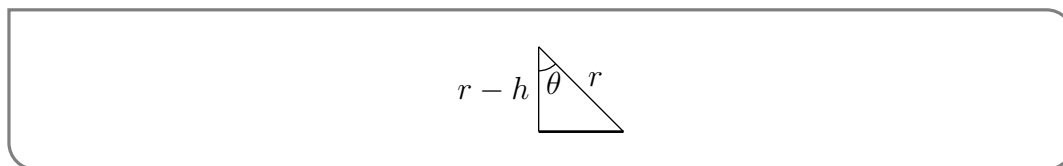
Substituting our expression for θ' into the expression for h' gives

$$h' = r \sin \theta \cdot \frac{q}{Lr^2(1 - \cos 2\theta)}$$

- We can clean this up a bit more — recall more double-angle formulas¹³

$$\begin{aligned} h' &= r \sin \theta \cdot \frac{q}{Lr^2(1 - \cos 2\theta)} && \text{substitute } \cos 2\theta = 1 - 2\sin^2 \theta \\ &= r \sin \theta \cdot \frac{q}{Lr^2 \cdot 2\sin^2 \theta} && \text{now cancel } r\text{'s and a } \sin \theta \\ &= \frac{q}{2Lr \sin \theta} \end{aligned}$$

- But we can clean this up even more — instead of writing this rate in terms of θ it is more natural to write it in terms of h (since the initial problem is stated in terms of h). From the triangle



and by Pythagoras we have

$$\sin \theta = \frac{\sqrt{r^2 - (r - h)^2}}{r} = \frac{\sqrt{2rh - h^2}}{r}$$

and hence

$$h' = \frac{q}{2L\sqrt{2rh - h^2}}.$$

- As a check, notice that h' becomes undefined when $h < 0$ and also when $h > 2r$, because then the argument of the square root in the denominator is negative. Both make sense — the fuel level in the tank must obey $0 \leq h \leq 2r$.

Example 3.2.6

3.3 ▲ Exponential Growth and Decay — a First Look at Differential Equations

A differential equation is an equation for an unknown function that involves the derivative of the unknown function. For example, Newton's law of cooling says:

¹³ Take another look at Appendix [A.14](#).

The rate of change of temperature of an object is proportional to the difference in temperature between the object and its surroundings.

We can write this more mathematically using a differential equation — an equation for the unknown function $T(t)$ that also involves its derivative $\frac{dT}{dt}(t)$. If we denote by $T(t)$ the temperature of the object at time t and by A the temperature of its surroundings, Newton's law of cooling says that there is some constant of proportionality, K , such that

$$\frac{dT}{dt}(t) = K[T(t) - A]$$

Differential equations play a central role in modelling a huge number of different phenomena, including the motion of particles, electromagnetic radiation, financial options, ecosystem populations and nerve action potentials. Most universities offer half a dozen different undergraduate courses on various aspects of differential equations. We are barely going to scratch the surface of the subject. At this point we are going to restrict ourselves to a few very simple differential equations for which we can just guess the solution. In particular, we shall learn how to solve systems obeying Newton's law of cooling in Section 3.3.2 below. But first, here is another slightly simpler example.

3.3.1 ► Carbon Dating

Scientists can determine the age of objects containing organic material by a method called *carbon dating* or *radiocarbon dating*¹⁴. Cosmic rays hitting the atmosphere convert nitrogen into a radioactive isotope of carbon, ^{14}C , with a half-life of about 5730 years¹⁵. Vegetation absorbs carbon dioxide from the atmosphere through photosynthesis and animals acquire ^{14}C by eating plants. When a plant or animal dies, it stops replacing its carbon and the amount of ^{14}C begins to decrease through radioactive decay. More precisely, let $Q(t)$ denote the amount of ^{14}C in the plant or animal t years after it dies. The number of radioactive decays per unit time, at time t , is proportional to the amount of ^{14}C present at time t , which is $Q(t)$. Thus

Equation 3.3.1 (Radioactive decay).

$$\frac{dQ}{dt}(t) = -kQ(t)$$

Here k is a constant of proportionality that is determined by the half-life. We shall explain what half-life is and also determine the value of k in Example 3.3.3 below. Before we do so, let's think about the sign in equation (3.3.1).

- Recall that $Q(t)$ denotes a quantity, namely the amount of ^{14}C present at time t . There cannot be a negative amount of ^{14}C , nor can this quantity be zero (otherwise we wouldn't use carbon dating, so we must have $Q(t) > 0$).

14 Willard Libby, of Chicago University was awarded the Nobel Prize in Chemistry in 1960, for developing radiocarbon dating.

15 A good question to ask yourself is "How can a scientist (who presumably doesn't live 60 centuries) measure this quantity?" One way exploits the little piece of calculus we are about to discuss.

- As the time t increases, $Q(t)$ decreases, because ^{14}C is being continuously converted into ^{14}N by radioactive decay¹⁶. Thus $\frac{dQ}{dt}(t) < 0$.
- The signs $Q(t) > 0$ and $\frac{dQ}{dt}(t) < 0$ are consistent with equation (3.3.1) provided the constant of proportionality $k > 0$.
- In equation (3.3.1), we chose to call the constant of proportionality “ $-k$ ”. We did so in order to make $k > 0$. We could just as well have chosen to call the constant of proportionality “ K ”. That is, we could have replaced equation (3.3.1) by $\frac{dQ}{dt}(t) = KQ(t)$. The constant of proportionality K would have to be negative, (and K and k would be related by $K = -k$).

Now, let's guess some solutions to equation (3.3.1). We wish to guess a function $Q(t)$ whose derivative is just a constant times itself. Here is a short table of derivatives. It is certainly not complete, but it contains the most important derivatives that we know.

$F(t)$	1	t^a	$\sin t$	$\cos t$	$\tan t$	e^t	$\log t$	$\arcsin t$	$\arctan t$
$\frac{d}{dt}F(t)$	0	at^{a-1}	$\cos t$	$-\sin t$	$\sec^2 t$	e^t	$\frac{1}{t}$	$\frac{1}{\sqrt{1-t^2}}$	$\frac{1}{1+t^2}$

There is exactly one function in this table whose derivative is just a (nonzero) constant times itself. Namely, the derivative of e^t is exactly $e^t = 1 \times e^t$. This is almost, but not quite what we want. We want the derivative of $Q(t)$ to be the constant $-k$ (rather than the constant 1) times $Q(t)$. We want the derivative to “pull a constant” out of our guess. That is exactly what happens when we differentiate e^{at} , where a is a constant. Differentiating gives

$$\frac{d}{dt}e^{at} = ae^{at}$$

i.e. “pulls the constant a out of e^{at} ”.

We have succeeded in guessing a single function, namely e^{-kt} , that obeys equation (3.3.1). Can we guess any other solutions? Yes. If C is any constant, Ce^{-kt} also obeys equation (3.3.1):

$$\frac{d}{dt}(Ce^{-kt}) = C \frac{d}{dt}e^{-kt} = Ce^{-kt}(-k) = -k(Ce^{-kt})$$

You can try guessing some more solutions, but you won't find any, because with a little trickery we can prove that a function $Q(t)$ obeys equation (3.3.1) if and only if $Q(t)$ is of the form Ce^{-kt} , where C is some constant.

The trick¹⁷ is to imagine that $Q(t)$ is any (at this stage, unknown) solution to (3.3.1) and to compare $Q(t)$ and our known solution e^{-kt} by studying the ratio $Q(t)/e^{-kt}$. We will show that $Q(t)$ obeys (3.3.1) if and only if the ratio $Q(t)/e^{-kt}$ is a constant, i.e. if and only if the derivative of the ratio is zero. By the product rule

$$\frac{d}{dt}[Q(t)/e^{-kt}] = \frac{d}{dt}[e^{kt}Q(t)] = ke^{kt}Q(t) + e^{kt}Q'(t)$$

16 The precise transition is $^{14}\text{C} \rightarrow ^{14}\text{N} + e^- + \bar{\nu}_e$ where e^- is an electron and $\bar{\nu}_e$ is an electron neutrino.

17 Notice that is very similar to what we needed in Example 3.1.2 except that here the constant is multiplicative rather than additive. That is $\text{const} \times f(t)$ rather than $\text{const} + f(t)$.

Since e^{kt} is never 0, the right hand side is zero if and only if $kQ(t) + Q'(t) = 0$; that is $Q'(t) = -kQ(t)$. Thus

$$\frac{d}{dt}Q(t) = -kQ(t) \iff \frac{d}{dt}[Q(t)/e^{-kt}] = 0$$

as required.

We have succeed in finding all functions that obey (3.3.1). That is, we have found the general solution to (3.3.1). This is worth stating as a theorem.

Theorem 3.3.2.

A differentiable function $Q(t)$ obeys the differential equation

$$\frac{dQ}{dt}(t) = -kQ(t)$$

if and only if there is a constant C such that

$$Q(t) = Ce^{-kt}$$

Before we start to apply the above theorem, we take this opportunity to remind the reader that in this text we will use $\log x$ with no base to indicate the natural logarithm. That is

$$\log x = \log_e x = \ln x$$

Both of the notations $\log(x)$ and $\ln(x)$ are used widely and the reader should be comfortable with both.

Example 3.3.3

In this example, we determine the value of the constant of proportionality k in equation (3.3.1) that corresponds to the half-life of ^{14}C , which is 5730 years.

- Imagine that some plant or animal contains a quantity Q_0 of ^{14}C at its time of death. Let's choose the zero point of time $t = 0$ to be the instant that the plant or animal died.
- Denote by $Q(t)$ the amount of ^{14}C in the plant or animal t years after it died. Then $Q(t)$ must obey both equation (3.3.1) and $Q(0) = Q_0$.
- Since $Q(t)$ must obey equation (3.3.1), Theorem 3.3.2 tells us that there must be a constant C such that $Q(t) = Ce^{-kt}$. To also have $Q_0 = Q(0) = Ce^{-k \times 0}$, the constant C must be Q_0 . That is, $Q(t) = Q_0e^{-kt}$ for all $t \geq 0$.
- By definition, the half-life of ^{14}C is the length of time that it takes for half of the ^{14}C to decay. That is, the half-life $t_{1/2}$ is determined by

$$\begin{aligned} Q(t_{1/2}) &= \frac{1}{2}Q(0) = \frac{1}{2}Q_0 \\ Q_0e^{-kt_{1/2}} &= \frac{1}{2}Q_0 \\ e^{-kt_{1/2}} &= \frac{1}{2} \end{aligned}$$

$$\begin{aligned} \text{but we know } Q(t) &= Q_0e^{-kt} \\ \text{now cancel } Q_0 \end{aligned}$$

Taking the logarithm of both sides gives

$$-kt_{1/2} = \log \frac{1}{2} = -\log 2 \quad \text{and so}$$

$$k = \frac{\log 2}{t_{1/2}}.$$

We are told that, for ^{14}C , the half-life $t_{1/2} = 5730$, so

$$k = \frac{\log 2}{5730} = 0.000121 \quad \text{to 6 digits}$$

Example 3.3.3

From the work in the above example we have accumulated enough new facts to make a corollary to Theorem [3.3.2](#).

Corollary 3.3.4.

The function $Q(t)$ satisfies the equation

$$\frac{dQ}{dt} = -kQ(t)$$

if and only if

$$Q(t) = Q(0) \cdot e^{-kt}.$$

The half-life is defined to be the time $t_{1/2}$ which obeys

$$Q(t_{1/2}) = \frac{1}{2} \cdot Q(0).$$

The half-life is related to the constant k by

$$t_{1/2} = \frac{\log 2}{k}$$

Now here is a typical problem that is solved using Corollary [3.3.4](#).

Example 3.3.5

A particular piece of parchment contains about 64% as much ^{14}C as plants do today. Estimate the age of the parchment.

Solution. Let $Q(t)$ denote the amount of ^{14}C in the parchment t years after it was first created. By equation [\(3.3.1\)](#) and Example [3.3.3](#),

$$\frac{dQ}{dt} = -kQ(t) \quad \text{with } k = \frac{\log 2}{5730} = 0.000121.$$

By Corollary 3.3.4

$$Q(t) = Q(0) \cdot e^{-kt}$$

The time at which $Q(t)$ reaches $0.64Q(0)$ is determined by

$$\begin{aligned} Q(t) &= 0.64Q(0) && \text{but } Q(t) = Q(0)e^{-kt} \\ Q(0)e^{-kt} &= 0.64Q(0) && \text{cancel } Q(0) \\ e^{-kt} &= 0.64 && \text{take logarithms} \\ -kt &= \log 0.64 \\ t &= \frac{\log 0.64}{-k} = \frac{\log 0.64}{-0.000121} = 3700 && \text{to 2 significant digits.} \end{aligned}$$

That is, the parchment¹⁸ is about 37 centuries old.

Example 3.3.5

We have stated that the half-life of ^{14}C is 5730 years. How can this be determined? We can explain this using the following example.

Example 3.3.6

A scientist in a B-grade science fiction film is studying a sample of the rare and fictitious element, implausium¹⁹. With great effort he has produced a sample of pure implausium. The next day — 17 hours later — he comes back to his lab and discovers that his sample is now only 37% pure. What is the half-life of the element?

Solution. We can again set up our problem using Corollary 3.3.4. Let $Q(t)$ denote the quantity of implausium at time t , measured in hours. Then we know

$$Q(t) = Q(0) \cdot e^{-kt}$$

We also know that

$$Q(17) = 0.37Q(0).$$

That enables us to determine k via

$$\begin{aligned} Q(17) &= 0.37Q(0) = Q(0)e^{-17k} && \text{divide both sides by } Q(0) \\ 0.37 &= e^{-17k} \end{aligned}$$

and so

$$k = -\frac{\log 0.37}{17} = 0.05849$$

We can then convert this to the half life using Corollary 3.3.4:

$$t_{1/2} = \frac{\log 2}{k} \approx 11.85 \text{ hours}$$

18 The British Museum has an Egyptian mathematical text from the seventeenth century B.C.

19 Implausium leads to even weaker plots than unobtainium.

While this example is entirely fictitious, one really can use this approach to measure the half-life of materials.

Example 3.3.6

3.3.2 ▶ Newton's Law of Cooling

Recall Newton's law of cooling from the start of this section:

The rate of change of temperature of an object is proportional to the difference in temperature between the object and its surroundings. The temperature of the surroundings is sometimes called the ambient temperature.

We translated this statement into the following differential equation

Equation 3.3.7 (Newton's law of cooling).

$$\frac{dT}{dt}(t) = K[T(t) - A]$$

where $T(t)$ is the temperature of the object at time t , A is the temperature of its surroundings, and K is a constant of proportionality. This mathematical model of temperature change works well when studying a small object in a large, fixed temperature, environment. For example, a hot cup of coffee in a large room²⁰.

Before we worry about solving this equation, let's think a little about the sign of the constant of proportionality. At any time t , there are three possibilities.

- If $T(t) > A$, that is, if the body is warmer than its surroundings, we would expect heat to flow from the body into its surroundings and so we would expect the body to cool off so that $\frac{dT}{dt}(t) < 0$. For this expectation to be consistent with equation (3.3.7), we need $K < 0$.
- If $T(t) < A$, that is the body is cooler than its surroundings, we would expect heat to flow from the surroundings into the body and so we would expect the body to warm up so that $\frac{dT}{dt}(t) > 0$. For this expectation to be consistent with equation (3.3.7), we again need $K < 0$.
- Finally if $T(t) = A$, that is the body and its environment have the same temperature, we would not expect any heat to flow between the two and so we would expect that $\frac{dT}{dt}(t) = 0$. This does not impose any condition on K .

In conclusion, we would expect $K < 0$. Of course, we could have chosen to call the constant of proportionality $-k$, rather than K . Then the differential equation would be $\frac{dT}{dt} = -k(T - A)$ and we would expect $k > 0$.

²⁰ It does not work so well when the object is of a similar size to its surroundings since the temperature of the surroundings will rise as the object cools. It also fails when there are phase transitions involved — for example, an ice-cube melting in a warm room does not obey Newton's law of cooling.

Now to find the general solution to equation (3.3.7). Since this equation is so similar in form to equation (3.3.1), we might expect a similar solution. Start by trying $T(t) = Ce^{Kt}$ and let's see what goes wrong. Substitute it into the equation:

$$\begin{aligned}\frac{dT}{dt} &= K(T(t) - A) \\ KCe^{Kt} &= KCe^{KT} - KA \\ 0 &= -KA\end{aligned}$$

the constant A causes problems!

Let's try something a little different — recall that the derivative of a constant is zero. So we can add or subtract a constant from $T(t)$ without changing its derivative. Set $Q(t) = T(t) + B$, then

$$\begin{aligned}\frac{dQ}{dt}(t) &= \frac{dT}{dt}(t) && \text{by Newton's law of cooling} \\ &= K(T(t) - A) = K(Q(t) - B - A)\end{aligned}$$

So if we choose $B = -A$ then we will have

$$\frac{dQ}{dt}(t) = KQ(t)$$

which is exactly the same form as equation (3.3.1), but with $K = -k$. So by Theorem 3.3.2

$$Q(t) = Q(0)e^{Kt}$$

We can translate back to $T(t)$, since $Q(t) = T(t) - A$ and $Q(0) = T(0) - A$. This gives us the solution.

Corollary 3.3.8.

A differentiable function $T(t)$ obeys the differential equation

$$\frac{dT}{dt}(t) = K[T(t) - A]$$

if and only if

$$T(t) = [T(0) - A]e^{Kt} + A$$

Just before we put this into action, we remind the reader that $\log x = \log_e x = \ln x$.

Example 3.3.9

The temperature of a glass of iced tea is initially 5° . After 5 minutes, the tea has heated to 10° in a room where the air temperature is 30° .

- (a) Determine the temperature as a function of time.
- (b) What is the temperature after 10 minutes?

(c) Determine when the tea will reach a temperature of 20° .

Solution. Part (a)

- Denote by $T(t)$ the temperature of the tea t minutes after it was removed from the fridge, and let $A = 30$ be the ambient temperature.
- By Newton's law of cooling,

$$\frac{dT}{dt} = K(T - A) = K(T - 30)$$

for some, as yet unknown, constant of proportionality K .

- By Corollary [3.3.8](#),

$$T(t) = [T(0) - 30]e^{Kt} + 30 = 30 - 25e^{Kt}$$

since the initial temperature $T(0) = 5$.

- This solution is not complete because it still contains an unknown constant, namely K . We have not yet used the given data that $T(5) = 10$. We can use it to determine K . At $t = 5$,

$$T(5) = 30 - 25e^{5K} = 10 \quad \text{rearrange}$$

$$e^{5K} = \frac{20}{25}$$

$$5K = \log \frac{20}{25} \quad \text{and so}$$

$$K = \frac{1}{5} \log \frac{4}{5} = -0.044629 \quad \text{to 6 digits}$$

Part (b)

- To find the temperature at 10 minutes we can just use the solution we have determined above.

$$\begin{aligned} T(10) &= 30 - 25e^{10K} \\ &= 30 - 25e^{10 \times \frac{1}{5} \log \frac{4}{5}} \\ &= 30 - 25e^{2 \log \frac{4}{5}} = 30 - 25e^{\log \frac{16}{25}} \\ &= 30 - 16 = 14^\circ \end{aligned}$$

Part (c)

- We can find when the temperature is 20° by solving $T(t) = 20$:

$$20 = 30 - 25e^{Kt} \quad \text{rearrange}$$

$$e^{Kt} = \frac{10}{25} = \frac{2}{5}$$

$$Kt = \log \frac{2}{5}$$

$$t = \frac{\log \frac{2}{5}}{K}$$

$$= 20.5 \text{ minutes} \quad \text{to 1 decimal place}$$

Example 3.3.9

A slightly more gruesome example.

Example 3.3.10

A dead body is discovered at 3:45pm in a room where the temperature is 20°C . At that time the temperature of the body is 27°C . Two hours later, at 5:45pm, the temperature of the body is 25.3°C . What was the time of death? Note that the normal (adult human) body temperature is 37° .

Solution. We will assume²¹ that the body's temperature obeys Newton's law of cooling.

- Denote by $T(t)$ the temperature of the body at time t , with $t = 0$ corresponding to 3:45pm. We wish to find the time of death — call it t_d .
- There is a lot of data in the statement of the problem; we are told that
 - the ambient temperature: $A = 20$
 - the temperature of the body when discovered: $T(0) = 27$
 - the temperature of the body 2 hours later: $T(2) = 25.3$
 - assuming the person was a healthy adult right up until he died, the temperature at the time of death: $T(t_d) = 37$.
- Since we assume the temperature of the body obeys Newton's law of cooling, we use Corollary 3.3.8 to find,

$$T(t) = [T(0) - A] e^{Kt} + A = 20 + 7e^{Kt}$$

Two unknowns remain, K and t_d .

- We can find the constant K by using $T(2) = 25.3$:

$$\begin{aligned} 25.3 &= T(2) = 20 + 7e^{2K} && \text{rearrange} \\ 7e^{2K} &= 5.3 && \text{rearrange a bit more} \\ 2K &= \log\left(\frac{5.3}{7}\right) \\ K &= \frac{1}{2} \log\left(\frac{5.3}{7}\right) = -0.139 && \text{to 3 decimal places} \end{aligned}$$

- Since we know²² that t_d is determined by $T(t_d) = 37$, we have

$$\begin{aligned} 37 &= T(t_d) = 20 + 7e^{-0.139t_d} && \text{rearrange} \\ e^{-0.139t_d} &= \frac{17}{7} \\ -0.139t_d &= \log\left(\frac{17}{7}\right) \\ t_d &= -\frac{1}{0.139} \log\left(\frac{17}{7}\right) \\ &= -6.38 && \text{to 2 decimal places} \end{aligned}$$

21 We don't know any other method!

22 Actually, we are assuming again.

Now 6.38 hours is 6 hours and $0.38 \times 60 = 23$ minutes. So the time of death was 6 hours and 23 minutes before 3:45pm, which is 9:22am.

Example 3.3.10

A slightly tricky example — we need to determine the ambient temperature from three measurements at different times.

Example 3.3.11

A glass of room-temperature water is carried out onto a balcony from an apartment where the temperature is 22°C . After one minute the water has temperature 26°C and after two minutes it has temperature 28°C . What is the outdoor temperature?

Solution. We will assume that the temperature of the thermometer obeys Newton's law of cooling.

- Let A be the outdoor temperature and $T(t)$ be the temperature of the water t minutes after it is taken outside.
- By Newton's law of cooling,

$$T(t) = A + (T(0) - A)e^{Kt}$$

by Corollary 3.3.8. Notice there are 3 unknowns here — A , $T(0)$ and K — so we need three pieces of information to find them all.

- We are told $T(0) = 22$, so

$$T(t) = A + (22 - A)e^{Kt}.$$

- We are also told $T(1) = 26$, which gives

$$\begin{aligned} 26 &= A + (22 - A)e^K && \text{rearrange things} \\ e^K &= \frac{26 - A}{22 - A} \end{aligned}$$

- Finally, $T(2) = 28$, so

$$\begin{aligned} 28 &= A + (22 - A)e^{2K} && \text{rearrange} \\ e^{2K} &= \frac{28 - A}{22 - A} && \text{but } e^K = \frac{26 - A}{22 - A}, \text{ so} \\ \left(\frac{26 - A}{22 - A}\right)^2 &= \frac{28 - A}{22 - A} && \text{multiply through by } (22 - A)^2 \\ (26 - A)^2 &= (28 - A)(22 - A) \end{aligned}$$

We can expand out both sides and collect up terms to get

$$\begin{aligned} \underbrace{26^2}_{=676} - 52A + A^2 &= \underbrace{28 \times 22}_{=616} - 50A + A^2 \\ 60 &= 2A \\ 30 &= A \end{aligned}$$

So the temperature outside is 30° .

3.3.3 ► Population Growth

Suppose that we wish to predict the size $P(t)$ of a population as a function of the time t . In the most naive model of population growth, each couple produces β offspring (for some constant β) and then dies. Thus over the course of one generation $\beta \frac{P(t)}{2}$ children are produced and $P(t)$ parents die so that the size of the population grows from $P(t)$ to

$$P(t + t_g) = \underbrace{P(t) + \beta \frac{P(t)}{2}}_{\text{parents+offspring}} - \underbrace{P(t)}_{\text{parents die}} = \frac{\beta}{2} P(t)$$

where t_g denotes the lifespan of one generation. The rate of change of the size of the population per unit time is

$$\frac{P(t + t_g) - P(t)}{t_g} = \frac{1}{t_g} \left[\frac{\beta}{2} P(t) - P(t) \right] = bP(t)$$

where $b = \frac{\beta-2}{2t_g}$ is the net birthrate per member of the population per unit time. If we approximate

$$\frac{P(t+t_g)-P(t)}{t_g} \approx \frac{dP}{dt}(t)$$

we get the differential equation

Equation 3.3.12 (Simple population model).

$$\frac{dP}{dt} = bP(t)$$

By Corollary 3.3.4, with $-k$ replaced by b ,

$$P(t) = P(0) \cdot e^{bt}$$

This is called the Malthusian²³ growth model. It is, of course, very simplistic. One of its main characteristics is that, since $P(t + T) = P(0) \cdot e^{b(t+T)} = P(t) \cdot e^{bT}$, every time you add T to the time, the population size is *multiplied* by e^{bT} . In particular, the population size doubles every $\frac{\log 2}{b}$ units of time. The Malthusian growth model can be a reasonably good model only when the population size is very small compared to its environment²⁴.

23 This is named after Rev. Thomas Robert Malthus. He described this model in a 1798 paper called “An essay on the principle of population”.

24 That is, the population has plenty of food and space to grow.

A more sophisticated model of population growth, that takes into account the “carrying capacity of the environment” is considered in the optional subsection below.

Example 3.3.13

In 1927 the population of the world was about 2 billion. In 1974 it was about 4 billion. Estimate when it reached 6 billion. What will the population of the world be in 2100, assuming the Malthusian growth model?

Solution. We follow our usual pattern for dealing with such problems.

- Let $P(t)$ be the world's population t years after 1927. Note that 1974 corresponds to $t = 1974 - 1927 = 47$.
- We are assuming that $P(t)$ obeys equation (3.3.12). So, by Corollary 3.3.4 with $-k$ replaced by b ,

$$P(t) = P(0) \cdot e^{bt}$$

Notice that there are 2 unknowns here — b and $P(0)$ — so we need two pieces of information to find them.

- We are told $P(0) = 2$, so

$$P(t) = 2 \cdot e^{bt}$$

- We are also told $P(47) = 4$, which gives

$$\begin{aligned} 4 &= 2 \cdot e^{47b} && \text{clean up} \\ e^{47b} &= 2 && \text{take the log and clean up} \\ b &= \frac{\log 2}{47} = 0.0147 && \text{to 3 significant digits} \end{aligned}$$

- We now know $P(t)$ completely, so we can easily determine the predicted population²⁵ in 2100, i.e. at $t = 2100 - 1927 = 173$.

$$P(173) = 2e^{173b} = 2e^{173 \times 0.0147} = 25.4 \text{ billion}$$

- Finally, our crude model predicts that the population is 6 billion at the time t that obeys

$$\begin{aligned} P(t) &= 2e^{bt} = 6 && \text{clean up} \\ e^{bt} &= 3 && \text{take the log and clean up} \\ t &= \frac{\log 3}{b} = 47 \frac{\log 3}{\log 2} = 74.5 \end{aligned}$$

which corresponds²⁶ to the middle of 2001.

Example 3.3.13

25 The 2015 Revision of World Population, a publication of the United Nations, predicts that the world's population in 2100 will be about 11 billion. They are predicting a reduction in the world population growth rate due to lower fertility rates, which the Malthusian growth model does not take into account.

26 The world population really reached 6 billion in about 1999.

►►► (Optional) — Logistic Population Growth

Logistic growth adds one more wrinkle to the simple population model. It assumes that the population only has access to limited resources. As the size of the population grows the amount of food available to each member decreases. This in turn causes the net birth rate b to decrease. In the logistic growth model $b = b_0 \left(1 - \frac{P}{K}\right)$, where K is called the carrying capacity of the environment, so that

$$P'(t) = b_0 \left(1 - \frac{P(t)}{K}\right) P(t)$$

We can learn quite a bit about the behaviour of solutions to differential equations like this, without ever finding formulae for the solutions, just by watching the sign of $P'(t)$. For concreteness, we'll look at solutions of the differential equation

$$\frac{dP}{dt}(t) = (6000 - 3P(t)) P(t)$$

We'll sketch the graphs of four functions $P(t)$ that obey this equation.

- For the first function, $P(0) = 0$.
- For the second function, $P(0) = 1000$.
- For the third function, $P(0) = 2000$.
- For the fourth function, $P(0) = 3000$.

The sketches will be based on the observation that $(6000 - 3P) P = 3(2000 - P) P$

- is zero for $P = 0, 2000$,
- is strictly positive for $0 < P < 2000$ and
- is strictly negative for $P > 2000$.

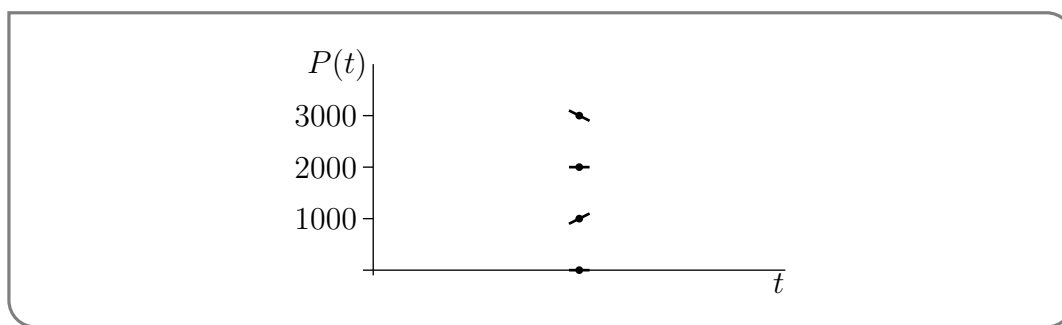
Consequently

$$\frac{dP}{dt}(t) \begin{cases} = 0 & \text{if } P(t) = 0 \\ > 0 & \text{if } 0 < P(t) < 2000 \\ = 0 & \text{if } P(t) = 2000 \\ < 0 & \text{if } P(t) > 2000 \end{cases}$$

Thus if $P(t)$ is some function that obeys $\frac{dP}{dt}(t) = (6000 - 3P(t))P(t)$, then as the graph of $P(t)$ passes through $(t, P(t))$

$$\text{the graph has } \begin{cases} \text{slope zero,} & \text{i.e. is horizontal, if } P(t) = 0 \\ \text{positive slope,} & \text{i.e. is increasing, if } 0 < P(t) < 2000 \\ \text{slope zero,} & \text{i.e. is horizontal, if } P(t) = 2000 \\ \text{negative slope,} & \text{i.e. is decreasing, if } P(t) > 2000 \end{cases}$$

as illustrated in the figure

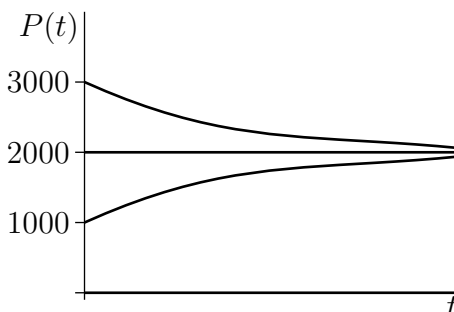


As a result,

- if $P(0) = 0$, the graph starts out horizontally. In other words, as t starts to increase, $P(t)$ remains at zero, so the slope of the graph remains at zero. The population size remains zero for all time. As a check, observe that the function $P(t) = 0$ obeys $\frac{dP}{dt}(t) = (6000 - 3P(t))P(t)$ for all t .
- Similarly, if $P(0) = 2000$, the graph again starts out horizontally. So $P(t)$ remains at 2000 and the slope remains at zero. The population size remains 2000 for all time. Again, the function $P(t) = 2000$ obeys $\frac{dP}{dt}(t) = (6000 - 3P(t))P(t)$ for all t .
- If $P(0) = 1000$, the graph starts out with positive slope. So $P(t)$ increases with t . As $P(t)$ increases towards 2000, the slope $(6000 - 3P(t))P(t)$, while remaining positive, gets closer and closer to zero. As the graph approaches height 2000, it becomes more and more horizontal. The graph cannot actually cross from below 2000 to above 2000, because to do so, it would have to have strictly positive slope for some value of P above 2000, which is not allowed.
- If $P(0) = 3000$, the graph starts out with negative slope. So $P(t)$ decreases with t . As $P(t)$ decreases towards 2000, the slope $(6000 - 3P(t))P(t)$, while remaining negative, gets closer and closer to zero. As the graph approaches height 2000, it becomes more and more horizontal. The graph cannot actually cross from above 2000 to below 2000, because to do so, it would have to have negative slope for some value of P below 2000, which is not allowed.

These curves are sketched in the figure below. We conclude that for any initial population size $P(0)$, except $P(0) = 0$, the population size approaches 2000 as $t \rightarrow \infty$.

Figure 3.3.1.



3.4 ▲ Approximating Functions Near a Specified Point — Taylor Polynomials

Suppose that you are interested in the values of some function $f(x)$ for x near some fixed point a . When the function is a polynomial or a rational function we can use some arithmetic (and maybe some hard work) to write down the answer. For example:

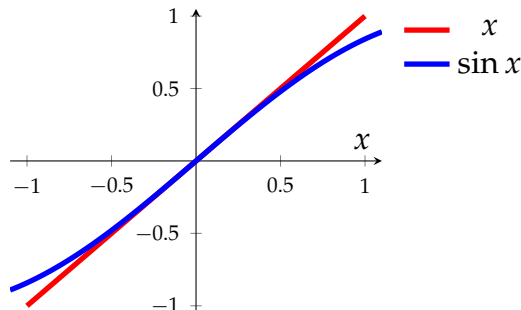
$$\begin{aligned} f(x) &= \frac{x^2 - 3}{x^2 - 2x + 4} \\ f(1/5) &= \frac{\frac{1}{25} - 3}{\frac{1}{25} - \frac{2}{5} + 4} = \frac{\frac{1-75}{25}}{\frac{1-10+100}{25}} \\ &= \frac{-74}{91} \end{aligned}$$

Tedious, but we can do it. On the other hand if you are asked to compute $\sin(1/10)$ then what can we do? We know that a calculator can work it out

$$\sin(1/10) = 0.09983341 \dots$$

but how does the calculator do this? How did people compute this before calculators²⁷? A hint comes from the following sketch of $\sin(x)$ for x around 0.

Figure 3.4.1.



The above figure shows that the curves $y = x$ and $y = \sin x$ are almost the same when x is close to 0. Hence if we want the value of $\sin(1/10)$ we could just use this approximation $y = x$ to get

$$\sin(1/10) \approx 1/10.$$

Of course, in this case we simply observed that one function was a good approximation of the other. We need to know how to find such approximations more systematically.

More precisely, say we are given a function $f(x)$ that we wish to approximate close to some point $x = a$, and we need to find another function $F(x)$ that

- is simple and easy to compute²⁸

27 Originally the word “calculator” referred not to the software or electronic (or even mechanical) device we think of today, but rather to a person who performed calculations.

28 It is no good approximating a function with something that is even more difficult to work with.

- is a good approximation to $f(x)$ for x values close to a .

Further, we would like to understand how good our approximation actually is. Namely we need to be able to estimate the error $|f(x) - F(x)|$.

There are many different ways to approximate a function and we will discuss one family of approximations: Taylor polynomials. This is an infinite family of ever improving approximations, and our starting point is the very simplest.

3.4.1 ► Zeroth Approximation — the Constant Approximation

The simplest functions are those that are constants. And our zeroth²⁹ approximation will be by a constant function. That is, the approximating function will have the form $F(x) = A$, for some constant A . Notice that this function is a polynomial of degree zero.

To ensure that $F(x)$ is a good approximation for x close to a , we choose A so that $f(x)$ and $F(x)$ take exactly the same value when $x = a$.

$$F(x) = A \quad \text{so} \quad F(a) = A = f(a) \implies A = f(a)$$

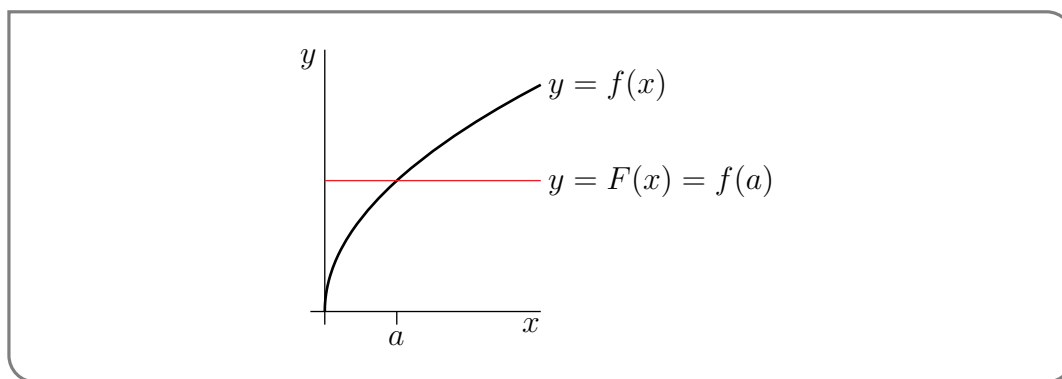
Our first, and crudest, approximation rule is

Equation 3.4.1 (Constant approximation).

$$f(x) \approx f(a)$$

An important point to note is that we need to know $f(a)$ — if we cannot compute that easily then we are not going to be able to proceed. We will often have to choose a (the point around which we are approximating $f(x)$) with some care to ensure that we can compute $f(a)$.

Here is a figure showing the graphs of a typical $f(x)$ and approximating function $F(x)$. At $x = a$, $f(x)$ and $F(x)$ take the same value. For x very near a , the values of $f(x)$ and $F(x)$



remain close together. But the quality of the approximation deteriorates fairly quickly as x moves away from a . Clearly we could do better with a straight line that follows the slope of the curve. That is our next approximation.

²⁹ It barely counts as an approximation at all, but it will help build intuition. Because of this, and the fact that a constant is a polynomial of degree 0, we'll start counting our approximations from zero rather than 1.

But before then, an example:

Example 3.4.2

Use the constant approximation to estimate $e^{0.1}$.

Solution. First set $f(x) = e^x$.

- Now we first need to pick a point $x = a$ to approximate the function. This point needs to be close to 0.1 and we need to be able to evaluate $f(a)$ easily. The obvious choice is $a = 0$.
- Then our constant approximation is just

$$\begin{aligned} F(x) &= f(0) = e^0 = 1 \\ F(0.1) &= 1 \end{aligned}$$

Note that $e^{0.1} = 1.105170918\dots$, so even this approximation isn't too bad..

Example 3.4.2

3.4.2 ▶ First Approximation — the Linear approximation

Our first³⁰ approximation improves on our zeroth approximation by allowing the approximating function to be a linear function of x rather than just a constant function. That is, we allow $F(x)$ to be of the form $A + Bx$, for some constants A and B .

To ensure that $F(x)$ is a good approximation for x close to a , we still require that $f(x)$ and $F(x)$ have the same value at $x = a$ (that was our zeroth approximation). Our additional requirement is that their tangent lines at $x = a$ have the same slope — that the derivatives of $f(x)$ and $F(x)$ are the same at $x = a$. Hence

$$\begin{aligned} F(x) &= A + Bx & \implies & & F(a) &= A + Ba = f(a) \\ F'(x) &= B & \implies & & F'(a) &= B = f'(a) \end{aligned}$$

So we must have $B = f'(a)$. Substituting this into $A + Ba = f(a)$ we get $A = f(a) - af'(a)$. So we can write

$$\begin{aligned} F(x) &= A + Bx = \overbrace{f(a) - af'(a)}^A + f'(a) \cdot x \\ &= f(a) + f'(a) \cdot (x - a) \end{aligned}$$

We write it in this form because we can now clearly see that our first approximation is just an extension of our zeroth approximation. This first approximation is also often called the linear approximation of $f(x)$ about $x = a$.

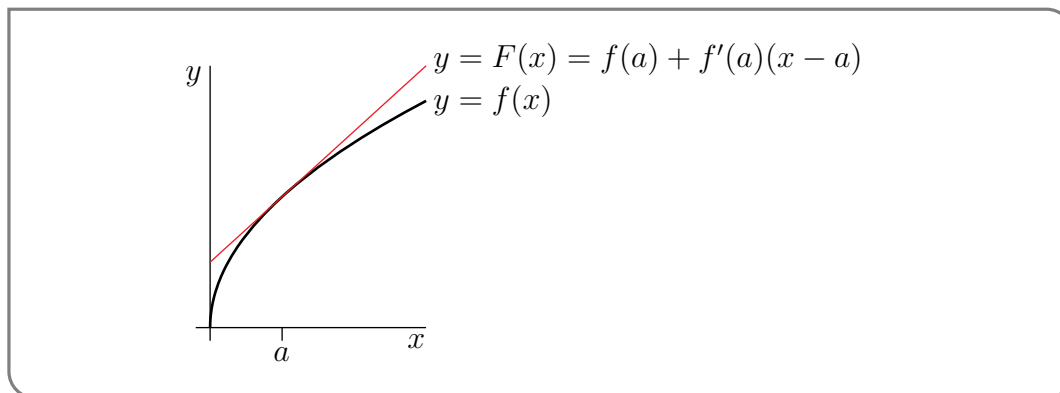
Equation 3.4.3 (Linear approximation).

$$f(x) \approx f(a) + f'(a)(x - a)$$

30 Recall that we started counting from zero.

We should again stress that in order to form this approximation we need to know $f(a)$ and $f'(a)$ — if we cannot compute them easily then we are not going to be able to proceed.

Recall, from Theorem 2.3.2, that $y = f(a) + f'(a)(x - a)$ is exactly the equation of the tangent line to the curve $y = f(x)$ at a . Here is a figure showing the graphs of a typical $f(x)$ and the approximating function $F(x)$. Observe that the graph of $f(a) + f'(a)(x - a)$



remains close to the graph of $f(x)$ for a much larger range of x than did the graph of our constant approximation, $f(a)$. One can also see that we can improve this approximation if we can use a function that curves down rather than being perfectly straight. That is our next approximation.

But before then, back to our example:

Example 3.4.4

Use the linear approximation to estimate $e^{0.1}$.

Solution. First set $f(x) = e^x$ and $a = 0$ as before.

- To form the linear approximation we need $f(a)$ and $f'(a)$:

$$\begin{aligned} f(x) &= e^x & f(0) &= 1 \\ f'(x) &= e^x & f'(0) &= 1 \end{aligned}$$

- Then our linear approximation is

$$\begin{aligned} F(x) &= f(0) + xf'(0) = 1 + x \\ F(0.1) &= 1.1 \end{aligned}$$

Recall that $e^{0.1} = 1.105170918\dots$, so the linear approximation is almost correct to 3 digits.

Example 3.4.4

It is worth doing another simple example here.

Example 3.4.5

Use a linear approximation to estimate $\sqrt{4.1}$.

Solution. First set $f(x) = \sqrt{x}$. Hence $f'(x) = \frac{1}{2\sqrt{x}}$. Then we are trying to approximate $f(4.1)$. Now we need to choose a sensible a value.

- We need to choose a so that $f(a)$ and $f'(a)$ are easy to compute.
 - We could try $a = 4.1$ — but then we need to compute $f(4.1)$ and $f'(4.1)$ — which is our original problem and more!
 - We could try $a = 0$ — then $f(0) = 0$ and $f'(0) = DNE$.
 - Setting $a = 1$ gives us $f(1) = 1$ and $f'(1) = \frac{1}{2}$. This would work, but we can get a better approximation by choosing a is closer to 4.1.
 - Indeed we can set a to be the square of any rational number and we'll get a result that is easy to compute.
 - Setting $a = 4$ gives $f(4) = 2$ and $f'(4) = \frac{1}{4}$. This seems good enough.
- Substitute this into equation (3.4.3) to get

$$\begin{aligned} f(4.1) &\approx f(4) + f'(4) \cdot (4.1 - 4) \\ &= 2 + \frac{0.1}{4} = 2 + 0.025 = 2.025 \end{aligned}$$

Notice that the true value is $\sqrt{4.1} = 2.024845673 \dots$

Example 3.4.5

3.4.3 ► Second Approximation — the Quadratic Approximation

We next develop a still better approximation by now allowing the approximating function be to a quadratic function of x . That is, we allow $F(x)$ to be of the form $A + Bx + Cx^2$, for some constants A , B and C . To ensure that $F(x)$ is a good approximation for x close to a , we choose A , B and C so that

- $f(a) = F(a)$ (just as in our zeroth approximation),
- $f'(a) = F'(a)$ (just as in our first approximation), and
- $f''(a) = F''(a)$ — this is a new condition.

These conditions give us the following equations

$$\begin{array}{lll} F(x) = A + Bx + Cx^2 & \implies & F(a) = A + Ba + Ca^2 = f(a) \\ F'(x) = B + 2Cx & \implies & F'(a) = B + 2Ca = f'(a) \\ F''(x) = 2C & \implies & F''(a) = 2C = f''(a) \end{array}$$

Solve these for C first, then B and finally A .

$$\begin{aligned} C &= \frac{1}{2}f''(a) && \text{substitute} \\ B &= f'(a) - 2Ca = f'(a) - af''(a) && \text{substitute again} \\ A &= f(a) - Ba - Ca^2 = f(a) - a[f'(a) - af''(a)] - \frac{1}{2}f''(a)a^2 \end{aligned}$$

Then put things back together to build up $F(x)$:

$$\begin{aligned}
 F(x) &= f(a) - f'(a)a + \frac{1}{2}f''(a)a^2 && \text{(this line is } A) \\
 &\quad + f'(a)x - f''(a)ax && \text{(this line is } Bx) \\
 &\quad + \frac{1}{2}f''(a)x^2 && \text{(this line is } Cx^2) \\
 &= f(a) + f'(a)(x-a) + \frac{1}{2}f''(a)(x-a)^2
 \end{aligned}$$

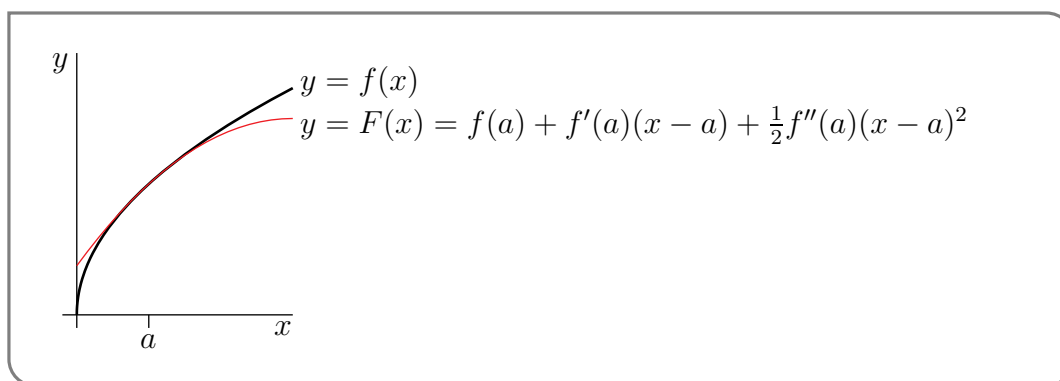
Oof! We again write it in this form because we can now clearly see that our second approximation is just an extension of our first approximation.

Our second approximation is called the quadratic approximation:

Equation 3.4.6 (Quadratic approximation).

$$f(x) \approx f(a) + f'(a)(x-a) + \frac{1}{2}f''(a)(x-a)^2$$

Here is a figure showing the graphs of a typical $f(x)$ and approximating function $F(x)$.



This new approximation looks better than both the first and second.

Now there is actually an easier way to derive this approximation, which we show you now. Let us rewrite ³¹ $F(x)$ so that it is easy to evaluate it and its derivatives at $x = a$:

$$F(x) = \alpha + \beta \cdot (x-a) + \gamma \cdot (x-a)^2$$

Then

$$\begin{aligned}
 F(x) &= \alpha + \beta \cdot (x-a) + \gamma \cdot (x-a)^2 && F(a) = \alpha = f(a) \\
 F'(x) &= \beta + 2\gamma \cdot (x-a) && F'(a) = \beta = f'(a) \\
 F''(x) &= 2\gamma && F''(a) = 2\gamma = f''(a)
 \end{aligned}$$

And from these we can clearly read off the values of α , β and γ and so recover our function $F(x)$. Additionally if we write things this way, then it is quite clear how to extend this to a cubic approximation and a quartic approximation and so on.

Return to our example:

³¹ Any polynomial of degree two can be written in this form. For example, when $a = 1$, $3 + 2x + x^2 = 6 + 4(x-1) + (x-1)^2$.

Example 3.4.7

Use the quadratic approximation to estimate $e^{0.1}$.

Solution. Set $f(x) = e^x$ and $a = 0$ as before.

- To form the quadratic approximation we need $f(a)$, $f'(a)$ and $f''(a)$:

$$\begin{array}{ll} f(x) = e^x & f(0) = 1 \\ f'(x) = e^x & f'(0) = 1 \\ f''(x) = e^x & f''(0) = 1 \end{array}$$

- Then our quadratic approximation is

$$\begin{aligned} F(x) &= f(0) + xf'(0) + \frac{1}{2}x^2f''(0) = 1 + x + \frac{x^2}{2} \\ F(0.1) &= 1.105 \end{aligned}$$

Recall that $e^{0.1} = 1.105170918\dots$, so the quadratic approximation is quite accurate with very little effort.

Example 3.4.7

Before we go on, let us first introduce (or revise) some notation that will make our discussion easier.

►► Whirlwind Tour of Summation Notation

In the remainder of this section we will frequently need to write sums involving a large number of terms. Writing out the summands explicitly can become quite impractical — for example, say we need the sum of the first 11 squares:

$$1 + 2^2 + 3^2 + 4^2 + 5^2 + 6^2 + 7^2 + 8^2 + 9^2 + 10^2 + 11^2$$

This becomes tedious. Where the pattern is clear, we will often skip the middle few terms and instead write

$$1 + 2^2 + \dots + 11^2.$$

A far more precise way to write this is using Σ (capital-sigma) notation. For example, we can write the above sum as

$$\sum_{k=1}^{11} k^2$$

This is read as

The sum from k equals 1 to 11 of k^2 .

More generally

Notation 3.4.8.

Let $m \leq n$ be integers and let $f(x)$ be a function defined on the integers. Then we write

$$\sum_{k=m}^n f(k)$$

to mean the sum of $f(k)$ for k from m to n :

$$f(m) + f(m+1) + f(m+2) + \cdots + f(n-1) + f(n).$$

Similarly we write

$$\sum_{i=m}^n a_i$$

to mean

$$a_m + a_{m+1} + a_{m+2} + \cdots + a_{n-1} + a_n$$

for some set of coefficients $\{a_m, \dots, a_n\}$.

Consider the example

$$\sum_{k=3}^7 \frac{1}{k^2} = \frac{1}{3^2} + \frac{1}{4^2} + \frac{1}{5^2} + \frac{1}{6^2} + \frac{1}{7^2}$$

It is important to note that the right hand side of this expression evaluates to a number³²; it does not contain “ k ”. The summation index k is just a “dummy” variable and it does not have to be called k . For example

$$\sum_{k=3}^7 \frac{1}{k^2} = \sum_{i=3}^7 \frac{1}{i^2} = \sum_{j=3}^7 \frac{1}{j^2} = \sum_{\ell=3}^7 \frac{1}{\ell^2}$$

Also the summation index has no meaning outside the sum. For example

$$k \sum_{k=3}^7 \frac{1}{k^2}$$

has no mathematical meaning; It is gibberish³³.

32 Some careful addition shows it is $\frac{46181}{176400}$.

33 Or possibly gobbledygook. For a discussion of statements without meaning and why one should avoid them we recommend the book “Bendable learnings: the wisdom of modern management” by Don Watson.

3.4.4 ► Still Better Approximations — Taylor Polynomials

We can use the same strategy to generate still better approximations by polynomials³⁴ of any degree we like. As was the case with the approximations above, we determine the coefficients of the polynomial by requiring, that at the point $x = a$, the approximation and its first n derivatives agree with those of the original function.

Rather than simply moving to a cubic polynomial, let us try to write things in a more general way. We will consider approximating the function $f(x)$ using a polynomial, $T_n(x)$, of degree n — where n is a non-negative integer. As we discussed above, the algebra is easier if we write

$$\begin{aligned} T_n(x) &= c_0 + c_1(x-a) + c_2(x-a)^2 + \cdots + c_n(x-a)^n \\ &= \sum_{k=0}^n c_k(x-a)^k \end{aligned} \quad \text{using } \Sigma \text{ notation}$$

The above form³⁵³⁶ makes it very easy to evaluate this polynomial and its derivatives at $x = a$. Before we proceed, we remind the reader of some notation (see Notation [2.2.8](#)):

- Let $f(x)$ be a function and k be a positive integer. We can denote its k^{th} derivative with respect to x by

$$\frac{d^k f}{dx^k} \qquad \left(\frac{d}{dx} \right)^k f(x) \qquad f^{(k)}(x)$$

Additionally we will need

-
- 34 Polynomials are generally a good choice for an approximating function since they are so easy to work with. Depending on the situation other families of functions may be more appropriate. For example if you are approximating a periodic function, then sums of sines and cosines might be a better choice; this leads to Fourier series.
- 35 Any polynomial in x of degree n can also be expressed as a polynomial in $(x-a)$ of the same degree n and vice versa. So $T_n(x)$ really still is a polynomial of degree n .
- 36 Furthermore when x is close to a , $(x-a)^k$ decreases very quickly as k increases, which often makes the “high k ” terms in $T_n(x)$ very small. This can be a considerable advantage when building up approximations by adding more and more terms. If we were to rewrite $T_n(x)$ in the form $\sum_{k=0}^n b_k x^k$ the “high k ” terms would typically not be very small when x is close to a .
-

Definition 3.4.9 (Factorial).

Let n be a positive integer³⁷, then n -factorial, denoted $n!$, is the product

$$n! = n \times (n-1) \times \cdots \times 3 \times 2 \times 1$$

Further, we use the convention that

$$0! = 1$$

The first few factorials are

$$1! = 1$$

$$2! = 2$$

$$3! = 6$$

$$4! = 24$$

$$5! = 120$$

$$6! = 720$$

Now consider $T_n(x)$ and its derivatives:

$$\begin{aligned} T_n(x) &= c_0 + c_1(x-a) + c_2(x-a)^2 + c_3(x-a)^3 + \cdots + c_n(x-a)^n \\ T'_n(x) &= c_1 + 2c_2(x-a) + 3c_3(x-a)^2 + \cdots + nc_n(x-a)^{n-1} \\ T''_n(x) &= 2c_2 + 6c_3(x-a) + \cdots + n(n-1)c_n(x-a)^{n-2} \\ T'''_n(x) &= 6c_3 + \cdots + n(n-1)(n-2)c_n(x-a)^{n-3} \\ &\vdots \\ T_n^{(n)}(x) &= n! \cdot c_n \end{aligned}$$

Now notice that when we substitute $x = a$ into the above expressions only the constant terms survive and we get

$$\begin{aligned} T_n(a) &= c_0 \\ T'_n(a) &= c_1 \\ T''_n(a) &= 2 \cdot c_2 \\ T'''_n(a) &= 6 \cdot c_3 \\ &\vdots \\ T_n^{(n)}(a) &= n! \cdot c_n \end{aligned}$$

So now if we want to set the coefficients of $T_n(x)$ so that it agrees with $f(x)$ at $x = a$ then we need

$$T_n(a) = c_0 = f(a) \qquad c_0 = f(a) = \frac{1}{0!}f(a)$$

³⁷ It is actually possible to define the factorial of positive real numbers and even negative numbers but it requires more advanced calculus and is outside the scope of this course. The interested reader should look up the Gamma function.

We also want the first n derivatives of $T_n(x)$ to agree with the derivatives of $f(x)$ at $x = a$, so

$$\begin{aligned} T'_n(a) &= c_1 = f'(a) & c_1 &= f'(a) = \frac{1}{1!}f'(a) \\ T''_n(a) &= 2 \cdot c_2 = f''(a) & c_2 &= \frac{1}{2}f''(a) = \frac{1}{2!}f''(a) \\ T'''_n(a) &= 6 \cdot c_3 = f'''(a) & c_3 &= \frac{1}{6}f'''(a) = \frac{1}{3!}f'''(a) \end{aligned}$$

More generally, making the k^{th} derivatives agree at $x = a$ requires :

$$T_n^{(k)}(a) = k! \cdot c_k = f^{(k)}(a) \quad c_k = \frac{1}{k!}f^{(k)}(a)$$

And finally the n^{th} derivative:

$$T_n^{(n)}(a) = n! \cdot c_n = f^{(n)}(a) \quad c_n = \frac{1}{n!}f^{(n)}(a)$$

Putting this all together we have

Equation 3.4.10 (Taylor polynomial).

$$\begin{aligned} f(x) \approx T_n(x) &= f(a) + f'(a)(x-a) + \frac{1}{2}f''(a) \cdot (x-a)^2 + \cdots + \frac{1}{n!}f^{(n)}(a) \cdot (x-a)^n \\ &= \sum_{k=0}^n \frac{1}{k!}f^{(k)}(a) \cdot (x-a)^k \end{aligned}$$

Let us formalise this definition.

Definition 3.4.11 (Taylor polynomial).

Let a be a constant and let n be a non-negative integer. The n^{th} order³⁸ Taylor polynomial for $f(x)$ about $x = a$ is

$$T_n(x) = \sum_{k=0}^n \frac{1}{k!}f^{(k)}(a) \cdot (x-a)^k.$$

The special case $a = 0$ is called a Maclaurin³⁹ polynomial.

Before we proceed with some examples, a couple of remarks are in order.

38 It is sometimes called the n^{th} degree Taylor polynomial, but its degree will actually be less than n if $f^{(n)}(a) = 0$.

39 The polynomials are named after Brook Taylor who devised a general method for constructing them in 1715. Slightly later, Colin Maclaurin made extensive use of the special case $a = 0$ (with attribution of the general case to Taylor) and it is now named after him. The special case of $a = 0$ was worked on previously by James Gregory and Isaac Newton, and some specific cases were known to the 14th century Indian mathematician Madhava of Sangamagrama.

- While we can compute a Taylor polynomial about any a -value (providing the derivatives exist), in order to be a *useful* approximation, we must be able to compute $f(a), f'(a), \dots, f^{(n)}(a)$ easily. This means we must choose the point a with care. Indeed for many functions the choice $a = 0$ is very natural — hence the prominence of Maclaurin polynomials.
- If we have computed the approximation $T_n(x)$, then we can readily extend this to the next Taylor polynomial $T_{n+1}(x)$ since

$$T_{n+1}(x) = T_n(x) + \frac{1}{(n+1)!} f^{(n+1)}(a) \cdot (x-a)^{n+1}$$

This is very useful if we discover that $T_n(x)$ is an insufficient approximation, because then we can produce $T_{n+1}(x)$ without having to start again from scratch.

3.4.5 ► Some Examples

Let us return to our running example of e^x :

Example 3.4.12

The constant, linear and quadratic approximations we used above were the first few Maclaurin polynomial approximations of e^x . That is

$$T_0(x) = 1 \qquad T_1(x) = 1 + x \qquad T_2(x) = 1 + x + \frac{x^2}{2}$$

Since $\frac{d}{dx}e^x = e^x$, the Maclaurin polynomials are very easy to compute. Indeed this invariance under differentiation means that

$$\begin{aligned} f^{(n)}(x) &= e^x & n = 0, 1, 2, \dots & \text{so} \\ f^{(n)}(0) &= 1 \end{aligned}$$

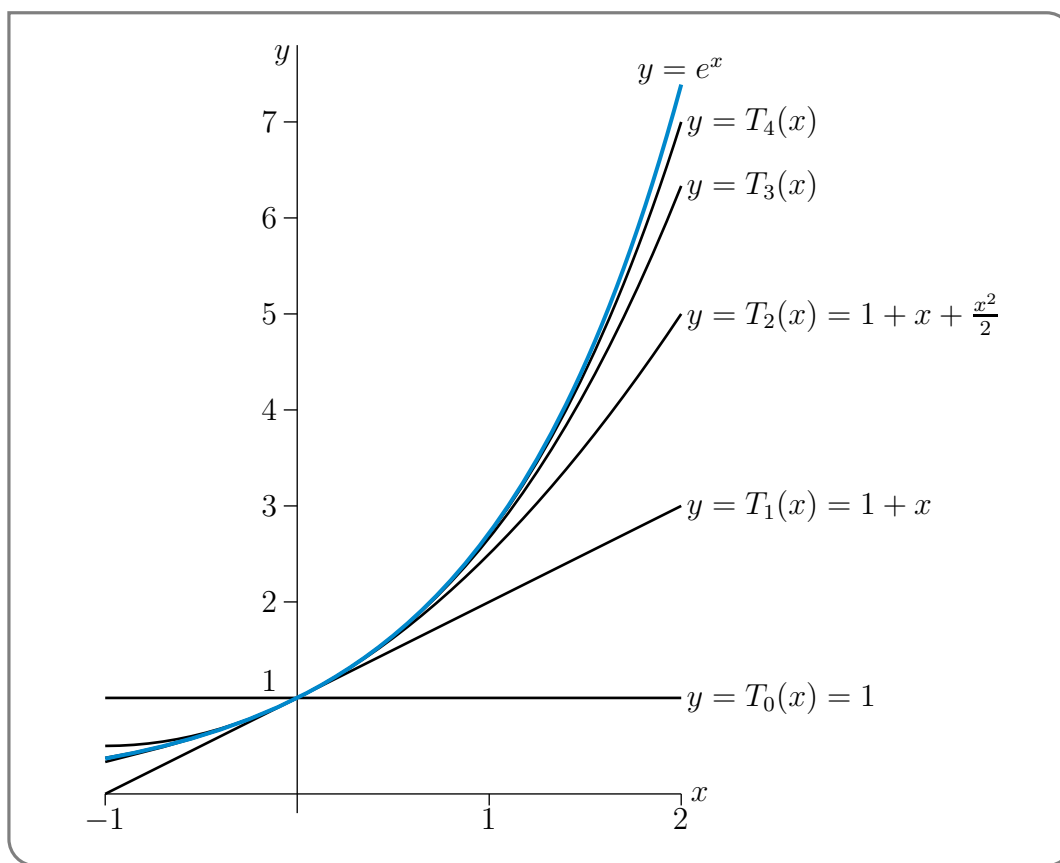
Substituting this into equation (3.4.10) we get

$$T_n(x) = \sum_{k=0}^n \frac{1}{k!} x^k$$

Thus we can write down the seventh Maclaurin polynomial very easily:

$$T_7(x) = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} + \frac{x^5}{120} + \frac{x^6}{720} + \frac{x^7}{5040}$$

The following figure contains sketches of the graphs of e^x and its Taylor polynomials $T_n(x)$ for $n = 0, 1, 2, 3, 4$.



Also notice that if we use $T_7(1)$ to approximate the value of e^1 we obtain:

$$\begin{aligned} e^1 \approx T_7(1) &= 1 + 1 + \frac{1}{2} + \frac{1}{6} + \frac{1}{24} + \frac{1}{120} + \frac{1}{720} + \frac{1}{5040} \\ &= \frac{685}{252} = 2.718253968 \dots \end{aligned}$$

The true value of e is $2.718281828 \dots$, so the approximation has an error of about 3×10^{-5} .

Under the assumption that the accuracy of the approximation improves with n (an assumption we examine in Subsection 3.4.8 below) we can see that the approximation of e above can be improved by adding more and more terms. Indeed this is how the expression for e in equation (2.7.3) in Section 2.7 comes about.

Example 3.4.12

Now that we have examined Maclaurin polynomials for e^x we should take a look at $\log x$. Notice that we cannot compute a Maclaurin polynomial for $\log x$ since it is not defined at $x = 0$.

Example 3.4.13

Compute the 5th order Taylor polynomial for $\log x$ about $x = 1$.

Solution. We have been told $a = 1$ and fifth order, so we should start by writing down the

function and its first five derivatives:

$$\begin{array}{ll}
 f(x) = \log x & f(1) = \log 1 = 0 \\
 f'(x) = \frac{1}{x} & f'(1) = 1 \\
 f''(x) = \frac{-1}{x^2} & f''(1) = -1 \\
 f'''(x) = \frac{2}{x^3} & f'''(1) = 2 \\
 f^{(4)}(x) = \frac{-6}{x^4} & f^{(4)}(1) = -6 \\
 f^{(5)}(x) = \frac{24}{x^5} & f^{(5)}(1) = 24
 \end{array}$$

Substituting this into equation (3.4.10) gives

$$\begin{aligned}
 T_5(x) &= 0 + 1 \cdot (x-1) + \frac{1}{2} \cdot (-1) \cdot (x-1)^2 + \frac{1}{6} \cdot 2 \cdot (x-1)^3 + \frac{1}{24} \cdot (-6) \cdot (x-1)^4 + \frac{1}{120} \cdot 24 \cdot (x-1)^5 \\
 &= (x-1) - \frac{1}{2}(x-1)^2 + \frac{1}{3}(x-1)^3 - \frac{1}{4}(x-1)^4 + \frac{1}{5}(x-1)^5
 \end{aligned}$$

Again, it is not too hard to generalise the above work to find the Taylor polynomial of order n : With a little work one can show that

$$T_n(x) = \sum_{k=1}^n \frac{(-1)^{k+1}}{k} (x-1)^k.$$

Example 3.4.13

For cosine:

Example 3.4.14

Find the 4th order Maclaurin polynomial for $\cos x$.

Solution. We have $a = 0$ and we need to find the first 4 derivatives of $\cos x$.

$$\begin{array}{ll}
 f(x) = \cos x & f(0) = 1 \\
 f'(x) = -\sin x & f'(0) = 0 \\
 f''(x) = -\cos x & f''(0) = -1 \\
 f'''(x) = \sin x & f'''(0) = 0 \\
 f^{(4)}(x) = \cos x & f^{(4)}(0) = 1
 \end{array}$$

Substituting this into equation (3.4.10) gives

$$\begin{aligned}
 T_4(x) &= 1 + 1 \cdot (0) \cdot x + \frac{1}{2} \cdot (-1) \cdot x^2 + \frac{1}{6} \cdot 0 \cdot x^3 + \frac{1}{24} \cdot (1) \cdot x^4 \\
 &= 1 - \frac{x^2}{2} + \frac{x^4}{24}
 \end{aligned}$$

Notice that since the 4th derivative of $\cos x$ is $\cos x$ again, we also have that the fifth derivative is the same as the first derivative, and the sixth derivative is the same as the second derivative and so on. Hence the next four derivatives are

$$\begin{array}{ll} f^{(4)}(x) = \cos x & f^{(4)}(0) = 1 \\ f^{(5)}(x) = -\sin x & f^{(5)}(0) = 0 \\ f^{(6)}(x) = -\cos x & f^{(6)}(0) = -1 \\ f^{(7)}(x) = \sin x & f^{(7)}(0) = 0 \\ f^{(8)}(x) = \cos x & f^{(8)}(0) = 1 \end{array}$$

Using this we can find the 8th order Maclaurin polynomial:

$$T_8(x) = 1 - \frac{x^2}{2} + \frac{x^4}{24} - \frac{x^6}{6!} + \frac{x^8}{8!}$$

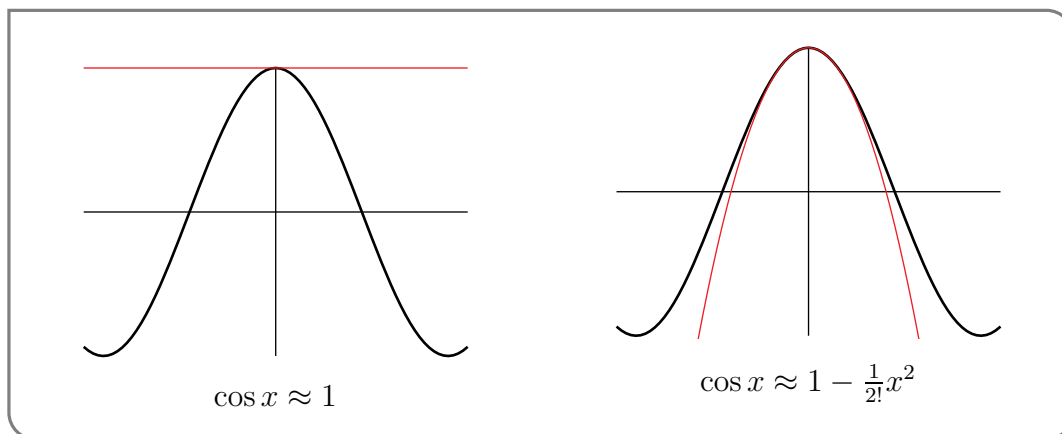
Continuing this process gives us the $2n^{\text{th}}$ Maclaurin polynomial

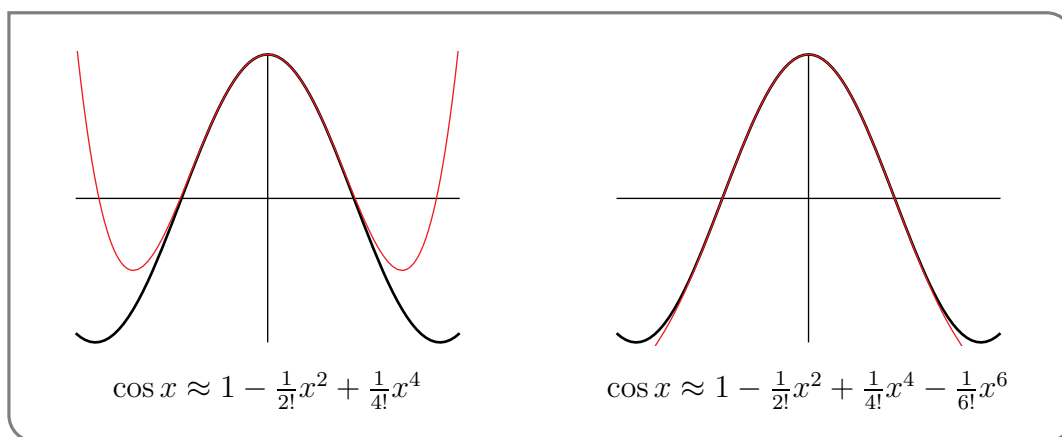
$$T_{2n}(x) = \sum_{k=0}^n \frac{(-1)^k}{(2k)!} \cdot x^{2k}$$

Warning 3.4.15.

The above formula only works when x is measured in radians, because all of our derivative formulae for trig functions were developed under the assumption that angles are measured in radians.

Below we plot $\cos x$ against its first few Maclaurin polynomial approximations:





Example 3.4.15

The above work is quite easily recycled to get the Maclaurin polynomial for sine:

Example 3.4.16

Find the 5th order Maclaurin polynomial for $\sin x$.

Solution. We could simply work as before and compute the first five derivatives of $\sin x$. But set $g(x) = \sin x$ and notice that $g(x) = -f'(x)$, where $f(x) = \cos x$. Then we have

$$\begin{aligned} g(0) &= -f'(0) = 0 \\ g'(0) &= -f''(0) = 1 \\ g''(0) &= -f'''(0) = 0 \\ g'''(0) &= -f^{(4)}(0) = -1 \\ g^{(4)}(0) &= -f^{(5)}(0) = 0 \\ g^{(5)}(0) &= -f^{(6)}(0) = 1 \end{aligned}$$

Hence the required Maclaurin polynomial is

$$T_5(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!}$$

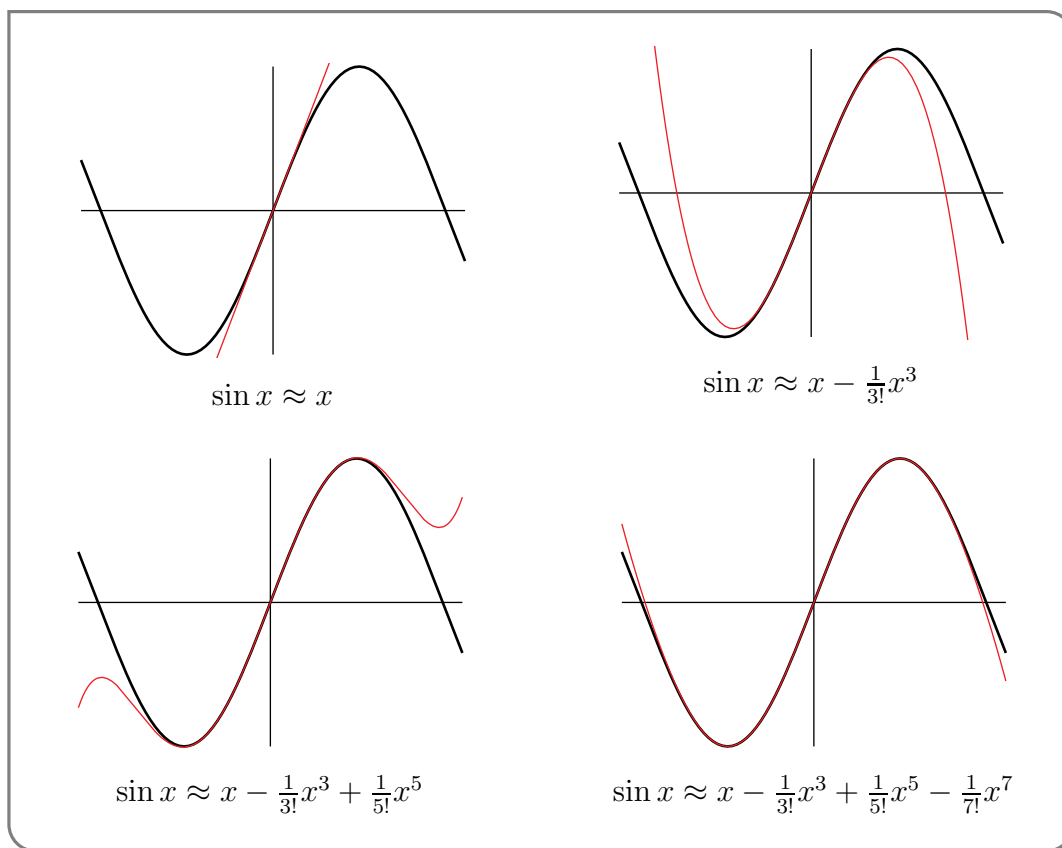
Just as we extended to the $2n^{\text{th}}$ Maclaurin polynomial for cosine, we can also extend our work to compute the $(2n+1)^{\text{th}}$ Maclaurin polynomial for sine:

$$T_{2n+1}(x) = \sum_{k=0}^n \frac{(-1)^k}{(2k+1)!} \cdot x^{2k+1}$$

Warning 3.4.17.

The above formula only works when x is measured in radians, because all of our derivative formulae for trig functions were developed under the assumption that angles are measured in radians.

Below we plot $\sin x$ against its first few Maclaurin polynomial approximations.



Example 3.4.17

To get an idea of how good these Taylor polynomials are at approximating $\sin$ and $\cos$, let's concentrate on $\sin x$ and consider x 's whose magnitude $|x| \leq 1$. There are tricks that you can employ⁴⁰ to evaluate sine and cosine at values of x outside this range.

If $|x| \leq 1$ radians⁴¹, then the magnitudes of the successive terms in the Taylor polynomials for $\sin x$ are bounded by

$$\begin{array}{lll} |x| \leq 1 & \frac{1}{3!}|x|^3 \leq \frac{1}{6} & \frac{1}{5!}|x|^5 \leq \frac{1}{120} \approx 0.0083 \\ \frac{1}{7!}|x|^7 \leq \frac{1}{7!} \approx 0.0002 & \frac{1}{9!}|x|^9 \leq \frac{1}{9!} \approx 0.000003 & \frac{1}{11!}|x|^{11} \leq \frac{1}{11!} \approx 0.000000025 \end{array}$$

From these inequalities, and the graphs on the previous pages, it certainly looks like, for x not too large, even relatively low order Taylor polynomials give very good approximations. In Section 3.4.8 we'll see how to get rigorous error bounds on our Taylor polynomial approximations.

- 40 If you are writing software to evaluate $\sin x$, you can always use the trig identity $\sin(x) = \sin(x - 2n\pi)$, to easily restrict to $|x| \leq \pi$. You can then use the trig identity $\sin(x) = -\sin(x \pm \pi)$ to reduce to $|x| \leq \frac{\pi}{2}$. Finally you can use the trig identity $\sin(x) = \mp \cos(\frac{\pi}{2} \pm x)$ to reduce to $|x| \leq \frac{\pi}{4} < 1$.
- 41 Recall that the derivative formulae that we used to derive the Taylor polynomials are valid only when x is in radians. The restriction $-1 \leq x \leq 1$ radians translates to angles bounded by $\frac{180}{\pi} \approx 57^\circ$.

3.4.6 ▶ Estimating Change and Δx , Δy Notation

Suppose that we have two variables x and y that are related by $y = f(x)$, for some function f . One of the most important applications of calculus is to help us understand what happens to y when we make a small change in x .

Notation 3.4.18.

Let x, y be variables related by a function f . That is $y = f(x)$. Then we denote a small change in the variable x by Δx (read as “delta x ”). The corresponding small change in the variable y is denoted Δy (read as “delta y ”).

$$\Delta y = f(x + \Delta x) - f(x)$$

In many situations we do not need to compute Δy exactly and are instead happy with an approximation. Consider the following example.

Example 3.4.19

Let x be the number of cars manufactured per week in some factory and let y the cost of manufacturing those x cars. Given that the factory currently produces a cars per week, we would like to estimate the increase in cost if we make a small change in the number of cars produced.

Solution. We are told that a is the number of cars currently produced per week; the cost of production is then $f(a)$.

- Say the number of cars produced is changed from a to $a + \Delta x$ (where Δx is some small number).
- As x undergoes this change, the costs change from $y = f(a)$ to $f(a + \Delta x)$. Hence

$$\Delta y = f(a + \Delta x) - f(a)$$

- We can estimate this change using a linear approximation. Substituting $x = a + \Delta x$ into the equation (3.4.3) yields the approximation

$$f(a + \Delta x) \approx f(a) + f'(a)(a + \Delta x - a)$$

and consequently the approximation

$$\Delta y = f(a + \Delta x) - f(a) \approx f(a) + f'(a)\Delta x - f(a)$$

simplifies to the following neat estimate of Δy :

Equation 3.4.20 (Linear approximation of Δy).

$$\Delta y \approx f'(a)\Delta x$$

- In the automobile manufacturing example, when the production level is a cars per week, increasing the production level by Δx will cost approximately $f'(a)\Delta x$. The additional cost per additional car, $f'(a)$, is called the “marginal cost” of a car.
- If we instead use the quadratic approximation (given by equation (3.4.6)) then we estimate

$$f(a + \Delta x) \approx f(a) + f'(a)\Delta x + \frac{1}{2}f''(a)\Delta x^2$$

and so

$$\Delta y = f(a + \Delta x) - f(a) \approx f(a) + f'(a)\Delta x + \frac{1}{2}f''(a)\Delta x^2 - f(a)$$

which simplifies to

Equation 3.4.21 (Quadratic approximation of Δy).

$$\Delta y \approx f'(a)\Delta x + \frac{1}{2}f''(a)\Delta x^2$$

Example 3.4.21

3.4.7 ► Further Examples

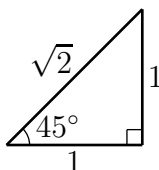
In this subsection we give further examples of computation and use of Taylor approximations.

Example 3.4.22

Estimate $\tan 46^\circ$, using the constant-, linear- and quadratic-approximations (equations (3.4.1), (3.4.3) and (3.4.6)).

Solution. Note that we need to be careful to translate angles measured in degrees to radians.

- Set $f(x) = \tan x$, $x = 46 \frac{\pi}{180}$ radians and $a = 45 \frac{\pi}{180} = \frac{\pi}{4}$ radians. This is a good choice for a because
 - $a = 45^\circ$ is close to $x = 46^\circ$. As noted above, it is generally the case that the closer x is to a , the better various approximations will be.
 - We know the values of all trig functions at 45° .
- Now we need to compute f and its first two derivatives at $x = a$. It is a good time to recall the special $1 : 1 : \sqrt{2}$ triangle



So

$$\begin{aligned} f(x) &= \tan x & f(\pi/4) &= 1 \\ f'(x) &= \sec^2 x = \frac{1}{\cos^2 x} & f'(\pi/4) &= \frac{1}{1/\sqrt{2}^2} = 2 \\ f''(x) &= \frac{2 \sin x}{\cos^3 x} & f''(\pi/4) &= \frac{2/\sqrt{2}}{1/\sqrt{2}^3} = 4 \end{aligned}$$

- As $x - a = 46 \frac{\pi}{180} - 45 \frac{\pi}{180} = \frac{\pi}{180}$ radians, the three approximations are

$$\begin{aligned} f(x) &\approx f(a) & &= 1 \\ f(x) &\approx f(a) + f'(a)(x - a) & = 1 + 2 \frac{\pi}{180} &= 1.034907 \\ f(x) &\approx f(a) + f'(a)(x - a) + \frac{1}{2} f''(a)(x - a)^2 = 1 + 2 \frac{\pi}{180} + \frac{1}{2} 4 \left(\frac{\pi}{180} \right)^2 &= 1.035516 \end{aligned}$$

For comparison purposes, $\tan 46^\circ$ really is 1.035530 to 6 decimal places.

Example 3.4.22

Warning 3.4.23.

All of our derivative formulae for trig functions were developed under the assumption that angles are measured in radians. Those derivatives appeared in the approximation formulae that we used in Example 3.4.22, so we were obliged to express $x - a$ in radians.

Example 3.4.24

Suppose that you are ten meters from a vertical pole. You were contracted to measure the height of the pole. You can't take it down or climb it. So you measure the angle subtended by the top of the pole. You measure $\theta = 30^\circ$, which gives

$$h = 10 \tan 30^\circ = \frac{10}{\sqrt{3}} \approx 5.77\text{m}$$

This is just standard trigonometry — if we know the angle exactly then we know the height exactly.

However, in the “real world” angles are hard to measure with such precision. If the contract requires you the measurement of the pole to be accurate within 10 cm, how accurate does your measurement of the angle θ need to be?

Solution. For simplicity⁴², we are going to assume that the pole is perfectly straight and perfectly vertical and that your distance from the pole was exactly 10 m.

- Write $\theta = \theta_0 + \Delta\theta$ where θ is the exact angle, θ_0 is the measured angle and $\Delta\theta$ is the error.

42 Mathematicians love assumptions that let us tame the real world.

- Similarly write $h = h_0 + \Delta h$, where h is the exact height and $h_0 = \frac{10}{\sqrt{3}}$ is the computed height. Their difference, Δh , is the error.
- Then

$$\begin{aligned} h_0 &= 10 \tan \theta_0 & h_0 + \Delta h &= 10 \tan(\theta_0 + \Delta \theta) \\ \Delta h &= 10 \tan(\theta_0 + \Delta \theta) - 10 \tan \theta_0 \end{aligned}$$

We could attempt to solve this equation for $\Delta \theta$ in terms of Δh — but it is far simpler to approximate Δh using the linear approximation in equation [3.4.20](#).

- To use equation [3.4.20](#), replace y with h , x with θ and a with θ_0 . Our function $f(\theta) = 10 \tan \theta$ and $\theta_0 = 30^\circ = \pi/6$ radians. Then

$$\Delta y \approx f'(a)\Delta x \quad \text{becomes} \quad \Delta h \approx f'(\theta_0)\Delta \theta$$

Since $f(\theta) = 10 \tan \theta$, $f'(\theta) = 10 \sec^2 \theta$ and

$$f'(\theta_0) = 10 \sec^2(\pi/6) = 10 \cdot \left(\frac{2}{\sqrt{3}}\right)^2 = \frac{40}{3}$$

- Putting things together gives

$$\Delta h \approx f'(\theta_0)\Delta \theta \quad \text{becomes} \quad \Delta h \approx \frac{40}{3}\Delta \theta$$

We can then solve this equation for $\Delta \theta$ in terms of Δh :

$$\Delta \theta \approx \frac{3}{40}\Delta h$$

- We are told that we must have $|\Delta h| < 0.1$, so we must have

$$|\Delta \theta| \leq \frac{3}{400}$$

This is measured in radians, so converting back to degrees

$$\frac{3}{400} \cdot \frac{180}{\pi} = 0.43^\circ$$

Example 3.4.24

Definition 3.4.25.

Suppose that you measure, approximately, some quantity. Suppose that the exact value of that quantity is Q_0 and that your measurement yielded $Q_0 + \Delta Q$. Then $|\Delta Q|$ is called the absolute error of the measurement and $100 \frac{|\Delta Q|}{Q_0}$ is called the percentage error of the measurement. As an example, if the exact value is 4 and the measured value is 5, then the absolute error is $|5 - 4| = 1$ and the percentage error is $100 \frac{|5-4|}{4} = 25$. That is, the error, 1, was 25% of the exact value, 4.

Example 3.4.26

Suppose that the radius of a sphere has been measured with a percentage error of at most $\varepsilon\%$. Find the corresponding approximate percentage errors in the surface area and volume of the sphere.

Solution. We need to be careful in this problem to convert between absolute and percentage errors correctly.

- Suppose that the exact radius is r_0 and that the measured radius is $r_0 + \Delta r$.
- Then the absolute error in the measurement is $|\Delta r|$ and, by definition, the percentage error is $100 \frac{|\Delta r|}{r_0}$. We are told that $100 \frac{|\Delta r|}{r_0} \leq \varepsilon$.
- The surface area⁴³ of a sphere of radius r is $A(r) = 4\pi r^2$. The error in the surface area computed with the measured radius is

$$\begin{aligned}\Delta A &= A(r_0 + \Delta r) - A(r_0) \approx A'(r_0)\Delta r \\ &= 8\pi r_0 \Delta r\end{aligned}$$

where we have made use of the linear approximation, equation (3.4.20).

- The corresponding percentage error is then

$$100 \frac{|\Delta A|}{A(r_0)} \approx 100 \frac{|A'(r_0)\Delta r|}{A(r_0)} = 100 \frac{8\pi r_0 |\Delta r|}{4\pi r_0^2} = 2 \times 100 \frac{|\Delta r|}{r_0} \leq 2\varepsilon$$

- The volume of a sphere⁴⁴ of radius r is $V(r) = \frac{4}{3}\pi r^3$. The error in the volume computed with the measured radius is

$$\begin{aligned}\Delta V &= V(r_0 + \Delta r) - V(r_0) \approx V'(r_0)\Delta r \\ &= 4\pi r_0^2 \Delta r\end{aligned}$$

where we have again made use of the linear approximation, equation (3.4.20).

⁴³ We do not expect you to remember the surface areas of solids for this course.

⁴⁴ We do expect you to remember the formula for the volume of a sphere.

- The corresponding percentage error is

$$100 \frac{|\Delta V|}{V(r_0)} \approx 100 \frac{|V'(r_0)\Delta r|}{V(r_0)} = 100 \frac{4\pi r_0^2 |\Delta r|}{4\pi r_0^3/3} = 3 \times 100 \frac{|\Delta r|}{r_0} \leq 3\varepsilon$$

We have just computed an approximation to ΔV . This problem is actually sufficiently simple that we can compute ΔV exactly:

$$\Delta V = V(r_0 + \Delta r) - V(r_0) = \frac{4}{3}\pi(r_0 + \Delta r)^3 - \frac{4}{3}\pi r_0^3$$

- Applying $(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$ with $a = r_0$ and $b = \Delta r$, gives

$$\begin{aligned} V(r_0 + \Delta r) - V(r_0) &= \frac{4}{3}\pi \left[r_0^3 + 3r_0^2\Delta r + 3r_0(\Delta r)^2 + (\Delta r)^3 \right] - \frac{4}{3}\pi r_0^3 \\ &= \frac{4}{3}\pi [3r_0^2\Delta r + 3r_0(\Delta r)^2 + (\Delta r)^3] \end{aligned}$$

- Thus the difference between the exact error and the linear approximation to the error is obtained by retaining only the last two terms in the square brackets. This has magnitude

$$\frac{4}{3}\pi |3r_0(\Delta r)^2 + (\Delta r)^3| = \frac{4}{3}\pi |3r_0 + \Delta r|(\Delta r)^2$$

or in percentage terms

$$\begin{aligned} 100 \cdot \frac{1}{\frac{4}{3}\pi r_0^3} \cdot \frac{4}{3}\pi |3r_0(\Delta r)^2 + (\Delta r)^3| &= 100 \left| 3 \frac{\Delta r^2}{r_0^2} + \frac{\Delta r^3}{r_0^3} \right| \\ &= \left(100 \frac{3\Delta r}{r_0} \right) \cdot \left(\frac{\Delta r}{r_0} \right) \left| 1 + \frac{\Delta r}{3r_0} \right| \\ &\leq 3\varepsilon \left(\frac{\varepsilon}{100} \right) \cdot \left(1 + \frac{\varepsilon}{300} \right) \end{aligned}$$

Since ε is small, we can assume that $1 + \frac{\varepsilon}{300} \approx 1$. Hence the difference between the exact error and the linear approximation of the error is roughly a factor of $\frac{\varepsilon}{100}$ smaller than the linear approximation 3ε .

- As an aside, notice that if we argue that Δr is very small and so we can ignore terms involving $(\Delta r)^2$ and $(\Delta r)^3$ as being really really small, then we obtain

$$\begin{aligned} V(r_0 + \Delta r) - V(r_0) &= \frac{4}{3}\pi [3r_0^2\Delta r + \underbrace{3r_0(\Delta r)^2 + (\Delta r)^3}_{\text{really really small}}] \\ &\approx \frac{4}{3}\pi \cdot 3r_0^2\Delta r = 4\pi r_0^2\Delta r \end{aligned}$$

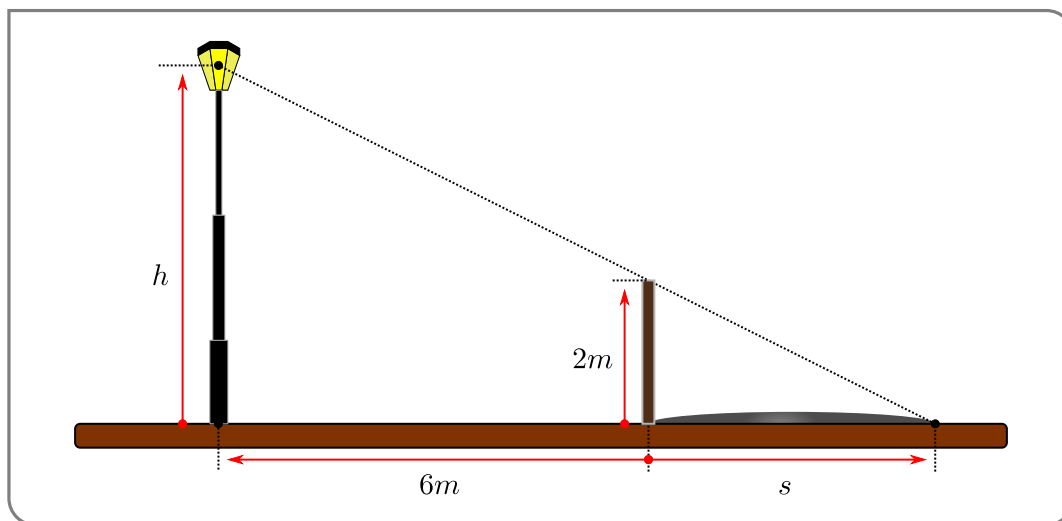
which is precisely the result of our linear approximation above.

Example 3.4.26

Example 3.4.27

To compute the height h of a lamp post, the length s of the shadow of a two meter pole is measured. The pole is 6 m from the lamp post. If the length of the shadow was measured to be 4 m, with an error of at most one cm, find the height of the lamp post and estimate the percentage error in the height.

Solution. We should first draw a picture [45](#)



- By similar triangles we see that

$$\frac{2}{s} = \frac{h}{6+s}$$

from which we can isolate h as a function of s :

$$h = \frac{2(6+s)}{s} = \frac{12}{s} + 2$$

- The length of the shadow was measured to be $s_0 = 4$ m. The corresponding height of the lamp post is

$$h_0 = \frac{12}{4} + 2 = 5m$$

- If the error in the measurement of the length of the shadow was Δs , then the exact shadow length was $s = s_0 + \Delta s$ and the exact lamp post height is $h = f(s_0 + \Delta s)$, where $f(s) = \frac{12}{s} + 2$. The error in the computed lamp post height is

$$\Delta h = h - h_0 = f(s_0 + \Delta s) - f(s_0)$$

- We can then make a linear approximation of this error using equation [\(3.4.20\)](#):

$$\Delta h \approx f'(s_0)\Delta s = -\frac{12}{s_0^2}\Delta s = -\frac{12}{4^2}\Delta s$$

45 We get to reuse that nice lamp post picture from Example [3.2.4](#)

- We are told that $|\Delta s| \leq \frac{1}{100}$ m. Consequently, approximately,

$$|\Delta h| \leq \frac{12}{4^2} \frac{1}{100} = \frac{3}{400}$$

The percentage error is then approximately

$$100 \frac{|\Delta h|}{h_0} \leq 100 \frac{3}{400 \times 5} = 0.15\%$$

Example 3.4.27

3.4.8 ► The Error in the Taylor Polynomial Approximations

Any time you make an approximation, it is desirable to have some idea of the size of the error you introduced. That is, we would like to know the difference $R(x)$ between the original function $f(x)$ and our approximation $F(x)$:

$$R(x) = f(x) - F(x).$$

Of course if we know $R(x)$ exactly, then we could recover $f(x) = F(x) + R(x)$ — so this is an unrealistic hope. In practice we would simply like to bound $R(x)$:

$$|R(x)| = |f(x) - F(x)| \leq M$$

where (hopefully) M is some small number. It is worth stressing that we do not need the tightest possible value of M , we just need a relatively easily computed M that isn't too far off the true value of $|f(x) - F(x)|$.

We will now develop a formula for the error introduced by the constant approximation, equation (3.4.1) (developed back in Section 3.4.1)

$$f(x) \approx f(a) = T_0(x) \quad 0^{\text{th}} \text{ Taylor polynomial}$$

The resulting formula can be used to get an upper bound on the size of the error $|R(x)|$.

The main ingredient we will need is the Mean-Value Theorem (Theorem 2.13.4) — so we suggest you quickly revise it. Consider the following obvious statement:

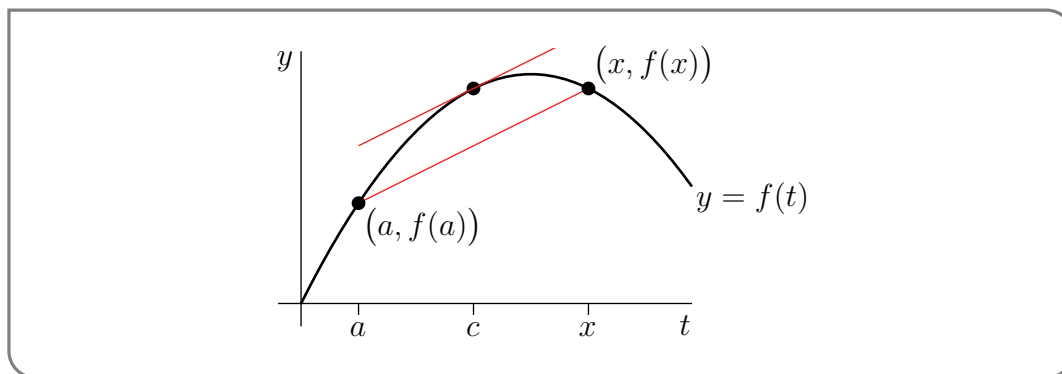
$$\begin{aligned} f(x) &= f(x) && \text{now some sneaky manipulations} \\ &= f(a) + (f(x) - f(a)) \\ &= \underbrace{f(a)}_{=T_0(x)} + (f(x) - f(a)) \cdot \underbrace{\frac{x-a}{x-a}}_{=1} \\ &= T_0(x) + \underbrace{\frac{f(x) - f(a)}{x-a}}_{\text{looks familiar}} \cdot (x-a) \end{aligned}$$

Indeed, this equation is important in the discussion that follows, so we'll highlight it

Equation 3.4.28 (We will need it again soon).

$$f(x) = T_0(x) + \left[\frac{f(x) - f(a)}{x - a} \right] (x - a)$$

The coefficient $\frac{f(x) - f(a)}{x - a}$ of $(x - a)$ is the average slope of $f(t)$ as t moves from $t = a$ to $t = x$. We can picture this as the slope of the secant joining the points $(a, f(a))$ and $(x, f(x))$ in the sketch below.



As t moves from a to x , the instantaneous slope $f'(t)$ keeps changing. Sometimes $f'(t)$ might be larger than the average slope $\frac{f(x) - f(a)}{x - a}$, and sometimes $f'(t)$ might be smaller than the average slope $\frac{f(x) - f(a)}{x - a}$. However, by the Mean-Value Theorem (Theorem 2.13.4), there must be some number c , strictly between a and x , for which $f'(c) = \frac{f(x) - f(a)}{x - a}$ exactly.

Substituting this into formula (3.4.28) gives

Equation 3.4.29 (Towards the error).

$$f(x) = T_0(x) + f'(c)(x - a) \quad \text{for some } c \text{ strictly between } a \text{ and } x$$

Notice that this expression as it stands is not quite what we want. Let us massage this around a little more into a more useful form

Equation 3.4.30 (The error in constant approximation).

$$f(x) - T_0(x) = f'(c) \cdot (x - a) \quad \text{for some } c \text{ strictly between } a \text{ and } x$$

Notice that the MVT doesn't tell us the value of c , however we do know that it lies strictly between x and a . So if we can get a good bound on $f'(c)$ on this interval then we can get a good bound on the error.

Example 3.4.31

Let us return to Example 3.4.2, and we'll try to bound the error in our approximation of $e^{0.1}$.

- Recall that $f(x) = e^x$, $a = 0$ and $T_0(x) = e^0 = 1$.
- Then by equation (3.4.30)

$$e^{0.1} - T_0(0.1) = f'(c) \cdot (0.1 - 0) \quad \text{with } 0 < c < 0.1$$

- Now $f'(c) = e^c$, so we need to bound e^c on $(0, 0.1)$. Since e^c is an increasing function, we know that

$$e^0 < f'(c) < e^{0.1} \quad \text{when } 0 < c < 0.1$$

So one is tempted to write that

$$\begin{aligned} |e^{0.1} - T_0(0.1)| &= |R(x)| = |f'(c)| \cdot (0.1 - 0) \\ &< e^{0.1} \cdot 0.1 \end{aligned}$$

And while this is true, it is rather circular. We have just bounded the error in our approximation of $e^{0.1}$ by $\frac{1}{10}e^{0.1}$ — if we actually knew $e^{0.1}$ then we wouldn't need to estimate it!

- While we don't know $e^{0.1}$ exactly, we do know⁴⁶ that $1 = e^0 < e^{0.1} < e^1 < 3$. This gives us

$$|R(0.1)| < 3 \times 0.1 = 0.3$$

That is — the error in our approximation of $e^{0.1}$ is no greater than 0.3. Recall that we don't need the error exactly, we just need a good idea of how large it actually is.

- In fact the real error here is

$$|e^{0.1} - T_0(0.1)| = |e^{0.1} - 1| = 0.1051709 \dots$$

so we have over-estimated the error by a factor of 3.

But we can actually go a little further here — we can bound the error above and below. If we do not take absolute values, then since

$$e^{0.1} - T_0(0.1) = f'(c) \cdot 0.1 \quad \text{and } 1 < f'(c) < 3$$

we can write

$$1 \times 0.1 \leq (e^{0.1} - T_0(0.1)) \leq 3 \times 0.1$$

so

$$\begin{aligned} T_0(0.1) + 0.1 &\leq e^{0.1} \leq T_0(0.1) + 0.3 \\ 1.1 &\leq e^{0.1} \leq 1.3 \end{aligned}$$

So while the upper bound is weak, the lower bound is quite tight.

46 Oops! Do we really know that $e < 3$? We haven't proved it. We will do so soon.

Example 3.4.31

There are formulae similar to equation (3.4.29), that can be used to bound the error in our other approximations; all are based on generalisations of the MVT. The next one — for linear approximations — is

$$f(x) = \underbrace{f(a) + f'(a)(x-a)}_{=T_1(x)} + \frac{1}{2}f''(c)(x-a)^2 \quad \text{for some } c \text{ strictly between } a \text{ and } x$$

which we can rewrite in terms of $T_1(x)$:

Equation 3.4.32 (The error in linear approximation).

$$f(x) - T_1(x) = \frac{1}{2}f''(c)(x-a)^2 \quad \text{for some } c \text{ strictly between } a \text{ and } x$$

It implies that the error that we make when we approximate $f(x)$ by $T_1(x) = f(a) + f'(a)(x-a)$ is exactly $\frac{1}{2}f''(c)(x-a)^2$ for some c strictly between a and x .

More generally

$$f(x) = \underbrace{f(a) + f'(a) \cdot (x-a) + \cdots + \frac{1}{n!}f^{(n)}(a) \cdot (x-a)^n}_{=T_n(x)} + \frac{1}{(n+1)!}f^{(n+1)}(c) \cdot (x-a)^{n+1}$$

for some c strictly between a and x . Again, rewriting this in terms of $T_n(x)$ gives

Equation 3.4.33.

$$f(x) - T_n(x) = \frac{1}{(n+1)!}f^{(n+1)}(c) \cdot (x-a)^{n+1} \quad \text{for some } c \text{ strictly between } a \text{ and } x$$

That is, the error introduced when $f(x)$ is approximated by its Taylor polynomial of order n , is precisely the last term of the Taylor polynomial of order $n+1$, but with the derivative evaluated at some point between a and x , rather than exactly at a . These error formulae are proven in the optional Section 3.4.9 later in this chapter.

Example 3.4.34

Approximate $\sin 46^\circ$ using Taylor polynomials about $a = 45^\circ$, and estimate the resulting error.

Solution.

- Start by defining $f(x) = \sin x$ and

$$a = 45^\circ = 45 \frac{\pi}{180} \text{ radians} \quad x = 46^\circ = 46 \frac{\pi}{180} \text{ radians} \quad x - a = \frac{\pi}{180} \text{ radians}$$

- The first few derivatives of f at a are

$$\begin{aligned} f(x) &= \sin x & f(a) &= \frac{1}{\sqrt{2}} \\ f'(x) &= \cos x & f'(a) &= \frac{1}{\sqrt{2}} \\ f''(x) &= -\sin x & f''(a) &= -\frac{1}{\sqrt{2}} \\ f^{(3)}(x) &= -\cos x & f^{(3)}(a) &= -\frac{1}{\sqrt{2}} \end{aligned}$$

- The constant, linear and quadratic Taylor approximations for $\sin(x)$ about $\frac{\pi}{4}$ are

$$\begin{aligned} T_0(x) &= f(a) = \frac{1}{\sqrt{2}} \\ T_1(x) &= T_0(x) + f'(a) \cdot (x - a) = \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} \left(x - \frac{\pi}{4}\right) \\ T_2(x) &= T_1(x) + \frac{1}{2}f''(a) \cdot (x - a)^2 = \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} \left(x - \frac{\pi}{4}\right) - \frac{1}{2\sqrt{2}} \left(x - \frac{\pi}{4}\right)^2 \end{aligned}$$

- So the approximations for $\sin 46^\circ$ are

$$\begin{aligned} \sin 46^\circ &\approx T_0\left(\frac{46\pi}{180}\right) = \frac{1}{\sqrt{2}} &&= 0.70710678 \\ \sin 46^\circ &\approx T_1\left(\frac{46\pi}{180}\right) = \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} \left(\frac{\pi}{180}\right) &&= 0.71944812 \\ \sin 46^\circ &\approx T_2\left(\frac{46\pi}{180}\right) = \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} \left(\frac{\pi}{180}\right) - \frac{1}{2\sqrt{2}} \left(\frac{\pi}{180}\right)^2 &&= 0.71934042 \end{aligned}$$

- The errors in those approximations are (respectively)

$$\begin{aligned} \text{error in } 0.70710678 &= f'(c)(x - a) = \cos c \cdot \left(\frac{\pi}{180}\right) \\ \text{error in } 0.71944812 &= \frac{1}{2}f''(c)(x - a)^2 = -\frac{1}{2} \cdot \sin c \cdot \left(\frac{\pi}{180}\right)^2 \\ \text{error in } 0.71934042 &= \frac{1}{3!}f^{(3)}(c)(x - a)^3 = -\frac{1}{3!} \cdot \cos c \cdot \left(\frac{\pi}{180}\right)^3 \end{aligned}$$

In each of these three cases c must lie somewhere between 45° and 46° .

- Rather than carefully estimating $\sin c$ and $\cos c$ for c in that range, we make use of a simpler (but much easier bound). No matter what c is, we know that $|\sin c| \leq 1$ and $|\cos c| \leq 1$. Hence

$$\begin{aligned} |\text{error in } 0.70710678| &\leq \left(\frac{\pi}{180}\right) < 0.018 \\ |\text{error in } 0.71944812| &\leq \frac{1}{2} \left(\frac{\pi}{180}\right)^2 < 0.00015 \\ |\text{error in } 0.71934042| &\leq \frac{1}{3!} \left(\frac{\pi}{180}\right)^3 < 0.0000009 \end{aligned}$$

Example 3.4.34

Example 3.4.35 (Showing $e < 3$)

In Example 3.4.31 above we used the fact that $e < 3$ without actually proving it. Let's do so now.

- Consider the linear approximation of e^x about $a = 0$.

$$T_1(x) = f(0) + f'(0) \cdot x = 1 + x$$

So at $x = 1$ we have

$$e \approx T_1(1) = 2$$

- The error in this approximation is

$$e^x - T_1(x) = \frac{1}{2}f''(c) \cdot x^2 = \frac{e^c}{2} \cdot x^2$$

So at $x = 1$ we have

$$e - T_1(1) = \frac{e^c}{2}$$

where $0 < c < 1$.

- Now since e^x is an increasing⁴⁷ function, it follows that $e^c < e$. Hence

$$e - T_1(1) = \frac{e^c}{2} < \frac{e}{2}$$

Moving the $\frac{e}{2}$ to the left hand side and the $T_1(1)$ to the right hand side gives

$$\frac{e}{2} < T_1(1) = 2$$

So $e < 4$.

- This isn't as tight as we would like — so now do the same with the quadratic approximation with $a = 0$:

$$e^x \approx T_2(x) = 1 + x + \frac{x^2}{2}$$

So when $x = 1$ we have

$$e \approx T_2(1) = 1 + 1 + \frac{1}{2} = \frac{5}{2}$$

⁴⁷ Since the derivative of e^x is e^x which is positive everywhere, the function is increasing everywhere.

- The error in this approximation is

$$e^x - T_2(x) = \frac{1}{3!} f'''(c) \cdot x^3 = \frac{e^c}{6} \cdot x^3$$

So at $x = 1$ we have

$$e - T_2(1) = \frac{e^c}{6}$$

where $0 < c < 1$.

- Again since e^x is an increasing function we have $e^c < e$. Hence

$$e - T_2(1) = \frac{e^c}{6} < \frac{e}{6}$$

That is

$$\frac{5e}{6} < T_2(1) = \frac{5}{2}$$

So $e < 3$ as required.

Example 3.4.35

Example 3.4.36 (More on e^x)

We wrote down the general n^{th} order Maclaurin polynomial approximation of e^x in Example 3.4.12 above.

- Recall that

$$T_n(x) = \sum_{k=0}^n \frac{1}{k!} x^k$$

- The error in this approximation is (by equation (3.4.33))

$$e^x - T_n(x) = \frac{1}{(n+1)!} e^c$$

where c is some number between 0 and x .

- So setting $x = 1$ in this gives

$$e - T_n(1) = \frac{1}{(n+1)!} e^c$$

where $0 < c < 1$.

- Since e^x is an increasing function we know that $1 = e^0 < e^c < e^1 < 3$, so the above expression becomes

$$\frac{1}{(n+1)!} \leq e - T_n(1) = \frac{1}{(n+1)!} e^c \leq \frac{3}{(n+1)!}$$

- So when $n = 9$ we have

$$\frac{1}{10!} \leq e - \left(1 + 1 + \frac{1}{2} + \cdots + \frac{1}{9!}\right) \leq \frac{3}{10!}$$

- Now $1/10! < 3/10! < 10^{-6}$, so the approximation of e by

$$e \approx 1 + 1 + \frac{1}{2} + \cdots + \frac{1}{9!} = \frac{98641}{36288} = 2.718281 \dots$$

is correct to 6 decimal places.

- More generally we know that using $T_n(1)$ to approximate e will have an error of at most $\frac{3}{(n+1)!}$ — so it converges very quickly.

Example 3.4.36

Example 3.4.37 (Example 3.4.24 Revisited)

Recall⁴⁸ that in Example 3.4.24 (measuring the height of the pole), we used the linear approximation

$$f(\theta_0 + \Delta\theta) \approx f(\theta_0) + f'(\theta_0)\Delta\theta$$

with $f(\theta) = 10 \tan \theta$ and $\theta_0 = 30 \frac{\pi}{180}$ to get

$$\Delta h = f(\theta_0 + \Delta\theta) - f(\theta_0) \approx f'(\theta_0)\Delta\theta \quad \text{which implies that} \quad \Delta\theta \approx \frac{\Delta h}{f'(\theta_0)}$$

- While this procedure is fairly reliable, it did involve an approximation. So that you could not 100% guarantee to your client's lawyer that an accuracy of 10 cm was achieved.
- On the other hand, if we use the *exact* formula (3.4.29), with the replacements $x \rightarrow \theta_0 + \Delta\theta$ and $a \rightarrow \theta_0$

$$f(\theta_0 + \Delta\theta) = f(\theta_0) + f'(c)\Delta\theta \quad \text{for some } c \text{ between } \theta_0 \text{ and } \theta_0 + \Delta\theta$$

in place of the approximate formula (3.4.3), this legality is taken care of:

$$\Delta h = f(\theta_0 + \Delta\theta) - f(\theta_0) = f'(c)\Delta\theta \quad \text{for some } c \text{ between } \theta_0 \text{ and } \theta_0 + \Delta\theta$$

We can clean this up a little more since in our example $f'(\theta) = 10 \sec^2 \theta$. Thus for some c between θ_0 and $\theta_0 + \Delta\theta$:

$$|\Delta h| = 10 \sec^2(c) |\Delta\theta|$$

48 Now is a good time to go back and re-read it.

- Of course we do not know exactly what c is. But suppose that we know that the angle was somewhere between 25° and 35° . In other words suppose that, even though we don't know precisely what our measurement error was, it was certainly no more than 5° .
- Now on the range $25^\circ < c < 35^\circ$, $\sec(c)$ is an increasing and positive function. Hence on this range

$$1.217 \dots = \sec^2 25^\circ \leq \sec^2 c \leq \sec^2 35^\circ = 1.490 \dots < 1.491$$

So

$$12.17 \cdot |\Delta\theta| \leq |\Delta h| = 10 \sec^2(c) \cdot |\Delta\theta| \leq 14.91 \cdot |\Delta\theta|$$

- Since we require $|\Delta h| < 0.1$, we need $14.91|\Delta\theta| < 0.1$, that is

$$|\Delta\theta| < \frac{0.1}{14.91} = 0.0067 \dots$$

So we must measure angles with an accuracy of no less than 0.0067 radians — which is

$$\frac{180}{\pi} \cdot 0.0067 = 0.38^\circ.$$

Hence a measurement error of 0.38° or less is acceptable.

Example 3.4.37

3.4.9 ► (Optional) — Derivation of the Error Formulae

In this section we will derive the formula for the error that we gave in equation (3.4.33) — namely

$$R_n(x) = f(x) - T_n(x) = \frac{1}{(n+1)!} f^{(n+1)}(c) \cdot (x-a)^{n+1}$$

for some c strictly between a and x , and where $T_n(x)$ is the n^{th} order Taylor polynomial approximation of $f(x)$ about $x = a$:

$$T_n(x) = \sum_{k=0}^n \frac{1}{k!} f^{(k)}(a).$$

Recall that we have already proved a special case of this formula for the constant approximation using the Mean-Value Theorem (Theorem 2.13.4). To prove the general case we need the following generalisation⁴⁹ of that theorem:

49 It is not a terribly creative name for the generalisation, but it is an accurate one.

Theorem 3.4.38 (Generalised Mean-Value Theorem).

Let the functions $F(x)$ and $G(x)$ both be defined and continuous on $a \leq x \leq b$ and both be differentiable on $a < x < b$. Furthermore, suppose that $G'(x) \neq 0$ for all $a < x < b$. Then, there is a number c obeying $a < c < b$ such that

$$\frac{F(b) - F(a)}{G(b) - G(a)} = \frac{F'(c)}{G'(c)}$$

Notice that setting $G(x) = x$ recovers the original Mean-Value Theorem. It turns out that this theorem is not too difficult to prove from the MVT using some sneaky algebraic manipulations:

Proof. • First we construct a new function $h(x)$ as a linear combination of $F(x)$ and $G(x)$ so that $h(a) = h(b) = 0$. Some experimentation yields

$$h(x) = [F(b) - F(a)] \cdot [G(x) - G(a)] - [G(b) - G(a)] \cdot [F(x) - F(a)]$$

- Since $h(a) = h(b) = 0$, the Mean-Value theorem (actually Rolle's theorem) tells us that there is a number c obeying $a < c < b$ such that $h'(c) = 0$:

$$\begin{aligned} h'(x) &= [F(b) - F(a)] \cdot G'(x) - [G(b) - G(a)] \cdot F'(x) & \text{so} \\ 0 &= [F(b) - F(a)] \cdot G'(c) - [G(b) - G(a)] \cdot F'(c) \end{aligned}$$

Now move the $G'(c)$ terms to one side and the $F'(c)$ terms to the other:

$$[F(b) - F(a)] \cdot G'(c) = [G(b) - G(a)] \cdot F'(c).$$

- Since we have $G'(x) \neq 0$, we know that $G'(c) \neq 0$. Further the Mean-Value theorem ensures⁵⁰ that $G(a) \neq G(b)$. Hence we can move terms about to get

$$\begin{aligned} [F(b) - F(a)] &= [G(b) - G(a)] \cdot \frac{F'(c)}{G'(c)} \\ \frac{F(b) - F(a)}{G(b) - G(a)} &= \frac{F'(c)}{G'(c)} \end{aligned}$$

as required. □

Armed with the above theorem we can now move on to the proof of the Taylor remainder formula.

Proof of equation (3.4.33). We begin by proving the remainder formula for $n = 1$. That is

$$f(x) - T_1(x) = \frac{1}{2}f''(c) \cdot (x - a)^2$$

50 Otherwise if $G(a) = G(b)$ the MVT tells us that there is some point c between a and b so that $G'(c) = 0$.

- Start by setting

$$F(x) = f(x) - T_1(x) \qquad G(x) = (x - a)^2$$

Notice that, since $T_1(a) = f(a)$ and $T_1'(x) = f'(a)$,

$$\begin{aligned} F(a) &= 0 & G(a) &= 0 \\ F'(x) &= f'(x) - f'(a) & G'(x) &= 2(x - a) \end{aligned}$$

- Now apply the generalised MVT with $b = x$: there exists a point q between a and x such that

$$\begin{aligned} \frac{F(x) - F(a)}{G(x) - G(a)} &= \frac{F'(q)}{G'(q)} \\ \frac{F(x) - 0}{G(x) - 0} &= \frac{f'(q) - f'(a)}{2(q - a)} \\ 2 \cdot \frac{F(x)}{G(x)} &= \frac{f'(q) - f'(a)}{q - a} \end{aligned}$$

- Consider the right-hand side of the above equation and set $g(x) = f'(x)$. Then we have the term $\frac{g(q) - g(a)}{q - a}$ — this is exactly the form needed to apply the MVT. So now apply the standard MVT to the right-hand side of the above equation — there is some c between q and a so that

$$\frac{f'(q) - f'(a)}{q - a} = \frac{g(q) - g(a)}{q - a} = g'(c) = f''(c)$$

Notice that here we have assumed that $f''(x)$ exists.

- Putting this together we have that

$$\begin{aligned} 2 \cdot \frac{F(x)}{G(x)} &= \frac{f'(q) - f'(a)}{q - a} = f''(c) \\ 2 \frac{f(x) - T_1(x)}{(x - a)^2} &= f''(c) \\ f(x) - T_1(x) &= \frac{1}{2!} f''(c) \cdot (x - a)^2 \end{aligned}$$

as required.

Oof! We have now proved the cases $n = 1$ (and we did $n = 0$ earlier).

To proceed — assume we have proved our result for $n = 1, 2, \dots, k$. We realise that we haven't done this yet, but bear with us. Using that assumption we will prove the result is true for $n = k + 1$. Once we have done that, then

- we have proved the result is true for $n = 1$, and
- we have shown if the result is true for $n = k$ then it is true for $n = k + 1$

Hence it must be true for all $n \geq 1$. This style of proof is called mathematical induction. You can think of the process as something like climbing a ladder:

- prove that you can get onto the ladder (the result is true for $n = 1$), and
- if I can stand on the current rung, then I can step up to the next rung (if the result is true for $n = k$ then it is also true for $n = k + 1$)

Hence I can climb as high as like.

- Let $k > 0$ and assume we have proved

$$f(x) - T_k(x) = \frac{1}{(k+1)!} f^{(k+1)}(c) \cdot (x-a)^{k+1}$$

for some c between a and x .

- Now set

$$F(x) = f(x) - T_{k+1}(x) \quad G(x) = (x-a)^{k+1}$$

and notice that, since $T_{k+1}(a) = f(a)$,

$$F(a) = f(a) - T_{k+1}(a) = 0 \quad G(a) = 0 \quad G'(x) = (k+1)(x-a)^k$$

and apply the generalised MVT with $b = x$: hence there exists a q between a and x so that

$$\begin{aligned} \frac{F(x) - F(a)}{G(x) - G(a)} &= \frac{F'(q)}{G'(q)} && \text{which becomes} \\ \frac{F(x)}{(x-a)^{k+1}} &= \frac{F'(q)}{(k+1)(q-a)^k} && \text{rearrange} \\ F(x) &= \frac{(x-a)^{k+1}}{(k+1)(q-a)^k} \cdot F'(q) \end{aligned}$$

- We now examine $F'(q)$. First carefully differentiate $F(x)$:

$$\begin{aligned} F'(x) &= \frac{d}{dx} \left[f(x) - \left(f(a) + f'(a)(x-a) + \frac{1}{2}f''(a)(x-a)^2 + \cdots + \frac{1}{k!}f^{(k)}(a)(x-a)^k \right) \right] \\ &= f'(x) - \left(f'(a) + \frac{2}{2}f''(a)(x-a) + \frac{3}{3!}f'''(a)(x-a)^2 + \cdots + \frac{k}{k!}f^{(k)}(a)(x-a)^{k-1} \right) \\ &= f'(x) - \left(f'(a) + f''(a)(x-a) + \frac{1}{2}f'''(a)(x-a)^2 + \cdots + \frac{1}{(k-1)!}f^{(k)}(a)(x-a)^{k-1} \right) \end{aligned}$$

Now notice that if we set $f'(x) = g(x)$ then this becomes

$$F'(x) = g(x) - \left(g(a) + g'(a)(x-a) + \frac{1}{2}g''(a)(x-a)^2 + \cdots + \frac{1}{(k-1)!}g^{(k-1)}(a)(x-a)^{k-1} \right)$$

So $F'(x)$ is then exactly the remainder formula but for an order $k-1$ approximation to the function $g(x) = f'(x)$.

- (b) Give an expression for the volume of the bottle as it is being smooshed, in terms of a and b (and no other variables).
- (c) Suppose the bottle was full when its cross section was a circle. How fast is the water spilling out when a is twice as big as b ?

Q[24]: The quantities A , B , C , and D all depend on time, and are related by the formula

$$AB = \log(C^2 + D^2 + 1).$$

At time $t = 10$, the following values are known:

- $A = 0$
- $\frac{dA}{dt} = 2$ units per second

What is B when $t = 10$?

3.3▲ Exponential Growth and Decay

3.3.1 ► Carbon Dating

Exercises

Jump to [HINTS](#), [ANSWERS](#), [SOLUTIONS](#) or [TABLE OF CONTENTS](#).

► Stage 1

Q[1]: Which of the following is a differential equation for an unknown function y of x ?

- (a) $y = \frac{dy}{dx}$ (b) $\frac{dy}{dx} = 3[y - 5]$ (c) $y = 3\left[y - \frac{dx}{dx}\right]$ (d) $e^x = e^y + 1$ (e) $y = 10e^x$

Q[2]: Which of the following functions $Q(t)$ satisfy the differential equation

$$Q(t) = 5 \frac{dQ}{dt}?$$

- (a) $Q(t) = 0$ (b) $Q(t) = 5e^t$ (c) $Q(t) = e^{5t}$ (d) $Q(t) = e^{t/5}$ (e) $Q(t) = e^{t/5} + 1$

Q[3]: Suppose a sample starts out with C grams of a radioactive isotope, and the amount of the radioactive isotope left in the sample at time t is given by

$$Q(t) = Ce^{-kt}$$

for some positive constant k . When will $Q(t) = 0$?

►► Stage 2

Q[4](*) : Consider a function of the form $f(x) = Ae^{kx}$ where A and k are constants. If $f(0) = 5$ and $f(7) = \pi$, find the constants A and k .

Q[5](*) : Find the function $y(t)$ if $\frac{dy}{dt} + 3y = 0$, $y(1) = 2$.

Q[6]: A sample of bone belongs to an animal that died 10,000 years ago. If the bone contained 5 μg of Carbon-14 when the animal died, how much Carbon-14 do you expect it to have now?

Q[7]: A sample containing one gram of Radium-226 was stored in a lab 100 years ago; now the sample only contains 0.9576 grams of Radium-226. What is the half-life of Radium-226?

Q[8](*) : The mass of a sample of Polonium-210, initially 6 grams, decreases at a rate proportional to the mass. After one year, 1 gram remains. What is the half-life (the time it takes for the sample to decay to half its original mass)?

Q[9]: Radium-221 has a half-life of 30 seconds. How long does it take for only 0.01% of an original sample to be left?

►► Stage 3

Q[10]: Polonium-210 has a half life of 138 days. What percentage of a sample of Polonium-210 decays in a day?

Q[11]: A sample of ore is found to contain $7.2 \pm 0.3 \mu\text{g}$ of Uranium-232, the half-life of which is between 68.8 and 70 years. How much Uranium-232 will remain undecayed in the sample in 10 years?

3.3.2 ►► Newton's Law of Cooling

Exercises

Jump to [HINTS](#), [ANSWERS](#), [SOLUTIONS](#) or [TABLE OF CONTENTS](#).

►► Stage 1

Q[1]: Which of the following functions $T(t)$ satisfy the differential equation $\frac{dT}{dt} = 5[T - 20]$?

- (a) $T(t) = 20$ (b) $T(t) = 20e^{5t} - 20$ (c) $T(t) = e^{5t} + 20$ (d) $T(t) = 20e^{5t} + 20$

Q[2]: At time $t = 0$, an object is placed in a room, of temperature A . After t seconds, Newton's Law of Cooling gives the temperature of the object is as

$$T(t) = 35e^{Kt} - 10$$

What is the temperature of the room? Is the room warmer or colder than the object?

Q[3]: A warm object is placed in a cold room. The temperature of the object, over time, approaches the temperature of the room it is in. The temperature of the object at time t is given by

$$T(t) = [T(0) - A]e^{Kt} + A.$$

Can K be a positive number? Can K be a negative number? Can K be zero?

Q[4]: Suppose an object obeys Newton's Law of Cooling, and its temperature is given by

$$T(t) = [T(0) - A]e^{kt} + A$$

for some constant k . At what time is $T(t) = A$?

► Stage 2

Q[5]: A piece of copper at room temperature (25°) is placed in a boiling pot of water. After 10 seconds, it has heated to 90° . When will it be 99.9° ?

Q[6]: Today is a chilly day. We heated up a stone to 500° C in a bonfire, then took it out and left it outside, where the temperature is 0° C. After 10 minutes outside of the bonfire, the stone had cooled to a still-untouchable 100° C. Now the stone is at a cozy 50° C. How long ago was the stone taken out of the fire?

► Stage 3

Q[7](*) : Isaac Newton drinks his coffee with cream. To be exact, 9 parts coffee to 1 part cream. His landlady pours him a cup of coffee at 95° C into which Newton stirs cream taken from the icebox at 5° C. When he drinks the mixture ten minutes later, he notes that it has cooled to 54° C. Newton wonders if his coffee would be hotter (and by how much) if he waited until just before drinking it to add the cream. Analyze this question, assuming that:

- (i) The temperature of the dining room is constant at 22° C.
- (ii) When a volume V_1 of liquid at temperature T_1 is mixed with a volume V_2 at temperature T_2 , the temperature of the mixture is $\frac{V_1 T_1 + V_2 T_2}{V_1 + V_2}$.
- (iii) Newton's Law of Cooling: The temperature of an object cools at a rate proportional to the difference in temperature between the object and its surroundings.
- (iv) The constant of proportionality is the same for the cup of coffee with cream as for the cup of pure coffee.

Q[8](*) : The temperature of a glass of iced tea is initially 5° . After 5 minutes, the tea has heated to 10° in a room where the air temperature is 30° .

- Use Newton's law of cooling to obtain a differential equation for the temperature $T(t)$ at time t .
- Determine when the tea will reach a temperature of 20° .

Q[9]: Suppose an object is changing temperature according to Newton's Law of Cooling, and its temperature at time t is given by

$$T(t) = 0.8^{kt} + 15$$

Is k positive or negative?

3.3.3 ► Population Growth

Exercises

Jump to [HINTS](#), [ANSWERS](#), [SOLUTIONS](#) or [TABLE OF CONTENTS](#).

► Stage 1

Q[1]: Let a population at time t be given by the Malthusian model,

$$P(t) = P(0)e^{bt} \text{ for some positive constant } b.$$

Evaluate $\lim_{t \rightarrow \infty} P(t)$. Does this model make sense for large values of t ?

► Stage 2

Q[2]: In the 1950s, pure-bred wood bison were thought to be extinct. However, a small population was found in Canada. For decades, a captive breeding program has been working to increase their numbers, and from time to time wood bison are released to the wild. Suppose in 2015, a released herd numbered 121 animals, and a year later, there were 136^{*}. If the wood bison adhere to the Malthusian model (a big assumption!), and if there are no more releases of captive animals, how many animals will the herd have in 2020?

* These numbers are loosely based on animals actually released near Shageluk, Alaska in 2015. Watch the first batch being released [here](#).

Q[3]: A founding colony of 1,000 bacteria is placed in a petri dish of yummy bacteria food. After an hour, the population has doubled. Assuming the Malthusian model, how long will it take for the colony to triple its original population?

Q[4]: A single pair of rats comes to an island after a shipwreck. They multiply according to the Malthusian model. In 1928, there were 1,000 rats on the island, and the next year there were 1500. When was the shipwreck?

Q[5]: A farmer wants to farm cochineals, which are insects used to make red dye. The farmer raises a small number of cochineals as a test. In three months, a test population of cochineals will increase from 200 individuals to 1000, given ample space and food.

The farmer's plan is to start with an initial population of $P(0)$ cochineals, and after a year have $1\,000\,000 + P(0)$ cochineals, so that one million can be harvested, and $P(0)$ saved to start breeding again. What initial population $P(0)$ does the Malthusian model suggest?

►► Stage 3

Q[6]: Let $f(t) = 100e^{kt}$, for some constant k .

- (a) If $f(t)$ is the amount of a decaying radioactive isotope in a sample at time t , what is the amount of the isotope in the sample when $t = 0$? What is the sign of k ?
- (b) If $f(t)$ is the number of individuals in a population that is growing according to the Malthusian model, how many individuals are there when $t = 0$? What is the sign of k ?
- (c) If $f(t)$ is the temperature of an object at time t , given by Newton's Law of Cooling, what is the ambient temperature surrounding the object? What is the sign of k ?

3.3.4 ►► Further problems

Exercises

Jump to [HINTS](#), [ANSWERS](#), [SOLUTIONS](#) or [TABLE OF CONTENTS](#).

►► More Section 3.3 Problems

Q[1](*) : Find $f(2)$ if $f'(x) = \pi f(x)$ for all x , and $f(0) = 2$.

Q[2]: Which functions $T(t)$ satisfy the differential equation $\frac{dT}{dt} = 7T + 9$?

Q[3](*) : It takes 8 days for 20% of a particular radioactive material to decay. How long does it take for 100 grams of the material to decay to 40 grams?

Q[4]: A glass of boiling water is left in a room. After 15 minutes, it has cooled to 85°C , and after 30 minutes it is 73°C . What temperature is the room?

Q[5](*) : A 25-year-old graduate of UBC is given \$50,000 which is invested at 5% per year compounded continuously. The graduate also intends to deposit money continuously at the rate of \$2000 per year. Assuming that the interest rate remains 5%, the amount $A(t)$ of money at time t satisfies the equation

$$\frac{dA}{dt} = 0.05A + 2000$$

- Solve this equation and determine the amount of money in the account when the graduate is 65.
- At age 65, the graduate will withdraw money continuously at the rate of W dollars per year. If the money must last until the person is 85, what is the largest possible value of W ?

Q[6](*) : An investor puts \$120,000 which into a bank account which pays 6% annual interest, compounded continuously. She plans to withdraw money continuously from the account at the rate of \$9000 per year. If $A(t)$ is the amount of money at time t , then

$$\frac{dA}{dt} = 0.06A - 9000$$

- Solve this equation for $A(t)$.
- When will the money run out?

Q[7](*) : A particular bacterial culture grows at a rate proportional to the number of bacteria present. If the size of the culture triples every nine hours, how long does it take the culture to double?

Q[8](*) : An object falls under gravity near the surface of the earth and its motion is impeded by air resistance proportional to its speed. Its velocity v satisfies the differential equation

$$\frac{dv}{dt} = -g - kv$$

where g and k are positive constants.

- Find the velocity of the object as a function of time t , given that it was v_0 at $t = 0$.
- Find $\lim_{t \rightarrow \infty} v(t)$.

3.4▲ Taylor polynomials

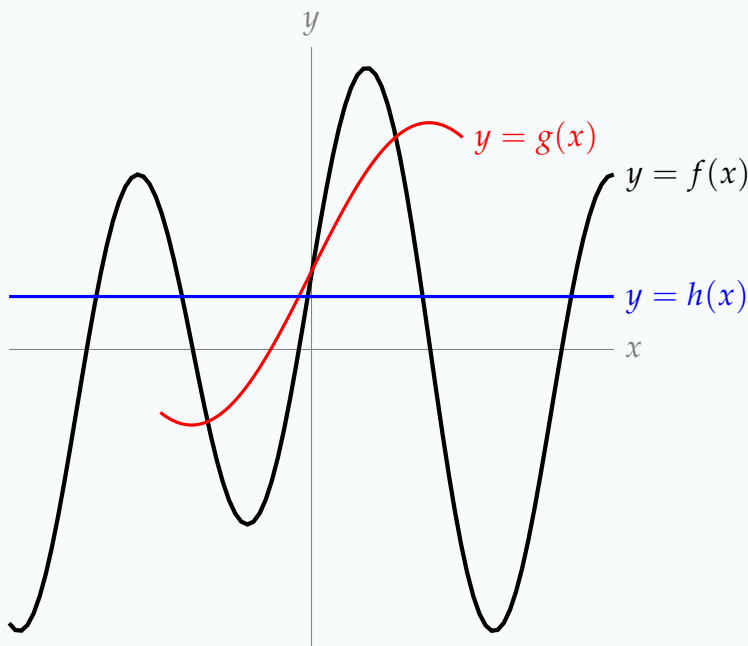
3.4.1 ► Zeroeth Approximation

Exercises

Jump to [HINTS](#), [ANSWERS](#), [SOLUTIONS](#) or [TABLE OF CONTENTS](#).

► Stage 1

Q[1]: The graph below shows three curves. The black curve is $y = f(x)$, the red curve is $y = g(x) = 1 + 2 \sin(1 + x)$, and the blue curve is $y = h(x) = 0.7$. If you want to estimate $f(0)$, what might cause you to use $g(0)$? What might cause you to use $h(0)$?



► Stage 2

In this and following sections, we will ask you to approximate the value of several constants, such as $\log(0.93)$. A valid question to consider is why we would ask for approximations of these constants that take lots of time, and are less accurate than what you get from a calculator.

One answer to this question is historical: people were approximating logarithms before they had calculators, and these are some of the ways they did that. Pretend you're on a desert island without any of your usual devices and that you want to make a number of quick and dirty approximate evaluations.

Another reason to make these approximations is technical: how does the calculator get such a good approximation of $\log(0.93)$? The techniques you will learn later on in this chapter give very accurate formulas for approximating functions like $\log x$ and $\sin x$, which are sometimes used in calculators.

A third reason to make simple approximations of expressions that a calculator could evaluate is to provide a reality check. If you have a ballpark guess for your answer, and your calculator gives you something wildly different, you know to double-check that you typed everything in correctly.

For now, questions like Question 2 through Question 4 are simply for you to practice the fundamental ideas we're learning.

Q[2]: Use a constant approximation to estimate the value of $\log(x)$ when $x = 0.93$. Sketch the curve $y = f(x)$ and your constant approximation.

(Remember that in CLP-1 we use $\log x$ to mean the natural logarithm of x , $\log_e x$.)

Q[3]: Use a constant approximation to estimate $\arcsin(0.1)$.

Q[4]: Use a constant approximation to estimate $\sqrt{3} \tan(1)$.

►► Stage 3

Q[5]: Use a constant approximation to estimate the value of 10.1^3 . Your estimation should be something you can calculate in your head.

3.4.2 ►► First Approximation

Exercises

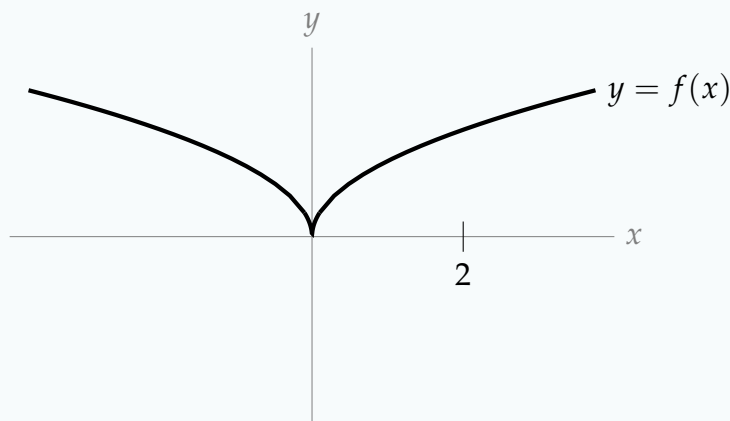
Jump to [HINTS](#), [ANSWERS](#), [SOLUTIONS](#) or [TABLE OF CONTENTS](#).

►► Stage 1

Q[1]: Suppose $f(x)$ is a function, and we calculated its linear approximation near $x = 5$ to be $f(x) \approx 3x - 9$.

- (a) What is $f(5)$?
- (b) What is $f'(5)$?
- (c) What is $f(0)$?

Q[2]: The curve $y = f(x)$ is shown below. Sketch the linear approximation of $f(x)$ about $x = 2$.



Q[3]: What is the linear approximation of the function $f(x) = 2x + 5$ about $x = a$?

►► Stage 2

Q[4]: Use a linear approximation to estimate $\log(x)$ when $x = 0.93$. Sketch the curve $y = f(x)$ and your linear approximation.
(Remember that in CLP-1 we use $\log x$ to mean the natural logarithm of x , $\log_e x$.)

Q[5]: Use a linear approximation to estimate $\sqrt{5}$.

Q[6]: Use a linear approximation to estimate $\sqrt[5]{30}$

►► Stage 3

Q[7]: Use a linear approximation to estimate 10.1^3 , then compare your estimation with the actual value.

Q[8]: Imagine $f(x)$ is some function, and you want to estimate $f(b)$. To do this, you choose a value a and take an approximation (linear or constant) of $f(x)$ about a . Give an example of a function $f(x)$, and values a and b , where the constant approximation gives a more accurate estimation of $f(b)$ than the linear approximation.

Q[9]: The function

$$L(x) = \frac{1}{4}x + \frac{4\pi - \sqrt{27}}{12}$$

is the linear approximation of $f(x) = \arctan x$ about what point $x = a$?

3.4.3 ►► Second Approximation**Exercises**

Jump to [HINTS](#), [ANSWERS](#), [SOLUTIONS](#) or [TABLE OF CONTENTS](#).

►► Stage 1

Q[1]: The quadratic approximation of a function $f(x)$ about $x = 3$ is

$$f(x) \approx -x^2 + 6x$$

What are the values of $f(3)$, $f'(3)$, $f''(3)$, and $f'''(3)$?

Q[2]: Give a quadratic approximation of $f(x) = 2x + 5$ about $x = a$.

► Stage 2

Q[3]: Use a quadratic approximation to estimate $\log(0.93)$.
(Remember that in CLP-1 we use $\log x$ to mean the natural logarithm of x , $\log_e x$.)

Q[4]: Use a quadratic approximation to estimate $\cos\left(\frac{1}{15}\right)$.

Q[5]: Calculate the quadratic approximation of $f(x) = e^{2x}$ about $x = 0$.

Q[6]: Use a quadratic approximation to estimate $5^{\frac{4}{3}}$.

Q[7]: Evaluate the expressions below.

(a) $\sum_{n=5}^{30} 1$

(b) $\sum_{n=1}^3 [2(n+3) - n^2]$

(c) $\sum_{n=1}^{10} \left[\frac{1}{n} - \frac{1}{n+1} \right]$

(d) $\sum_{n=1}^4 \frac{5 \cdot 2^n}{4^{n+1}}$

Q[8]: Write the following in sigma notation:

(a) $1 + 2 + 3 + 4 + 5$

(b) $2 + 4 + 6 + 8$

(c) $3 + 5 + 7 + 9 + 11$

(d) $9 + 16 + 25 + 36 + 49$

(e) $9 + 4 + 16 + 5 + 25 + 6 + 36 + 7 + 49 + 8$

(f) $8 + 15 + 24 + 35 + 48$

(g) $3 - 6 + 9 - 12 + 15 - 18$

► Stage 3

Q[9]: Use a quadratic approximation of $f(x) = 2 \arcsin x$ about $x = 0$ to approximate $f(1)$. What number are you approximating?

Q[10]: Use a quadratic approximation of e^x to estimate e as a decimal.

Q[11]: Group the expressions below into collections of equivalent expressions.

(a) $\sum_{n=1}^{10} 2n$

(b) $\sum_{n=1}^{10} 2^n$

(c) $\sum_{n=1}^{10} n^2$

(d) $2 \sum_{n=1}^{10} n$

(e) $2 \sum_{n=2}^{11} (n-1)$

(f) $\sum_{n=5}^{14} (n-4)^2$

(g) $\frac{1}{4} \sum_{n=1}^{10} \left(\frac{4^{n+1}}{2^n} \right)$

3.4.4 ► Still Better Approximations

Exercises

Jump to [HINTS](#), [ANSWERS](#), [SOLUTIONS](#) or [TABLE OF CONTENTS](#).

► Stage 1

Q[1]: The 3rd order Taylor polynomial for a function $f(x)$ about $x = 1$ is

$$T_3(x) = x^3 - 5x^2 + 9x$$

What is $f''(1)$?

Q[2]: The n th order Taylor polynomial for $f(x)$ about $x = 5$ is

$$T_n(x) = \sum_{k=0}^n \frac{2k+1}{3k-9} (x-5)^k$$

What is $f^{(10)}(5)$?

► Stage 3

Q[3]: The 4th order Maclaurin polynomial for $f(x)$ is

$$T_4(x) = x^4 - x^3 + x^2 - x + 1$$

What is the third order Maclaurin polynomial for $f(x)$?

Q[4]: The 4th order Taylor polynomial for $f(x)$ about $x = 1$ is

$$T_4(x) = x^4 + x^3 - 9$$

What is the third order Taylor polynomial for $f(x)$ about $x = 1$?

Q[5]: For any even number n , suppose the n th order Taylor polynomial for $f(x)$ about $x = 5$ is

$$\sum_{k=0}^{n/2} \frac{2k+1}{3k-9} (x-5)^{2k}$$

What is $f^{(10)}(5)$?

Q[6]: The third order Taylor polynomial for $f(x) = x^3 \left[2 \log x - \frac{11}{3} \right]$ about $x = a$ is

$$T_3(x) = -\frac{2}{3}\sqrt{e^3} + 3ex - 6\sqrt{e}x^2 + x^3$$

What is a ?

3.4.5 ► Some Examples

Exercises

Jump to [HINTS](#), [ANSWERS](#), [SOLUTIONS](#) or [TABLE OF CONTENTS](#).

► Stage 2

Q[1]: Give the 16th order Maclaurin polynomial for $f(x) = \sin x + \cos x$.

Q[2]: Give the 100th order Taylor polynomial for $s(t) = 4.9t^2 - t + 10$ about $t = 5$.

Q[3]: Write the n th order Taylor polynomial for $f(x) = 2^x$ about $x = 1$ in sigma notation.

Q[4]: Find the 6th order Taylor polynomial of $f(x) = x^2 \log x + 2x^2 + 5$ about $x = 1$, remembering that $\log x$ is the natural logarithm of x , $\log_e x$.

Q[5]: Give the n th order Maclaurin polynomial for $\frac{1}{1-x}$ in sigma notation.

►► Stage 3

Q[6]: Calculate the 3rd order Taylor Polynomial for $f(x) = x^x$ about $x = 1$.

Q[7]: Use a 5th order Maclaurin polynomial for $6 \arctan x$ to approximate π .

Q[8]: Write the 100th order Taylor polynomial for $f(x) = x(\log x - 1)$ about $x = 1$ in sigma notation.

Q[9]: Write the $(2n)$ th order Taylor polynomial for $f(x) = \sin x$ about $x = \frac{\pi}{4}$ in sigma notation.

Q[10]: Estimate the sum below

$$1 + \frac{1}{2} + \frac{1}{3!} + \frac{1}{4!} + \cdots + \frac{1}{157!}$$

by interpreting it as a Maclaurin polynomial.

Q[11]: Estimate the sum below

$$\sum_{k=0}^{100} \frac{(-1)^k}{(2k)!} \left(\frac{5\pi}{4} \right)^{2k}$$

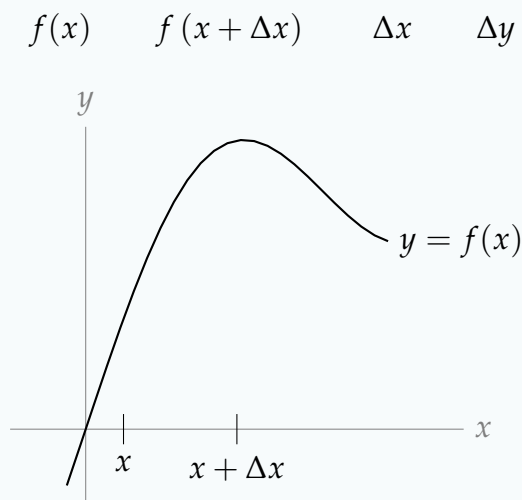
by interpreting it as a Maclaurin polynomial.

3.4.6 ►► Estimating Changes and Δx , Δy notation**Exercises**

Jump to [HINTS](#), [ANSWERS](#), [SOLUTIONS](#) or [TABLE OF CONTENTS](#).

►► Stage 1

Q[1]: In the picture below, label the following:



Q[2]: At this point in the book, every homework problem takes you about 5 minutes. Use the terms you learned in this section to answer the question: if you spend 15 minutes more, how many more homework problems will you finish?

►► Stage 2

Q[3]: Let $f(x) = \arctan x$.

- (a) Use a linear approximation to estimate $f(5.1) - f(5)$.
- (b) Use a quadratic approximation to estimate $f(5.1) - f(5)$.

Q[4]: When diving off a cliff from x metres above the water, your speed as you hit the water is given by

$$s(x) = \sqrt{19.6x} \frac{\text{m}}{\text{sec}}$$

Your last dive was from a height of 4 metres.

- (a) Use a linear approximation of Δy to estimate how much faster you will be falling when you hit the water if you jump from a height of 5 metres.
- (b) A diver makes three jumps: the first is from x metres, the second from $x + \Delta x$ metres, and the third from $x + 2\Delta x$ metres, for some fixed positive values of x and Δx . Which is bigger: the increase in terminal speed from the first to the second jump, or the increase in terminal speed from the second to the third jump?

3.4.7 ►► Further Examples

Exercises

Jump to [HINTS](#), [ANSWERS](#), [SOLUTIONS](#) or [TABLE OF CONTENTS](#).

►► Stage 1

Q[1]: Let $f(x) = 7x^2 - 3x + 4$. Suppose we measure x to be $x_0 = 2$ but that the real value of x is $x_0 + \Delta x$. Suppose further that the error in our measurement is $\Delta x = 1$. Let Δy be the change in $f(x)$ corresponding to a change of Δx in x_0 . That is, $\Delta y = f(x_0 + \Delta x) - f(x_0)$.

True or false: $\Delta y = f'(2)(1) = 25$

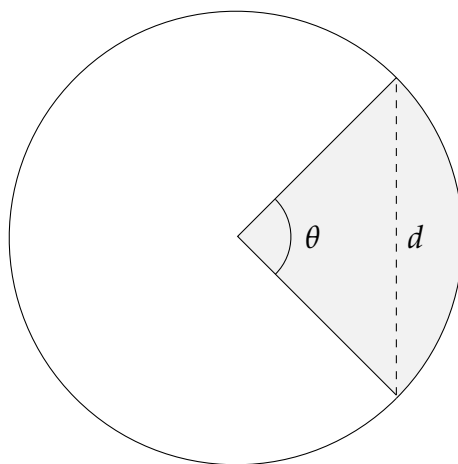
Q[2]: Suppose the exact amount you are supposed to tip is \$5.83, but you approximate and tip \$6. What is the absolute error in your tip? What is the percent error in your tip?

Q[3]: Suppose $f(x) = 3x^2 - 5$. If you measure x to be 10, but its actual value is 11, estimate the resulting error in $f(x)$ using the linear approximation, and then the quadratic approximation.

►► Stage 2

Q[4]: A circular pen is being built on a farm. The pen must contain A_0 square metres, with an error of no more than 2%. Estimate the largest percentage error allowable on the radius.

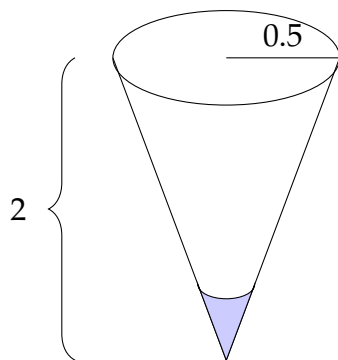
Q[5]: A circle with radius 3 has a sector cut out of it. It's a smallish sector, no more than a quarter of the circle. You want to find out the area of the sector.



- Suppose the angle of the sector is θ . What is the area of the sector?
- Unfortunately, you don't have a protractor, only a ruler. So, you measure the chord made by the sector (marked d in the diagram above). What is θ in terms of d ?
- Suppose you measured $d = 0.7$, but actually $d = 0.68$. Estimate the absolute error in your calculation of the area removed.

Q[6]: A conical tank, standing on its pointy end, has height 2 metres and radius 0.5 metres. Estimate change in volume of the water in the tank associated to a change in the

height of the water from 50 cm to 45 cm.



►► Stage 3

Q[7]: A sample begins with precisely $1 \mu\text{g}$ of a radioactive isotope, and after 3 years is measured to have $0.9 \mu\text{g}$ remaining. If this measurement is correct to within $0.05 \mu\text{g}$, estimate the corresponding accuracy of the half-life calculated using it.

3.4.8 ►► The error in Taylor polynomial approximation

Exercises

Jump to [HINTS](#), [ANSWERS](#), [SOLUTIONS](#) or [TABLE OF CONTENTS](#).

►► Stage 1

Q[1]: Suppose $f(x)$ is a function that we approximated by $F(x)$. Further, suppose $f(10) = -3$, while our approximation was $F(10) = 5$. Let $R(x) = f(x) - F(x)$.

- (a) True or false: $|R(10)| \leq 7$
- (b) True or false: $|R(10)| \leq 8$
- (c) True or false: $|R(10)| \leq 9$
- (d) True or false: $|R(10)| \leq 100$

Q[2]: Let $f(x) = e^x$, and let $T_3(x)$ be the third order Maclaurin polynomial for $f(x)$,

$$T_3(x) = 1 + x + \frac{1}{2}x^2 + \frac{1}{3!}x^3$$

Use Equation 3.4.33 to give a reasonable bound on the error $|f(2) - T_3(2)|$. Then, find the error $|f(2) - T_3(2)|$ using a calculator.

Q[3]: Let $f(x) = 5x^3 - 24x^2 + ex - \pi^4$, and let $T_5(x)$ be the fifth order Taylor polynomial for $f(x)$ about $x = 1$. Give the best bound you can on the error $|f(37) - T(37)|$.

Q[4]: You and your friend both want to approximate $\sin(33)$. Your friend uses the first order Maclaurin polynomial for $f(x) = \sin x$, while you use the zeroth order (constant)

Maclaurin polynomial for $f(x) = \sin x$. Who has a better approximation, you or your friend?

►► Stage 2

Q[5]: Suppose a function $f(x)$ has sixth derivative

$$f^{(6)}(x) = \frac{6!(2x-5)}{x+3}.$$

Let $T_5(x)$ be the 5th order Taylor polynomial for $f(x)$ about $x = 11$. Give a bound for the error $|f(11.5) - T_5(11.5)|$.

Q[6]: Let $f(x) = \tan x$, and let $T_2(x)$ be the second order Taylor polynomial for $f(x)$ about $x = 0$. Give a reasonable bound on the error $|f(0.1) - T_2(0.1)|$ using Equation 3.4.33.

Q[7]: Let $f(x) = \log(1-x)$, and let $T_5(x)$ be the fifth order Maclaurin polynomial for $f(x)$. Use Equation 3.4.33 to give a bound on the error $|f(-\frac{1}{4}) - T_5(-\frac{1}{4})|$. (Remember $\log x = \log_e x$, the natural logarithm of x .)

Q[8]: Let $f(x) = \sqrt[5]{x}$, and let $T_3(x)$ be the third order Taylor polynomial for $f(x)$ about $x = 32$. Give a bound on the error $|f(30) - T_3(30)|$.

Q[9]: Let

$$f(x) = \sin\left(\frac{1}{x}\right),$$

and let $T_1(x)$ be the first order Taylor polynomial for $f(x)$ about $x = \frac{1}{\pi}$. Give a bound on the error $|f(0.01) - T_1(0.01)|$, using Equation 3.4.33. You may leave your answer in terms of π .

Then, give a *reasonable* bound on the error $|f(0.01) - T_1(0.01)|$.

Q[10]: Let $f(x) = \arcsin x$, and let $T_2(x)$ be the second order Maclaurin polynomial for $f(x)$. Give a reasonable bound on the error $|f(\frac{1}{2}) - T_2(\frac{1}{2})|$ using Equation 3.4.33. What is the exact value of the error $|f(\frac{1}{2}) - T_2(\frac{1}{2})|$?

►► Stage 3

Q[11]: Let $f(x) = \log(x)$, and let $T_n(x)$ be the n th order Taylor polynomial for $f(x)$ about $x = 1$. You use $T_n(1.1)$ to estimate $\log(1.1)$. If your estimation needs to have an error of no more than 10^{-4} , what is an acceptable value of n to use?

Q[12]: Give an estimation of $\sqrt[7]{2200}$ using a Taylor polynomial. Your estimation should have an error of less than 0.001.

Q[13]: Use Equation 3.4.33 to show that

$$\frac{4241}{5040} \leq \sin(1) \leq \frac{4243}{5040}$$

Q[14]: In this question, we use the remainder of a Maclaurin polynomial to approximate e .

- (a) Write out the 4th order Maclaurin polynomial $T_4(x)$ of the function e^x .
- (b) Compute $T_4(1)$.
- (c) Use your answer from **b** to conclude $\frac{326}{120} < e < \frac{325}{119}$.

3.4.9 ► Further problems

Exercises

Jump to [HINTS](#), [ANSWERS](#), [SOLUTIONS](#) or [TABLE OF CONTENTS](#)

► More Problems for Section 3.4

► Stage 1

Q[1](*) : Consider a function $f(x)$ whose third order Maclaurin polynomial is $4 + 3x^2 + \frac{1}{2}x^3$. What is $f'(0)$? What is $f''(0)$?

Q[2](*) : Consider a function $h(x)$ whose third order Maclaurin polynomial is $1 + 4x - \frac{1}{3}x^2 + \frac{2}{3}x^3$. What is $h^{(3)}(0)$?

Q[3](*) : The third order Taylor polynomial of $h(x)$ about $x = 2$ is $3 + \frac{1}{2}(x - 2) + 2(x - 2)^3$.

What is $h'(2)$? What is $h''(2)$?

► Stage 2

Q[4](*) : The function $f(x)$ has the property that $f(3) = 2$, $f'(3) = 4$ and $f''(3) = -10$.

- (a) Use the linear approximation to $f(x)$ centred at $x = 3$ to approximate $f(2.98)$.
- (b) Use the quadratic approximation to $f(x)$ centred at $x = 3$ to approximate $f(2.98)$.

Q[5](*) : Use the tangent line to the graph of $y = x^{1/3}$ at $x = 8$ to find an approximate value for $10^{1/3}$. Is the approximation too large or too small?

Q[6](*) : Estimate $\sqrt{2}$ using a linear approximation.

Q[7](*) : Estimate $\sqrt[3]{26}$ using a linear approximation.

Q[8](*) : Estimate $(10.1)^5$ using a linear approximation.

Q[9](*) : Estimate $\sin\left(\frac{101\pi}{100}\right)$ using a linear approximation. (Leave your answer in terms of π .)

Q[10](*) : Use a linear approximation to estimate $\arctan(1.1)$, using $\arctan 1 = \frac{\pi}{4}$.

Q[11](*) : Use a linear approximation to estimate $(2.001)^3$. Write your answer in the form $n/1000$ where n is an integer.

Q[12](*) : Using a suitable linear approximation, estimate $(8.06)^{2/3}$. Give your answer as a fraction in which both the numerator and denominator are integers.

Q[13](*) : Find the third-order Taylor polynomial for $f(x) = (1 - 3x)^{-1/3}$ around $x = 0$.

Q[14](*) : Consider a function $f(x)$ which has $f^{(3)}(x) = \frac{x}{22 - x^2}$. Show that when we approximate $f(2)$ using its second order Taylor polynomial at $a = 1$, the absolute value of the error is less than $\frac{1}{50} = 0.02$.

Q[15](*) : Consider a function $f(x)$ which has $f^{(4)}(x) = \frac{\cos(x^2)}{3 - x}$. Show that when we approximate $f(0.5)$ using its third order Maclaurin polynomial, the absolute value of the error is less than $\frac{1}{500} = 0.002$.

Q[16](*) : Consider a function $f(x)$ which has $f^{(3)}(x) = \frac{e^{-x}}{8 + x^2}$. Show that when we approximate $f(1)$ using its second order Maclaurin polynomial, the absolute value of the error is less than $1/40$.

Q[17](*) :

(a) By using an appropriate linear approximation for $f(x) = x^{1/3}$, estimate $5^{2/3}$.

(b) Improve your answer in (a) by making a quadratic approximation.

(c) Obtain an error estimate for your answer in (a) (not just by comparing with your calculator's answer for $5^{2/3}$).

► Stage 3

Q[18]: The 4th order Maclaurin polynomial for $f(x)$ is

$$T_4(x) = 5x^2 - 9$$

This will probably be useful information. Since we're also given the value of a derivative, let's differentiate the equation relating the variables implicitly with respect to t . For ease of notation, we will write $\frac{dA}{dt} = A'$, etc.

$$A'B + AB' = \frac{2CC' + 2DD'}{C^2 + D^2 + 1}$$

At $t = 10$, $A = C = D = 0$:

$$\begin{aligned} A'B + 0 &= \frac{0 + 0}{0 + 0 + 1} \\ A'B &= 0 \end{aligned}$$

at $t = 10$, $A' = 2$ units per second:

$$\begin{aligned} 2B &= 0 \\ B &= 0. \end{aligned}$$

Solutions to Exercises [3.3.1](#) — Jump to [TABLE OF CONTENTS](#)

S-1: In the beginning of this section, the text says “A differential equation is an equation for an unknown function that involves the derivative of the unknown function.” Our unknown function is y , so a differential equation is an equation that relates y and $\frac{dy}{dx}$. This applies to (a) and (b), but not (c), (d), or (e).

Note that $\frac{dx}{dx} = 1$: this is the derivative of x with respect to x .

S-2: Theorem 3.3.2 tells us that a function is a solution to the differential equation $\frac{dQ}{dt} = kQ(t)$ if and only if the function has the form $Q(t) = Ce^{kt}$ for some constant C . In our case, we want $Q(t) = 5\frac{dQ}{dt}$, so $\frac{dQ}{dt} = \frac{1}{5}Q(t)$. So, the theorem tells us that the solutions are the functions of the form $Q(t) = Ce^{t/5}$. This applies to (a) (with $C = 0$) and (d) (with $C = 1$), but none of the other functions.

We don't actually need a theorem to answer this question, though: we can just test every option.

- (a) $\frac{dQ}{dt} = 0$, so $Q(t) = 0 = 5 \cdot 0 = 5\frac{dQ}{dt}$, so (a) is a solution.
- (b) $\frac{dQ}{dt} = 5e^t = Q(t)$, so $Q(t) = \frac{dQ}{dt} \neq 5\frac{dQ}{dt}$, so (b) is not a solution.
- (c) $\frac{dQ}{dt} = 5e^{5t} = 5Q(t)$, so $Q(t) = \frac{1}{5}\frac{dQ}{dt} \neq 5\frac{dQ}{dt}$, so (c) is not a solution.
- (d) $\frac{dQ}{dt} = \frac{1}{5}e^{t/5} = \frac{1}{5}Q(t)$, so $Q(t) = 5\frac{dQ}{dt}$, so (d) is a solution.
- (e) $\frac{dQ}{dt} = \frac{1}{5}e^{t/5} = \frac{1}{5}(Q(t) - 1)$, so $Q(t) = 5\frac{dQ}{dt} + 1$, so (e) is not a solution.

S-3: What we're asked to find is when

$$Q(t) = 0$$

That is,

$$Ce^{-kt} = 0$$

If $C = 0$, then this is the case for all t . There was no isotope to begin with, and there will continue not being any undecayed isotope forever.

If $C > 0$, then since $e^{-kt} > 0$, also $Q(t) > 0$: so $Q(t)$ is never 0 for any value of t . (But as t gets bigger and bigger, $Q(t)$ gets closer and closer to 0.)

Remark: The last result is somewhat disturbing: surely at some point the last atom has decayed. The differential equation we use is a model that assumes Q runs continuously. This is a good approximation only when there is a very large number of atoms. In practice, that is almost always the case.

S-4: The two pieces of information give us

$$f(0) = A = 5$$

$$f(7) = Ae^{7k} = \pi$$

Thus we know that $A = 5$ and so $\pi = f(7) = 5e^{7k}$. Hence

$$\begin{aligned} e^{7k} &= \frac{\pi}{5} \\ 7k &= \log(\pi/5) \\ k &= \frac{1}{7} \cdot \log(\pi/5). \end{aligned}$$

where we use $\log$ to mean natural logarithm, $\log_e$.

S-5: In Theorem 3.3.2, we saw that if y is a function of t , and $\frac{dy}{dt} = -ky$, then $y = Ce^{-kt}$ for some constant C .

Our equation y satisfies $\frac{dy}{dt} = -3y$, so the theorem tells us $y = Ce^{-3t}$ for some constant C .

We are also told that $y(1) = 2$. So, $2 = Ce^{-3 \times 1}$ tells us $C = 2e^3$. Then:

$$y = 2e^3 \cdot e^{-3t} = 2e^{-3(t-1)}.$$

S-6: The amount of Carbon-14 in the sample t years after the animal died will be

$$Q(t) = 5e^{-kt}$$

for some constant k (where 5 is the amount of Carbon-14 in the sample at time $t = 0$). So, the answer we're looking for is $Q(10000)$. We need to replace k with an actual number to evaluate

$Q(10000)$, and the key to doing this is the half-life. The text tells us that the half-life of Carbon-14 is 5730 years, so we know:

$$\begin{aligned} Q(5730) &= \frac{5}{2} \\ 5e^{-k \cdot 5730} &= \frac{5}{2} \\ (e^{-k})^{5730} &= \frac{1}{2} \\ e^{-k} &= \sqrt[5730]{\frac{1}{2}} = 2^{-\frac{1}{5730}} \end{aligned}$$

So:

$$\begin{aligned} Q(t) &= 5(e^{-k})^t \\ &= 5 \cdot 2^{-\frac{t}{5730}} \end{aligned}$$

Now, we can evaluate:

$$Q(10000) = 5 \cdot 2^{-\frac{10000}{5730}} \approx 1.5 \mu g$$

Remark: after $2(5730) = 11,460$ years, the sample will have been sitting for two half-lives, so its remaining Carbon-14 will be a quarter of its original amount, or $1.25 \mu g$. It makes sense that at 10,000 years, the sample will contain slightly more Carbon-14 than at 11,460 years. Indeed, 1.5 is slightly larger than 1.25, so our answer seems plausible.

It's a good habit to look for ways to quickly check whether your answer seems plausible, since a small algebra error can easily turn into a big error in your solution.

S-7: Let 100 years ago be the time $t = 0$. Then if $Q(t)$ is the amount of Radium-226 in the sample, $Q(0) = 1$, and

$$Q(t) = e^{-kt}$$

for some positive constant k . When $t = 100$, the amount of Radium-226 left is 0.9576 grams, so

$$\begin{aligned} 0.9576 &= Q(100) = e^{-k \cdot 100} = (e^{-k})^{100} \\ e^{-k} &= 0.9576^{\frac{1}{100}} \end{aligned}$$

This tells us

$$Q(t) = 0.9576^{\frac{t}{100}}$$

So, if half the original amount of Radium-226 is left,

$$\begin{aligned} \frac{1}{2} &= 0.9576^{\frac{t}{100}} \\ \log\left(\frac{1}{2}\right) &= \log\left(0.9576^{\frac{t}{100}}\right) \\ -\log 2 &= \frac{t}{100} \log(0.9576) \\ t &= -100 \frac{\log 2}{\log 0.9576} \approx 1600 \end{aligned}$$

So, the half life of Radium-226 is about 1600 years.

S-8: Let $Q(t)$ denote the mass at time t . Then $\frac{dQ}{dt}$ is the rate at which the mass is changing. Since the rate the mass is decreasing is proportional to the mass remaining, we know $\frac{dQ}{dt} = -kQ(t)$, where k is a positive constant. (Remark: since Q is decreasing, $\frac{dQ}{dt}$ is negative. Since we cannot have a negative mass, if we choose k to be positive, then k and Q are both positive—this is why we added the negative sign.)

The information given in the question is:

$$Q(0) = 6 \quad \frac{dQ}{dt} = -kQ(t) \quad Q(1) = 1$$

for some constant $k > 0$. By Theorem 3.3.2, we know

$$Q(t) = Ce^{-kt}$$

for some constant C . Since $Q(0) = Ce^0 = C$, the given information tells us $6 = C$. (This is the initial mass of our sample.) So, $Q(t) = 6e^{-kt}$. To get the full picture of the behaviour of Q , we should find k . We do this using the given information $Q(1) = 1$:

$$\begin{aligned} 1 &= Q(1) = 6e^{-k(1)} \\ 6^{-1} &= \frac{1}{6} = e^{-k} \end{aligned}$$

So, all together,

$$Q(t) = 6 \left(e^{-k} \right)^t = 6 \cdot \left(6^{-1} \right)^t = 6^{1-t}$$

The question asks us to determine the time t_h which obeys $Q(t_h) = \frac{6}{2} = 3$. Now that we know the equation for $Q(t)$, we simply solve:

$$\begin{aligned} Q(t) &= 6^{1-t} \\ 3 &= Q(t_h) = 6^{1-t_h} \\ \log 3 &= \log \left(6^{1-t_h} \right) = (1 - t_h) \log 6 \\ \frac{\log 3}{\log 6} &= 1 - t_h \\ t_h &= 1 - \frac{\log 3}{\log 6} = \frac{\log 6 - \log 3}{\log 6} = \frac{\log 2}{\log 6} \end{aligned}$$

The half-life of Polonium-210 is $\frac{\log 2}{\log 6}$ years, or about 141 days.

Remark: The actual half-life of Polonium-210 is closer to 138 days. The numbers in the question are made to work out nicely, at the expense of some accuracy.

S-9: The amount of Radium-221 in a sample will be

$$Q(t) = Ce^{-kt}$$

where C is the amount in the sample at time $t = 0$, and k is some positive constant. We know the half-life of the isotope, so we can find e^{-k} :

$$\begin{aligned}\frac{C}{2} &= Q(30) = Ce^{-k \cdot 30} \\ \frac{1}{2} &= (e^{-k})^{30} \\ 2^{-\frac{1}{30}} &= e^{-k}\end{aligned}$$

So,

$$Q(t) = C(e^{-k})^t = C \cdot 2^{-\frac{t}{30}}$$

When only 0.01% of the original sample is left, $Q(t) = 0.0001C$:

$$\begin{aligned}0.0001C &= Q(t) = C \cdot 2^{-\frac{t}{30}} \\ 0.0001 &= 2^{-\frac{t}{30}} \\ \log(0.0001) &= \log\left(2^{-\frac{t}{30}}\right) \\ \log(10^{-4}) &= -\frac{t}{30} \log 2 \\ -4 \log 10 &= -\frac{t}{30} \log 2 \\ t &= 120 \cdot \frac{\log 10}{\log 2} \approx 398.6\end{aligned}$$

It takes about 398.6 seconds (that is, roughly 6 and a half minutes) for all but 0.01% of the sample to decay.

Remark: we can do another reality check here. The half-life is 30 seconds. 6 and a half minutes represents 13 half-lives. So, the sample is halved 13 times: $\left(\frac{1}{2}\right)^{13} \approx 0.00012 = 0.012\%$. So these 13 half-lives should reduce the sample to about 0.01% of its original amount, as desired.

S-10: We know that the amount of Polonium-210 in a sample after t days is given by

$$Q(t) = Ce^{-kt}$$

where C is the original amount of the sample, and k is some positive constant.

The question asks us what percentage of the sample decays in a day. Since t is measured in days, the amount that decays in a day is $Q(t) - Q(t+1)$. The percentage of $Q(t)$ that this represents is $100 \frac{Q(t) - Q(t+1)}{Q(t)}$. (For example, if there were two grams at time t , and one gram at time $t+1$, then $100 \frac{2-1}{2} = 50$: 50% of the sample decayed in a day.)

In order to simplify, we should figure out a better expression for $Q(t)$. As usual, we make use of the half-life.

$$\begin{aligned} Q(138) &= \frac{C}{2} \\ Ce^{-k \cdot 138} &= \frac{C}{2} \\ (e^{-k})^{138} &= \frac{1}{2} = 2^{-1} \\ e^{-k} &= 2^{-\frac{1}{138}} \end{aligned}$$

Now, we have a better formula for $Q(t)$:

$$\begin{aligned} Q(t) &= C \left(e^{-k} \right)^t \\ Q(t) &= C \cdot 2^{-\frac{t}{138}} \end{aligned}$$

Finally, we can evaluate what percentage of the sample decays in a day.

$$\begin{aligned} 100 \frac{Q(t) - Q(t+1)}{Q(t)} &= 100 \frac{C \cdot 2^{-\frac{t}{138}} - C \cdot 2^{-\frac{t+1}{138}}}{C \cdot 2^{-\frac{t}{138}}} \left(\frac{\frac{1}{C}}{\frac{1}{C}} \right) \\ &= 100 \frac{2^{-\frac{t}{138}} - 2^{-\frac{t+1}{138}}}{2^{-\frac{t}{138}}} \\ &= 100 \left(2^{-\frac{t}{138}} - 2^{-\frac{t+1}{138}} \right) 2^{\frac{t}{138}} \\ &= 100 \left(1 - 2^{-\frac{1}{138}} \right) \approx 0.5 \end{aligned}$$

About 0.5% of the sample decays in a day.

Remark: when we say that half a percent of the sample decays in a day, we don't mean half a percent of the *original* sample. If a day starts out with, say, 1 microgram, then what decays in the next 24 hours is about half a percent of that 1 microgram, regardless of what the "original" sample (at some time $t = 0$) held.

In particular, since the sample is getting smaller and smaller, that half of a percent that decays every day represents fewer and fewer actual atoms decaying. That's why we can't say that half of the sample (50%) will decay after about 100 days, even though 0.5% decays every day and $100 \times 0.5 = 50$.

S-11: The amount of Uranium-232 in the sample of ore at time t will be

$$Q(t) = Q(0)e^{-kt}$$

where $6.9 \leq Q(0) \leq 7.5$. We don't exactly know $Q(0)$, and we don't exactly know the half-life, so we also won't exactly know $Q(10)$: we can only say that it is between two numbers. Our strategy is to find the highest and lowest possible values of $Q(10)$, given the information in the problem.

In order for the most possible Uranium-232 to be in the sample after 10 years, we should start with the most and have the longest half-life (since this represents the slowest decay). So, we take $Q(0) = 7.5$ and $Q(70) = \frac{1}{2}(7.5)$.

$$\begin{aligned} Q(t) &= 7.5e^{-kt} \\ \frac{1}{2}(7.5) &= Q(70) = 7.5e^{-k(70)} \\ \frac{1}{2} &= \left(e^{-k}\right)^{70} \\ 2^{-\frac{1}{70}} &= e^{-k} \end{aligned}$$

So, in this scenario,

$$Q(t) = 7.5 \cdot 2^{-\frac{t}{70}}$$

After ten years,

$$Q(10) = 7.5 \cdot 2^{-\frac{10}{70}} \approx 6.79$$

So after ten years, the sample contains *at most* $6.8 \mu\text{g}$.

Now, let's think about the least possible amount of Uranium-232 that could be left after 10 years. We should start with as little as possible, so take $Q(0) = 6.9$, and the sample should decay quickly, so take the half-life to be 68.8 years.

$$\begin{aligned} Q(t) &= 6.9e^{-kt} \\ \frac{1}{2}6.9 &= Q(68.8) = 6.9e^{-k(68.8)} \\ \frac{1}{2} &= \left(e^{-k}\right)^{68.8} \\ 2^{-\frac{1}{68.8}} &= e^{-k} \end{aligned}$$

In this scenario,

$$Q(t) = 6.9 \cdot 2^{-\frac{t}{68.8}}$$

After ten years,

$$Q(10) = 6.9 \cdot 2^{-\frac{10}{68.8}} \approx 6.24$$

So after ten years, the sample contains *at least* $6.2 \mu\text{g}$.

After ten years, the sample contains between 6.2 and $6.8 \mu\text{g}$ of Uranium-232.

Solutions to Exercises [3.3.2](#) — Jump to [TABLE OF CONTENTS](#)

S-1: Using Corollary 3.3.8 (with $A = 20$ and $K = 5$), solutions to the differential equation all have the form

$$T(t) = [T(0) - 20]e^{5t} + 20$$

for some constant $T(0)$. This fits (a) (with $T(0) = 20$), (c) (with $T(0) = 21$), and (d) (with $T(0) = 40$), but not (b) (since the constant has the wrong sign).

Instead of using the corollary, we can also just check each function for ourselves.

- (a) $\frac{dT}{dt} = 0 = 5 \cdot 0 = 5[T(t) - 20]$, so (a) gives a solution to the differential equation.
- (b) $\frac{dT}{dt} = 5[20e^{5t}] = 5[T + 20] \neq 5[T - 20]$, so (b) does not give a solution to the differential equation.
- (c) $\frac{dT}{dt} = 5[e^{5t}] = 5[T - 20]$, so (c) gives a solution to the differential equation.
- (d) $\frac{dT}{dt} = 5[20e^{5t}] = 5[T - 20]$, so (d) gives a solution to the differential equation.

S-2: From Newton's Law of Cooling and Corollary 3.3.8, the temperature of the object will be

$$T(t) = [T(0) - A]e^{Kt} + A$$

where A is the ambient temperature (the temperature of the room), $T(0)$ is the initial temperature of the copper, and K is some constant. So, the ambient temperature—the temperature of the room—is -10 degrees. Since the coefficient of the exponential part of the function is positive, the temperature of the object is higher than the temperature of the room.

S-3: As t grows very large, $T(t)$ approaches A . That is:

$$\begin{aligned}\lim_{t \rightarrow \infty} T(t) &= A \\ \lim_{t \rightarrow \infty} [T(0) - A]e^{Kt} + A &= A \\ \lim_{t \rightarrow \infty} [T(0) - A]e^{Kt} &= 0\end{aligned}$$

Since the object is warmer than the room, $T(0) - A$ is a nonzero constant. So,

$$\lim_{t \rightarrow \infty} e^{Kt} = 0$$

This tells us that K is a negative number. So, K must be negative—not zero, and not positive.

Remark: in our work, we used the fact that the object and the room have different temperatures (but it didn't matter which one was hotter). If not, then $T(0) = A$, and $T(t) = A$: that is, the temperature of the object is constant. In this case, our usual form for the temperature of the object looks like this:

$$T(t) = 0e^{Kt} + A$$

Keeping the exponential piece in there is overkill: the temperature isn't changing, the function is simply constant. If the object and the room have the same temperature, K could be any real number since we're multiplying e^{Kt} by zero.

Remark: contrast this to Question **9**.

S-4: We want to know when

$$[T(0) - A]e^{kt} + A = A$$

That is, when

$$[T(0) - A]e^{kt} = 0$$

Since $e^{kt} > 0$ for all values of k and t , this happens exactly when

$$T(0) - A = 0$$

So: if the initial temperature of the object is not the same as the ambient temperature, then according to this model, it never will be! (However, as t gets larger and larger, $T(t)$ gets closer and closer to A —it just never exactly reaches there.)

If the initial temperature of the object starts out the same as the ambient temperature, then $T(t) = A$ for all values of t .

S-5: From Newton's Law of Cooling and Corollary 3.3.8, we know the temperature of the copper will be

$$T(t) = [T(0) - A]e^{Kt} + A$$

where A is the ambient temperature (100°), $T(0)$ is the temperature of the copper at time 0 (let's make this the instant it was dumped in the water, so $T(0) = 25^\circ$), and K is some constant. That is:

$$\begin{aligned} T(t) &= [25 - 100]e^{Kt} + 100 \\ &= -75e^{Kt} + 100 \end{aligned}$$

The information given tells us that $T(10) = 90$, so

$$\begin{aligned} 90 &= -75e^{10K} + 100 \\ 75(e^K)^{10} &= 10 \\ (e^K)^{10} &= \frac{2}{15} \\ e^K &= \left(\frac{2}{15}\right)^{\frac{1}{10}} \end{aligned}$$

This lets us describe $T(t)$ without any unknown constants.

$$\begin{aligned} T(t) &= -75(e^K)^t + 100 \\ &= -75\left(\frac{2}{15}\right)^{\frac{t}{10}} + 100 \end{aligned}$$

The question asks what value of t gives $T(t) = 99.9$.

$$\begin{aligned}99.9 &= -75 \left(\frac{2}{15} \right)^{\frac{t}{10}} + 100 \\75 \left(\frac{2}{15} \right)^{\frac{t}{10}} &= 0.1 \\ \left(\frac{2}{15} \right)^{\frac{t}{10}} &= \frac{1}{750} \\ \log \left(\left(\frac{2}{15} \right)^{\frac{t}{10}} \right) &= \log \left(\frac{1}{750} \right) \\ \frac{t}{10} \log \left(\frac{2}{15} \right) &= -\log(750) \\ t &= \frac{-10 \log(750)}{\log \left(\frac{2}{15} \right)} \approx 32.9\end{aligned}$$

It takes about 32.9 seconds.

S-6: The temperature of the stone t minutes after taking it from the bonfire is

$$\begin{aligned}T(t) &= [T(0) - A]e^{Kt} + A \\ &= [500 - 0]e^{Kt} + 0 \\ &= 500e^{Kt}\end{aligned}$$

for some constant K . We are given that $T(10) = 100$.

$$\begin{aligned}100 &= T(10) = 500e^{10K} \\ e^{10K} &= \frac{1}{5} \\ e^K &= 5^{-\frac{1}{10}}\end{aligned}$$

This gives us the more complete picture for the temperature of the stone.

$$T(t) = 500 \left(e^K \right)^t = 500 \cdot 5^{-\frac{t}{10}}$$

If $T(t) = 50$:

$$\begin{aligned}50 &= T(t) = 500 \cdot 5^{-\frac{t}{10}} \\ \frac{1}{10} &= 10^{-1} = 5^{-\frac{t}{10}} \\ 10 &= 5^{\frac{t}{10}} \\ \log(10) &= \frac{t}{10} \log(5) \\ t &= 10 \frac{\log(10)}{\log(5)} \approx 14.3\end{aligned}$$

So the stone has been out of the fire for about 14.3 minutes.

S-7:

- First scenario: At time 0, Newton mixes 9 parts coffee at temperature 95°C with 1 part cream at temperature 5°C . The resulting mixture has temperature

$$\frac{9 \times 95 + 1 \times 5}{9 + 1} = 86^\circ$$

The mixture cools according to Newton's Law of Cooling, with initial temperature 86° and ambient temperature 22° :

$$T(t) = [86 - 22]e^{-kt} + 22$$

$$T(t) = 64e^{-kt} + 22$$

After 10 minutes,

$$54 = T(10) = 22 + 64e^{-10k}$$

$$e^{-10k} = \frac{54 - 22}{64} = \frac{1}{2}$$

We could compute k from this, but we don't need it.

- Second scenario: At time 0, Newton gets hot coffee at temperature 95°C . It cools according to Newton's Law of Cooling

$$T(t) = [T(0) - 22]e^{-kt} + 22$$

In this second scenario, $T(0) = 95$, so

$$T(t) = [95 - 22]e^{-kt} + 22 = 73e^{-kt} + 22$$

The value of k is the same as in the first scenario, so after 10 minutes

$$T(10) = 22 + 73e^{-10k} = 22 + 73\frac{1}{2} = 58.5$$

This cooled coffee is mixed with cold cream to yield a mixture of temperature

$$\frac{9 \times 58.5 + 1 \times 5}{9 + 1} = 53.15$$

Under the second (add cream just before drinking) scenario, the coffee ends up cooler by 0.85°C .

S-8:

- (a) By Newton's law of cooling, the rate of change of temperature is proportional to the difference between $T(t)$ and the ambient temperature, which in this case is 30° . Thus

$$\frac{dT}{dt} = k[T(t) - 30]$$

for some constant of proportionality k . To answer part (a), all we have to do is find k .

Under Newton's Law of Cooling, the temperature at time t will be given by

$$\begin{aligned}T(t) &= [T(0) - A]e^{kt} + A \\&= [5 - 30]e^{kt} + 30 \\&= -25e^{kt} + 30\end{aligned}$$

We are told $T(5) = 10$:

$$\begin{aligned}10 &= -25e^{5k} + 30 \\25e^{5k} &= 20 \\e^{5k} &= \frac{4}{5} \\5k &= \log(4/5) \\k &= \frac{1}{5} \log(4/5)\end{aligned}$$

So, the differential equation is

$$\frac{dT}{dt}(t) = \frac{1}{5} \log(4/5) [T(t) - 30]$$

(b) Since $T(t) = 30 - 25e^{kt}$, the temperature of the tea is 20° when

$$\begin{aligned}30 - 25e^{kt} &= 20 \\e^{kt} &= \frac{10}{25} \\kt &= \log\left(\frac{10}{25}\right) \\t &= \frac{1}{k} \log \frac{2}{5} \\&= \frac{5 \log(2/5)}{\log(4/5)} \\&\approx 20.53 \text{ min}\end{aligned}$$

S-9: As time goes on, temperatures that follow Newton's Law of Cooling get closer and closer to the ambient temperature. So, $\lim_{t \rightarrow \infty} T(t)$ exists. In particular, $\lim_{t \rightarrow \infty} 0.8^{kt}$ exists.

- If $k < 0$, then $\lim_{t \rightarrow \infty} 0.8^{kt} = \infty$, since $0.8 < 1$. So, $k \geq 0$.
- If $k = 0$, then $T(t) = 16$ for all values of t . But, in the statement of the question, the object is changing temperature. So, $k > 0$.

Therefore, k is positive.

Remark: contrast this to Question **3**. The reason we get a different answer is that our base in this question (0.8) is less than one, while the base in Question **3**(e) is greater than one.

Although the given equation $T(t)$ does not exactly look like the Newton's Law equations we're used to, it is equivalent. Remembering $e^{\log(0.8)} = 0.8$:

$$\begin{aligned} T(t) &= 0.8^{kt} + 15 \\ &= (e^{\log 0.8})^{kt} + 15 \\ &= e^{(k \log 0.8)t} + 15 \\ &= [16 - 15]e^{(k \log 0.8)t} + 15 \\ &= [16 - 15]e^{Kt} + 15 \end{aligned}$$

with $K = k \log 0.8$. This is now the more familiar form for Newton's Law of Cooling (with $A = 15$ and $T(0) = 16$).

Since $0.8 < 1$, $\log(0.8)$ is negative, so k and K have opposite signs.

Solutions to Exercises [3.3.3](#) — Jump to [TABLE OF CONTENTS](#)

S-1: Since b is a positive constant, $\lim_{t \rightarrow \infty} e^{bt} = \infty$. Therefore:

$$\lim_{t \rightarrow \infty} P(t) = \lim_{t \rightarrow \infty} P(0)e^{bt} = \begin{cases} 0 & \text{if } P(0) = 0 \\ \infty & \text{if } P(0) > 0 \end{cases}$$

If $P(0) = 0$, then the model simply says “a population that starts with no individuals continues to have no individuals indefinitely,” which certainly makes sense. If $P(0) \neq 0$, then (since we can't have a negative population) $P(0) > 0$, and the model says “a population that starts out with some individuals will end up with any gigantically huge number you can think of, given enough time.” This one doesn't make so much sense. Populations only grow to a certain finite amount, due to scarcity of resources and such. In the derivation of the Malthusian model, we assume a constant net birth rate—that the birth and death rates (per individual) don't depend on the population, which is not a reasonable assumption long-term.

S-2: The assumption that the animals grow according to the Malthusian model tells us that their population t years after 2015 is given by $P(t) = 121e^{bt}$ for some constant b , since $121 = P(0)$, the population 0 years after 2015. The information about 2016 tells us

$$\begin{aligned} 136 &= P(1) = 121e^b \\ \frac{136}{121} &= e^b \end{aligned}$$

This gives us a better idea of $P(t)$:

$$P(t) = 121e^{bt} = 121 \left(\frac{136}{121} \right)^t$$

2020 is 5 years after 2015, so in 2020 (assuming the population keeps growing according to the Malthusian model) the population will be

$$P(5) = 121 \left(\frac{136}{121} \right)^5 \approx 217$$

In 2020, the Malthusian model predicts the herd will number 217 individuals.

S-3: Since the initial population of bacteria is 1000 individuals, the Malthusian model says that the population of bacteria t hours after being placed in the dish will be $P(t) = 1000e^{bt}$ for some constant b . Since $P(1) = 2000$,

$$\begin{aligned} 2000 &= P(1) = 1000e^b \\ 2 &= e^b \end{aligned}$$

So, the population at time t is

$$P(t) = 1000 \cdot 2^t$$

We want to know at what time the population triples, to 3,000 individuals.

$$\begin{aligned} 3000 &= 1000 \cdot 2^t \\ 3 &= 2^t \\ \log(3) &= \log(2^t) = t \log(2) \\ t &= \frac{\log(3)}{\log(2)} \approx 1.6 \end{aligned}$$

The population triples in about 1.6 hours. Note that the tripling time depends only on the constant b . In particular, it does not depend on the initial condition $P(0)$.

S-4: According to the Malthusian Model, if the ship wrecked at year $t = 0$ and 2 rats washed up on the island, then t years after the wreck, the population of rats will be

$$P(t) = 2e^{bt}$$

for some constant b . We want to get rid of this extraneous variable b , so we use the given information. If 1928 is a years after the wreck:

$$\begin{aligned} 1000 &= P(a) = 2e^{ba} \\ 1500 &= P(a+1) = 2e^{b(a+1)} = 2e^{ba}e^b \end{aligned}$$

So,

$$(1000) (e^b) = (2e^{ba}) (e^b) = 1500$$

Which tells us

$$e^b = \frac{1500}{1000} = 1.5$$

Now, our model is complete:

$$P(t) = 2 (e^b)^t = 2 \cdot 1.5^t$$

Since $P(a) = 1000$, we can find a :

$$\begin{aligned} 1000 &= P(a) = 2 \cdot 1.5^a \\ 500 &= 1.5^a \\ \log(500) &= \log(1.5^a) = a \log(1.5) \\ a &= \frac{\log(500)}{\log(1.5)} \approx 15.3 \end{aligned}$$

So, the year 1928 was about 15.3 years after the shipwreck. Since we aren't given a month when the rats reached exactly 1000 in number, that puts the shipwreck at the year 1912 or 1913.

S-5: The Malthusian model suggests that, if we start with $P(0)$ cochineals, their population after 3 months will be

$$P(t) = P(0)e^{bt}$$

for some constant b . The constant b is the net birthrate per population member per unit time. Assuming that the net birthrate for a larger population will be the same as the test population, we can use the data from the test to find e^b . Let $Q(t)$ be the number of individuals in the test population at time t .

$$Q(t) = Q(0)e^{bt} = 200e^{bt}$$

$$1000 = Q(3) = 200e^{3b}$$

$$5 = e^{3b}$$

$$e^b = 5^{1/3}$$

Now that we have an idea of the birthrate, we predict

$$P(t) = P(0) \left(e^b\right)^t = P(0) \cdot 5^{t/3}$$

We want $P(12) = 10^6 + P(0)$.

$$10^6 + P(0) = P(12) = P(0) \cdot 5^{12/3} = P(0) \cdot 5^4$$

$$10^6 = P(0) \cdot 5^4 - P(0) = P(0) [5^4 - 1]$$

$$P(0) = \frac{10^6}{5^4 - 1} \approx 1603$$

The farmer should use an initial population of (at least) about 1603 individuals.

Remark: if we hadn't specified that we need to save $P(0)$ individuals to start next year's population, the number of individual cochineals we would want to start with to get a million in a year would be 1600—almost the same!

S-6:

- (a) Since $f(t)$ gives the amount of the radioactive isotope in the sample at time t , the amount of the radioactive isotope in the sample when $t = 0$ is $f(0) = 100e^0 = 100$ units. Since the sample is decaying, $f(t)$ is decreasing, so $f'(t)$ is negative. Differentiating, $f'(t) = k(100e^{kt})$. Since $100e^{kt}$ is positive and $f'(t)$ is negative, k is negative.
- (b) Since $f(t)$ gives the size of the population at time t , the number of individuals in the population when $t = 0$ is $f(0) = 100e^0 = 100$. Since the population is growing, $f(t)$ is increasing, so $f'(t)$ is positive. Differentiating, $f'(t) = k(100e^{kt})$. Since $100e^{kt}$ is positive and $f'(t)$ is positive, k is also positive.
- (c) Newton's Law of Cooling gives the temperature of an object at time t as $f(t) = [f(0) - A]e^{kt} + A$, where A is the ambient temperature surrounding the object. In our case, the ambient temperature is 0 degrees. In an object whose temperature is being modelled by Newton's Law of Cooling, it doesn't matter whether the object is heating or

cooling, k is negative. We saw this in Question 3 of Section 3.3.2, but it bears repeating. Since $f(t)$ approaches the ambient temperature (in this case, 0) as t goes to infinity:

$$\lim_{t \rightarrow \infty} 100e^{kt} = 0$$

so k is negative.

Solutions to Exercises 3.3.4 — Jump to [TABLE OF CONTENTS](#)

S-1: The first piece of information given tells us $\frac{df}{dx} = \pi f(x)$. Then by Theorem 3.3.2,

$$f(x) = Ce^{\pi x}$$

for some constant C . The second piece of given information tells us $f(0) = 2$. Using this, we can solve for C :

$$2 = f(0) = Ce^0 = C$$

Now, we know $f(x)$ entirely:

$$f(x) = 2e^{\pi x}$$

So, we can evaluate $f(2)$

$$f(2) = 2e^{2\pi}$$

S-2: To use Corollary 3.3.8, we re-write the differential equation as

$$\frac{dT}{dt} = 7 \left[T - \left(-\frac{9}{7} \right) \right].$$

Now, $A = -\frac{9}{7}$ and $K = 7$, so we see that the solutions to the differential equation have the form

$$T(t) = \left[T(0) + \frac{9}{7} \right] e^{7t} - \frac{9}{7}$$

for some constant $T(0)$.

We can check that this is reasonable: if

$$T(t) = \left[T(0) + \frac{9}{7} \right] e^{7t} - \frac{9}{7}$$

then

$$\begin{aligned} \frac{dT}{dt} &= 7 \left[T(0) + \frac{9}{7} \right] e^{7t} \\ &= 7 \left[T + \frac{9}{7} \right] \\ &= 7T + 9. \end{aligned}$$

S-3: Let $Q(t)$ denote the amount of radioactive material after t days. Then $Q(t) = Q(0)e^{kt}$. We are told

$$Q(8) = 0.8 Q(0)$$

So,

$$Q(0)e^{8k} = 0.8 Q(0)$$

$$e^{8k} = 0.8$$

$$e^k = 0.8^{\frac{1}{8}}$$

If $Q(0) = 100$, the time t at which $Q(t) = 40$ is determined by

$$40 = Q(t) = Q(0)e^{kt} = 100e^{kt} = 100 \left(0.8^{\frac{1}{8}}\right)^t = 100 \cdot 0.8^{\frac{t}{8}}$$

Solving for t :

$$\frac{40}{100} = 0.8^{\frac{t}{8}}$$

$$\log(0.4) = \log\left(0.8^{\frac{t}{8}}\right) = \frac{t}{8} \log(0.8)$$

$$t = \frac{8 \log(0.4)}{\log(0.8)} \approx 32.85 \text{ days}$$

100 grams will decay to 40 grams in about 32.85 days.

S-4: Let $t = 0$ be the time the boiling water is left in the room, and let $T(t)$ be the temperature of the water t minutes later, so $T(0) = 100$. Using Newton's Law of Cooling, we model the temperature of the water at time t as

$$T(t) = [100 - A]e^{Kt} + A$$

where A is the room temperature, and K is some constant. We are told that $T(15) = 85$ and $T(30) = 73$, so:

$$85 = T(15) = [100 - A]e^{15K} + A$$

$$73 = T(30) = [100 - A]e^{30K} + A$$

Rearranging both equations:

$$\begin{aligned} \frac{85 - A}{100 - A} &= e^{15K} \\ \frac{73 - A}{100 - A} &= e^{30K} = \left(e^{15K}\right)^2 \end{aligned}$$

Using these equations:

$$\begin{aligned}
 \left(\frac{85-A}{100-A}\right)^2 &= (e^{15K})^2 = e^{30K} = \frac{73-A}{100-A} \\
 \frac{(85-A)^2}{100-A} &= 73-A \\
 (85-A)^2 &= (73-A)(100-A) \\
 85^2 - 170A + A^2 &= 7300 - 173A + A^2 \\
 173A - 170A &= 7300 - 85^2 \\
 3A &= 75 \\
 A &= 25
 \end{aligned}$$

The room temperature is 25° C.

S-5:

(a) The amount of money at time t obeys

$$\frac{dA}{dt} = 0.05A + 2,000 = 0.05[A - (-40,000)]$$

Using Corollary 3.3.8,

$$\begin{aligned}
 A(t) &= [A(0) + 40,000]e^{0.05t} - 40,000 \\
 &= 90,000 \cdot e^{0.05t} - 40,000
 \end{aligned}$$

where $t = 0$ corresponds to the year when the graduate is 25.

When the graduate is 65 years old, $t = 40$, so

$$A(40) = 90,000 e^{0.05 \times 40} - 40,000 \approx \$625,015.05$$

(b) When the graduate stops depositing money and instead starts withdrawing money at a rate W , the equation for A becomes

$$\frac{dA}{dt} = 0.05A - W = 0.05[A(t) - 20W]$$

Using Corollary 3.3.8, and assuming that the interest rate remains 5%,

$$\begin{aligned}
 A(t) &= [A(0) - 20W]e^{0.05t} + 20W \\
 &= [625,015.05 - 20W]e^{0.05t} + 20W
 \end{aligned}$$

Note that, for part (b), we only care about what happens when the graduate starts withdrawing money. We take $t = 0$ to correspond to the year when the graduate is 65—so we're using a different t from part (a). Then from part (a), $A(0) = 625,015.05$.

We want the account to be depleted when the graduate is 85. So, we want

$$\begin{aligned}
 0 &= A(20) \\
 0 &= 20W + e^{0.05 \times 20}(625,015.05 - 20W) \\
 0 &= 20W + e(625,015.05 - 20W) \\
 20(e-1)W &= 625,015.05e \\
 W &= \frac{625,015.05e}{20(e-1)} \approx \$49,437.96
 \end{aligned}$$

S-6: (a) The amount of money at time t obeys

$$\frac{dA}{dt} = 0.06A - 9,000 = 0.06[A - 150,000]$$

Using Corollary 3.3.8,

$$\begin{aligned} A(t) &= [A(0) - 150,000]e^{0.06t} + 150,000 \\ &= [120,000 - 150,000]e^{0.06t} + 150,000 \\ &= -30,000e^{0.06t} + 150,000 \end{aligned}$$

(b) The money runs out when $A(t) = 0$.

$$\begin{aligned} A(t) &= 0 \\ 150,000 - 30,000e^{0.06t} &= 0 \\ 30,000e^{0.06t} &= 150,000 \\ e^{0.06t} &= 5 \\ 0.06t &= \log 5 \\ t &= \frac{\log 5}{0.06} \approx 26.8 \text{ yrs} \end{aligned}$$

The money runs out in about 26.8 years.

Remark: without earning any interest, the money would have run out in about 13.3 years.

S-7: Let $Q(t)$ denote the number of bacteria at time t . We are told that $Q'(t) = kQ(t)$ for some constant of proportionality k . Consequently, $Q(t) = Q(0)e^{kt}$ (Corollary 3.3.2). We are also told

$$\begin{aligned} Q(9) &= 3Q(0) \\ \text{So, } Q(0)e^{9k} &= 3Q(0) \\ e^{9k} &= 3 \\ e^k &= 3^{\frac{1}{9}} \end{aligned}$$

The doubling time t obeys:

$$\begin{aligned} Q(t) &= 2Q(0) \\ \text{So, } Q(0)e^{kt} &= 2Q(0) \\ e^{kt} &= 2 \\ 3^{\frac{t}{9}} &= 2 \\ \frac{t}{9} \log 3 &= \log 2 \\ t &= 9 \frac{\log 2}{\log 3} \approx 5.68 \text{ hr} \end{aligned}$$

S-8: (a) We want our differential equation to have the format of the equation in Corollary 3.3.8:

$$\begin{aligned}\frac{dv}{dt}(t) &= -g - kv(t) \\ &= -k \left(v(t) + \frac{g}{k} \right) \\ &= -k \left(v(t) - \left(-\frac{g}{k} \right) \right)\end{aligned}$$

So, we can use the corollary, with $K = -k$, $T = v$, and $A = -\frac{g}{k}$.

$$\begin{aligned}v(t) &= \left(v(0) - \left(-\frac{g}{k} \right) \right) e^{-kt} - \frac{g}{k} \\ &= \left(v_0 + \frac{g}{k} \right) e^{-kt} - \frac{g}{k}\end{aligned}$$

(b)

$$\begin{aligned}\lim_{t \rightarrow \infty} v(t) &= \lim_{t \rightarrow \infty} \left[\left(v_0 + \frac{g}{k} \right) e^{-kt} - \frac{g}{k} \right] \\ &= \left(v_0 + \frac{g}{k} \right) \left(\lim_{t \rightarrow \infty} e^{-kt} \right) - \frac{g}{k}\end{aligned}$$

Since k is *positive*:

$$\begin{aligned}&= \left(v_0 + \frac{g}{k} \right) (0) - \frac{g}{k} \\ &= -\frac{g}{k}\end{aligned}$$

Remark: This means, as the object falls, instead of accelerating without bound, it approaches some maximum speed. The velocity is negative because the object is moving in the negative direction—downwards.

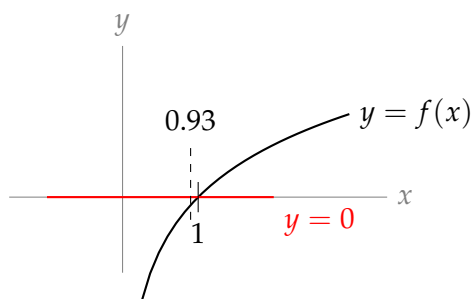
Solutions to Exercises 3.4.1 — Jump to [TABLE OF CONTENTS](#)

S-1: Since $f(0)$ is closer to $g(0)$ than it is to $h(0)$, you would probably want to estimate $f(0) \approx g(0) = 1 + 2 \sin(1)$ if you had the means to efficiently figure out what $\sin(1)$ is, and if you were concerned with accuracy. If you had a calculator, you could use this estimation. Also, later in this chapter we will learn methods of approximating $\sin(1)$ that do not require a calculator, but they do require time.

Without a calculator, or without a lot of time, using $f(0) \approx h(0) = 0.7$ probably makes the most sense. It isn't as accurate as $f(0) \approx g(0)$, but you get an estimate very quickly, without worrying about figuring out what $\sin(1)$ is.

Remark: when you're approximating something in real life, there probably won't be an obvious "correct" way to do it. There's usually a trade-off between accuracy and ease.

S-2: 0.93 is pretty close to 1, and we know $\log(1) = 0$, so we estimate $\log(0.93) \approx \log(1) = 0$.



S-3: We don't know $\arcsin(0.1)$, but 0.1 is reasonably close to 0, and $\arcsin(0) = 0$. So, we estimate $\arcsin(0.1) \approx 0$.

S-4: We don't know $\tan(1)$, but we do know $\tan\left(\frac{\pi}{3}\right) = \sqrt{3}$. Since $\frac{\pi}{3} \approx 1.047$ is pretty close to 1, we estimate $\sqrt{3} \tan(1) \approx \sqrt{3} \tan\left(\frac{\pi}{3}\right) = (\sqrt{3})^2 = 3$.

S-5: Since 10.1 is pretty close to 10, we estimate $10.1^3 \approx 10^3 = 1000$.

Remark: these kinds of approximations are very useful when you are doing computations. It's easy to make a mistake in your work, and having in mind that 10.1^3 should be *about* a thousand is a good way to check that whatever answer you have makes sense.

Solutions to Exercises **3.4.2** — Jump to [TABLE OF CONTENTS](#)

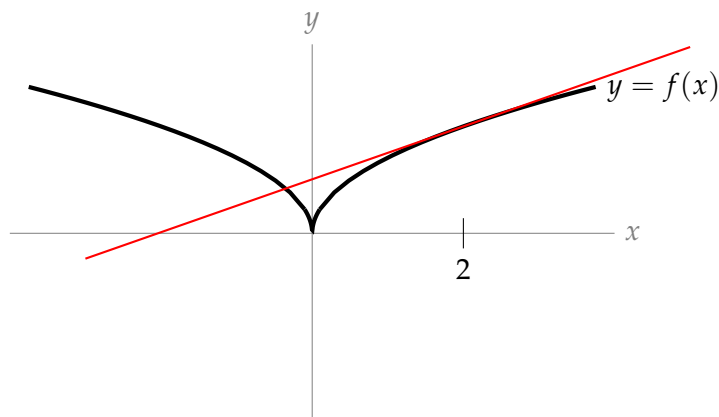
S-1: The linear approximation is $L(x) = 3x - 9$. Since we're approximating at $x = 5$, $f(5) = L(5)$, and $f'(5) = L'(5)$. However, there is no guarantee that $f(x)$ and $L(x)$ have the same value when $x \neq 5$. So:

(a) $f(5) = L(5) = 6$

(b) $f'(5) = L'(5) = 3$

(c) there is not enough information to find $f(0)$.

S-2: The linear approximation is a line, passing through $(2, f(2))$, with slope $f'(2)$. That is, the linear approximation to $f(x)$ about $x = 2$ is the tangent line to $f(x)$ at $x = 2$. It is shown below in red.



S-3: For any constant a , $f(a) = (2a + 5)$, and $f'(a) = 2$, so our approximation gives us

$$f(x) \approx (2a + 5) + 2(x - a) = 2x + 5$$

Since $f(x)$ itself is a linear function, the linear approximation is actually just $f(x)$ itself. As a consequence, the linear approximation is perfectly accurate for all values of x .

S-4: We have no idea what $f(0.93)$ is, but 0.93 is pretty close to 1, and we definitely know $f(1)$. The linear approximation of $f(x)$ about $x = 1$ is given by

$$f(x) \approx f(1) + f'(1)(x - 1)$$

So, we calculate:

$$f(1) = \log(1) = 0$$

$$f'(x) = \frac{1}{x}$$

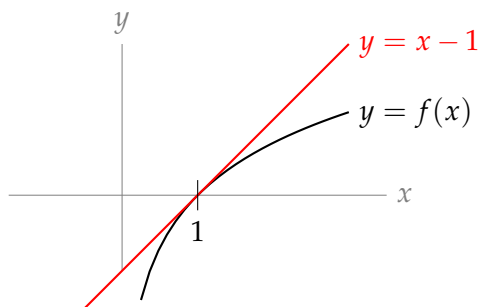
$$f'(1) = \frac{1}{1} = 1$$

Therefore,

$$f(x) \approx 0 + 1(x - 1) = x - 1$$

When $x = 0.93$:

$$f(0.93) \approx 0.93 - 1 = -0.07$$



S-5: We approximate the function $f(x) = \sqrt{x}$ about $x = 4$, since 4 is a perfect square and it is close to 5.

$$f(4) = \sqrt{4} = 2$$

$$f'(x) = \frac{1}{2\sqrt{x}} \quad \Rightarrow \quad f'(4) = \frac{1}{2\sqrt{4}} = \frac{1}{4}$$

$$f(x) \approx f(4) + f'(4)(x - 4) = 2 + \frac{1}{4}(x - 4)$$

$$f(5) \approx 2 + \frac{1}{4}(5 - 4) = \frac{9}{4}$$

We estimate $\sqrt{5} \approx \frac{9}{4}$.

Remark: $\left(\frac{9}{4}\right)^2 = \frac{81}{16}$, which is pretty close to $\frac{80}{16} = 5$. Our approximation seems pretty good.

S-6: We approximate the function $f(x) = \sqrt[5]{x}$. We need to centre the approximation about some value $x = a$ such that we know $f(a)$ and $f'(a)$, and a is not too far from 30.

$$f(x) = \sqrt[5]{x} = x^{\frac{1}{5}}$$
$$f'(x) = \frac{1}{5}x^{-\frac{4}{5}} = \frac{1}{5\sqrt[5]{x^4}}$$

a needs to be a number whose fifth root we know. Since $\sqrt[5]{32} = 2$, and 32 is reasonably close to 30, $a = 32$ is a great choice.

$$f(32) = \sqrt[5]{32} = 2$$
$$f'(32) = \frac{1}{5 \cdot 2^4} = \frac{1}{80}$$

The linear approximation of $f(x)$ about $x = 32$ is

$$f(x) \approx 2 + \frac{1}{80}(x - 32)$$

When $x = 30$:

$$f(30) \approx 2 + \frac{1}{80}(30 - 32) = 2 - \frac{1}{40} = \frac{79}{40}$$

We estimate $\sqrt[5]{30} \approx \frac{79}{40}$.

Remark: $\frac{79}{40} = 1.975$, while $\sqrt[5]{30} \approx 1.97435$. This is a decent estimation.

S-7: If $f(x) = x^3$, then $f(10.1) = 10.1^3$, which is the value we want to estimate. Let's take the linear approximation of $f(x)$ about $x = 10$:

$$f(10) = 10^3 = 1000$$
$$f'(x) = 3x^2$$
$$f'(10) = 3(10^2) = 300$$
$$f(a) \approx f(10) + f'(10)(x - 10)$$
$$= 1000 + 300(x - 10)$$
$$f(10.1) \approx 1000 + 300(10.1 - 10) = 1030$$

We estimate $10.1^3 \approx 1030$. If we calculate 10.1^3 exactly (which is certainly possible to do by hand), we get 1030.301.

Remark: in the previous subsection, we used a constant approximation to estimate $10.1^3 \approx 1000$. That approximation was easy to do in your head, in a matter of seconds. The linear approximation is more accurate, but not much faster than simply calculating 10.1^3 .

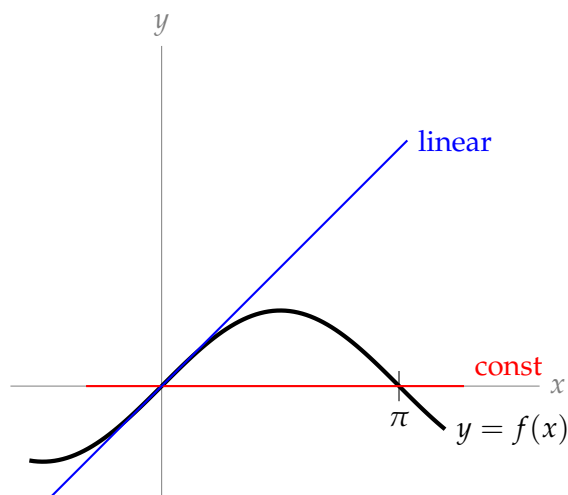
S-8: There are many possible answers. One is:

$$f(x) = \sin x \quad a = 0 \quad b = \pi$$

We know that $f(\pi) = 0$ and $f(0) = 0$. Using a constant approximation of $f(x)$ about $x = 0$, our estimation is $f(\pi) \approx f(0) = 0$, which is exactly the correct value. However, if we make a linear approximation of $f(x)$ about $x = 0$, we get

$$f(\pi) \approx f(0) + f'(0)(\pi - 0) = \sin(0) + \cos(0)\pi = \pi$$

which is not exactly the correct value.



Remark: in reality, we wouldn't estimate $\sin(\pi)$, because we know its value exactly. The purpose of this problem is to demonstrate that fancier approximations are not *always* more accurate. At the end of this section, we'll talk about how to bound the error of your estimations, to make sure that you are finding something sufficiently accurate.

S-9: The linear approximation $L(x)$ of $f(x)$ about $x = a$ is chosen so that $L(a) = f(a)$ and $L'(a) = f'(a)$. So,

$$\begin{aligned} L'(a) &= f'(a) = \frac{1}{1+a^2} \\ \frac{1}{4} &= \frac{1}{1+a^2} \\ a &= \pm\sqrt{3} \end{aligned}$$

We've narrowed down a to $\sqrt{3}$ or $-\sqrt{3}$. Recall the linear approximation of $f(x)$ about $x = a$ is $f(a) + f'(a)(x - a)$, so its constant term is $f(a) - af'(a) = \arctan(a) - \frac{a}{1+a^2}$. We compute this for $a = \sqrt{3}$ and $a = -\sqrt{3}$.

$$\begin{aligned} a = \sqrt{3}: \quad \arctan(a) - \frac{a}{1+a^2} &= \arctan(\sqrt{3}) - \frac{\sqrt{3}}{1+(\sqrt{3})^2} = \frac{\pi}{3} - \frac{\sqrt{3}}{4} = \frac{4\pi - \sqrt{27}}{12} \\ a = -\sqrt{3}: \quad \arctan(a) - \frac{a}{1+a^2} &= \arctan(-\sqrt{3}) - \frac{-\sqrt{3}}{1+(-\sqrt{3})^2} = -\frac{\pi}{3} + \frac{\sqrt{3}}{4} = \frac{-4\pi + \sqrt{27}}{12} \end{aligned}$$

So, when $a = \sqrt{3}$,

$$L(x) = \frac{1}{4}x + \frac{4\pi - \sqrt{27}}{12}$$

and this does not hold when $a = -\sqrt{3}$. We conclude $a = \sqrt{3}$.

Solutions to Exercises 3.4.3 — Jump to [TABLE OF CONTENTS](#)

S-1: If $Q(x)$ is the quadratic approximation of f about 3, then $Q(3) = f(3)$, $Q'(3) = f'(3)$, and $Q''(3) = f''(3)$. There is no guarantee that $f(x)$ and $Q(x)$ share the same third derivative, though, so we do not have enough information to know $f'''(3)$.

$$\begin{aligned}f(3) &= -3^2 + 6(3) = 9 \\f'(3) &= \left. \frac{d}{dx} \{-x^2 + 6x\} \right|_{x=3} = -2x + 6|_{x=3} = 0 \\f''(3) &= \left. \frac{d^2}{dx^2} \{-x^2 + 6x\} \right|_{x=3} = \left. \frac{d}{dx} \{-2x + 6\} \right|_{x=3} = -2\end{aligned}$$

S-2: The quadratic approximation of $f(x)$ about $x = a$ is

$$f(x) \approx f(a) + f'(a)(x - a) + \frac{1}{2}f''(a)(x - a)^2$$

We substitute $f(a) = 2a + 5$, $f'(a) = 2$, and $f''(a) = 0$:

$$f(x) \approx (2a + 5) + 2(x - a) = 2x + 5$$

So, our approximation is $f(x) \approx 2x + 5$.

Remark: Our approximation is exact for every value of x . This will always happen with a quadratic approximation of a function that is quadratic, linear, or constant.

S-3: We approximate the function $f(x) = \log x$ about the point $x = 1$. We choose 1 because it is close to 0.93, and we can evaluate $f(x)$ and its first two derivatives at $x = 1$.

$$\begin{aligned}f(1) &= 0 \\f'(x) &= \frac{1}{x} \quad \Rightarrow \quad f'(1) = 1 \\f''(x) &= \frac{-1}{x^2} \quad \Rightarrow \quad f''(1) = -1\end{aligned}$$

So,

$$\begin{aligned}f(x) &\approx f(1) + f'(1)(x - 1) + \frac{1}{2}f''(1)(x - 1)^2 \\&= 0 + (x - 1) - \frac{1}{2}(x - 1)^2\end{aligned}$$

When $x = 0.93$:

$$f(0.93) \approx (0.93 - 1) - \frac{1}{2}(0.93 - 1)^2 = -0.07 - \frac{1}{2}(0.0049) = -0.07245$$

We estimate $\log(0.93) \approx -0.07245$.

Remark: a calculator approximates $\log(0.93) \approx -0.07257$. We're pretty close.

S-4: We approximate the function $f(x) = \cos x$. We can easily evaluate $\cos x$ and $\sin x$ ($\sin x$ will appear in the first derivative) at $x = 0$, and 0 is quite close to $\frac{1}{15}$, so we centre our approximation about $x = 0$.

$$\begin{aligned}f(0) &= 1 \\f'(x) &= -\sin x \\f'(0) &= -\sin(0) = 0 \\f''(x) &= -\cos x \\f''(0) &= -\cos(0) = -1\end{aligned}$$

Using the quadratic approximation $f(x) \approx f(0) + f'(0)(x - 0) + \frac{1}{2}f''(0)(x - 0)^2$:

$$\begin{aligned}f(x) &\approx 1 - \frac{1}{2}x^2 \\f\left(\frac{1}{15}\right) &\approx 1 - \frac{1}{2 \cdot 15^2} = \frac{449}{450}\end{aligned}$$

We approximate $\cos\left(\frac{1}{15}\right) \approx \frac{449}{450}$.

Remark: $\frac{449}{450} = 0.997\overline{7}$, while a calculator gives $\cos\left(\frac{1}{15}\right) \approx 0.9977786$. Our approximation has an error of about 0.000001.

S-5: The quadratic approximation of a function $f(x)$ about $x = a$ is

$$f(x) \approx f(a) + f'(a)(x - a) + \frac{1}{2}f''(a)(x - a)^2$$

We compute derivatives.

$$\begin{aligned}f(0) &= e^0 = 1 \\f'(x) &= 2e^{2x} \\f'(0) &= 2e^0 = 2 \\f''(x) &= 4e^{2x} \\f''(0) &= 4e^0 = 4\end{aligned}$$

Substituting:

$$\begin{aligned}f(x) &\approx 1 + 2(x - 0) + \frac{4}{2}(x - 0)^2 \\f(x) &\approx 1 + 2x + 2x^2\end{aligned}$$

S-6: There are a few functions we could choose to approximate. For example:

- $f(x) = x^{4/3}$. In this case, we would probably choose to approximate $f(x)$ about $x = 8$ (since 8 is a cube, $8^{4/3} = 2^4 = 16$ is something we can evaluate) or $x = 1$.

- $f(x) = 5^x$. We can easily figure out $f(x)$ when x is a whole number, so we would want to centre our approximation around some whole number $x = a$, but then $f'(a) = 5^a \log(5)$ gives us a problem: without a calculator, it's hard to know what $\log(5)$ is.
- Since $5^{4/3} = 5\sqrt[3]{5}$, we can use $f(x) = 5\sqrt[3]{x}$. As in the first bullet, we would centre about $x = 8$, or $x = 1$.

There isn't much difference between the first and third bullets. We'll go with $f(x) = 5\sqrt[3]{x}$, centred about $x = 8$.

$$\begin{aligned} f(x) &= 5x^{1/3} & \Rightarrow & f(8) = 5 \cdot 2 = 10 \\ f'(x) &= \frac{5}{3}x^{-2/3} & \Rightarrow & f'(8) = \frac{5}{3}(2^{-2}) = \frac{5}{12} \\ f''(x) &= \frac{5}{3}\left(-\frac{2}{3}\right)x^{-5/3} = -\frac{10}{9}x^{-5/3} & \Rightarrow & f''(8) = -\frac{10}{9}(2^{-5}) = -\frac{5}{144} \end{aligned}$$

Using the quadratic approximation $f(x) \approx f(a) + f'(a)(x-a) + \frac{1}{2}f''(a)(x-a)^2$:

$$\begin{aligned} f(x) &\approx 10 + \frac{5}{12}(x-8) - \frac{5}{288}(x-8)^2 \\ f(5) &\approx 10 + \frac{5}{12}(-3) - \frac{5}{288}(9) = \frac{275}{32} \end{aligned}$$

We estimate $5^{4/3} \approx \frac{275}{32}$

Remark: $\frac{275}{32} = 8.59375$, and a calculator gives $5^{4/3} \approx 8.5499$. Although 5 and 8 are somewhat far apart, our estimate is only off by about 0.04.

S-7:

(a) For every value of n , the term being added is simply the constant 1. So,

$$\sum_{n=5}^{30} 1 = 1 + 1 + \cdots + 1. \text{ The trick is figuring out how many 1s are added. Our index } n \text{ takes}$$

on all integers from 5 to 30, *including* 5 and 30, which is 26 numbers. So, $\sum_{n=5}^{30} = 26$.

If you're having a hard time seeing why the sum is 26, instead of 25, think of it this way: there are thirty numbers in the collection $\{1, 2, 3, 4, 5, 6, \dots, 29, 30\}$. If we remove the first four, we get $30 - 4 = 26$ numbers in the collection $\{5, 6, \dots, 30\}$.

(b)

$$\begin{aligned} \sum_{n=1}^3 [2(n+3) - n^2] &= \underbrace{2(1+3) - 1^2}_{n=1} + \underbrace{2(2+3) - 2^2}_{n=2} + \underbrace{2(3+3) - 3^2}_{n=3} \\ &= 8 - 1 + 10 - 4 + 12 - 9 = 16 \end{aligned}$$

(c)

$$\begin{aligned}\sum_{n=1}^{10} \left[\frac{1}{n} - \frac{1}{n+1} \right] &= \underbrace{\frac{1}{1} - \frac{1}{1+1}}_{n=1} + \underbrace{\frac{1}{2} - \frac{1}{2+1}}_{n=2} + \underbrace{\frac{1}{3} - \frac{1}{3+1}}_{n=3} + \underbrace{\frac{1}{4} - \frac{1}{4+1}}_{n=4} + \underbrace{\frac{1}{5} - \frac{1}{5+1}}_{n=5} \\ &\quad + \underbrace{\frac{1}{6} - \frac{1}{6+1}}_{n=6} + \underbrace{\frac{1}{7} - \frac{1}{7+1}}_{n=7} + \underbrace{\frac{1}{8} - \frac{1}{8+1}}_{n=8} + \underbrace{\frac{1}{9} - \frac{1}{9+1}}_{n=9} + \underbrace{\frac{1}{10} - \frac{1}{10+1}}_{n=10}\end{aligned}$$

Most of these cancel!

$$\begin{aligned}&= \frac{1}{1} - \underbrace{\frac{1}{2} + \frac{1}{2}}_0 - \underbrace{\frac{1}{3} + \frac{1}{3}}_0 - \underbrace{\frac{1}{4} + \frac{1}{4}}_0 - \underbrace{\frac{1}{5} + \frac{1}{5}}_0 - \underbrace{\frac{1}{6} + \frac{1}{6}}_0 \\ &\quad - \underbrace{\frac{1}{7} + \frac{1}{7}}_0 - \underbrace{\frac{1}{8} + \frac{1}{8}}_0 - \underbrace{\frac{1}{9} + \frac{1}{9}}_0 - \underbrace{\frac{1}{10} + \frac{1}{10}}_0 - \frac{1}{11} \\ &= 1 - \frac{1}{11} = \frac{10}{11}\end{aligned}$$

(d)

$$\begin{aligned}\sum_{n=1}^4 \frac{5 \cdot 2^n}{4^{n+1}} &= 5 \sum_{n=1}^4 \frac{2^n}{4 \cdot 4^n} = \frac{5}{4} \sum_{n=1}^4 \frac{2^n}{4^n} = \frac{5}{4} \sum_{n=1}^4 \frac{1}{2^n} \\ &= \frac{5}{4} \left(\underbrace{\frac{1}{2}}_{n=1} + \underbrace{\frac{1}{4}}_{n=2} + \underbrace{\frac{1}{8}}_{n=3} + \underbrace{\frac{1}{16}}_{n=4} \right) = \frac{75}{64}\end{aligned}$$

S-8: For each of these, there are many solutions. We provide some below.

(a) $1 + 2 + 3 + 4 + 5 = \sum_{n=1}^5 n$

(b) $2 + 4 + 6 + 8 = \sum_{n=1}^4 2n$

(c) $3 + 5 + 7 + 9 + 11 = \sum_{n=1}^5 (2n + 1)$

(d) $9 + 16 + 25 + 36 + 49 = \sum_{n=3}^7 n^2$

(e) $9 + 4 + 16 + 5 + 25 + 6 + 36 + 7 + 49 + 8 = \sum_{n=3}^7 (n^2 + n + 1)$

(f) $8 + 15 + 24 + 35 + 48 = \sum_{n=3}^7 (n^2 - 1)$

(g) $3 - 6 + 9 - 12 + 15 - 18 = \sum_{n=1}^6 (-1)^{n+1} 3n$

Remark: if we had written $(-1)^n$ instead of $(-1)^{n+1}$, with everything else the same, the signs would have been reversed.

S-9: Let's start by taking the first two derivative of $f(x)$.

$$\begin{aligned} f(x) &= 2 \arcsin x & \Rightarrow & f(0) = 2(0) = 0 \\ f'(x) &= \frac{2}{\sqrt{1-x^2}} & \Rightarrow & f'(0) = \frac{2}{1} = 2 \\ f''(x) &= \frac{d}{dx} \left\{ 2(1-x^2)^{-\frac{1}{2}} \right\} \\ &= 2 \left(-\frac{1}{2} \right) (1-x^2)^{-\frac{3}{2}} (-2x) & (\text{chain rule}) \\ &= \frac{2x}{(\sqrt{1-x^2})^3} & \Rightarrow & f''(0) = 0 \end{aligned}$$

Now, we can find the quadratic approximation about $x = 0$.

$$\begin{aligned} f(x) &\approx f(0) + f'(0)x + \frac{1}{2}f''(0)x^2 \\ &= 2x \\ f(1) &\approx 2 \end{aligned}$$

Our quadratic approximation gives $2 \arcsin(1) \approx 2$. However, $2 \arcsin(1)$ is exactly equal to $2 \left(\frac{\pi}{2} \right) = \pi$. We've just made the rather unfortunate approximation $\pi \approx 2$.

S-10: From the text, the quadratic approximation of e^x about $x = 0$ is

$$e^x \approx 1 + x + \frac{1}{2}x^2$$

So,

$$e = e^1 \approx 1 + 1 + \frac{1}{2} = 2.5$$

We estimate $e \approx 2.5$.

Remark: actually, $e \approx 2.718$.

S-11:

- First, we'll show that

$$\text{(a)}, \text{(d)}, \text{(e)}$$

are equivalent:

$$\text{(d)} = 2 \sum_{n=1}^{10} n = 2(1 + 2 + \cdots + 10) = 2(1) + 2(2) + \cdots + 2(10) = \sum_{n=1}^{10} 2n = \text{(a)}$$

So (a) and (d) are equivalent.

$$(e) = 2 \sum_{n=2}^{11} (n-1) = 2(1+2+\cdots+10) = (d)$$

So (e) and (d) are equivalent.

- Second, we'll show that

$$(b), (g)$$

are equivalent.

$$(g) = \frac{1}{4} \sum_{n=1}^{10} \left(\frac{4^{n+1}}{2^n} \right) = \frac{1}{4} \sum_{n=1}^{10} \left(\frac{4 \cdot 4^n}{2^n} \right) = \frac{4}{4} \sum_{n=1}^{10} \left(\frac{4^n}{2^n} \right) = \sum_{n=1}^{10} \left(\frac{4}{2} \right)^n = \sum_{n=1}^{10} 2^n = (b)$$

- Third, we'll show that

$$(c), (f)$$

are equivalent.

$$(f) = \sum_{n=5}^{14} (n-4)^2 = 1^2 + 2^2 + \cdots + 10^2 = \sum_{n=1}^{10} n^2 = (c)$$

- Now, we have three groups, where each group consists of equivalent expressions. To be quite thorough, we should show that no two of these groups contain expressions that are secretly equivalent. They would be hard to evaluate, but we can bound them and show that no two expressions in two separate groups could possibly be equivalent. Notice that

$$\begin{aligned} \sum_{n=1}^{10} 2^n &= 2^1 + 2^2 + \cdots + 2^{10} > 2^{10} = 1024 \\ \sum_{n=1}^{10} n^2 &< \sum_{n=1}^{10} 10^2 = 10(100) = 1000 \\ \sum_{n=1}^{10} n^2 &= 1^2 + 2^2 + \cdots + 8^2 + 9^2 + 10^2 > 8^2 + 9^2 + 10^2 = 245 \\ \sum_{n=1}^{10} 2n &< \sum_{n=1}^{10} 20 = 200 \end{aligned}$$

The expressions in the blue group add to less than 200, but the expressions in the green group add to more than 245, and the expressions in the red group add to more than 1024, so the blue groups expressions can't possibly simplify to the same number as the red and green group expressions.

The expressions in the green group add to less than 1000. Since the expressions in the red group add to more than 1024, the expressions in the green and red groups can't possibly simplify to the same numbers.

We group our expressions in to collections of equivalent expressions as follows:

$\{\textcolor{blue}{a}, \textcolor{blue}{d}, \textcolor{blue}{e}\}, \{\textcolor{blue}{b}, \textcolor{blue}{g}\}, \{\textcolor{green}{c}, \textcolor{green}{f}\}$

Solutions to Exercises [3.4.4](#) — Jump to [TABLE OF CONTENTS](#)

S-1: Since $T_3(x)$ is the third order Taylor polynomial for $f(x)$ about $x = 1$:

- $T_3(1) = f(1)$
- $T'_3(1) = f'(1)$
- $T''_3(1) = f''(1)$
- $T'''_3(1) = f'''(1)$

In particular, $f''(1) = T''_3(1)$.

$$T'_3(x) = 3x^2 - 10x + 9$$

$$T''_3(x) = 6x - 10$$

$$T''_3(1) = 6 - 10 = -4$$

So, $f''(1) = -4$.

S-2: In Question [1](#), we differentiated the Taylor polynomial to find its derivative. We don't really want to differentiate this ten times, though, so let's look for another way. Unlike Question [1](#), our Taylor polynomial is given to us in a form very similar to its definition. The n th order Taylor polynomial for $f(x)$ about $x = 5$ is

$$T_n(x) = \sum_{k=0}^n \frac{f^{(k)}(5)}{k!} (x-5)^k$$

So,

$$\sum_{k=0}^n \frac{f^{(k)}(5)}{k!} (x-5)^k = \sum_{k=0}^n \frac{2k+1}{3k-9} (x-5)^k$$

For any k from 0 to n ,

$$\frac{f^{(k)}(5)}{k!} = \frac{2k+1}{3k-9}$$

In particular, when $k = 10$,

$$\begin{aligned} \frac{f^{(10)}(5)}{10!} &= \frac{20+1}{30-9} = 1 \\ f^{(10)}(5) &= 10! \end{aligned}$$

S-3: The fourth order Maclaurin polynomial for $f(x)$ is

$$T_4(x) = f(0) + f'(0)x + \frac{1}{2}f''(0)x^2 + \frac{1}{3!}f'''(0)x^3 + \frac{1}{4!}f^{(4)}(0)x^4$$

while the third order Maclaurin polynomial for $f(x)$ is

$$T_3(x) = f(0) + f'(0)x + \frac{1}{2}f''(0)x^2 + \frac{1}{3!}f'''(0)x^3$$

So, we simply “chop off” the part of $T_4(x)$ that includes x^4 :

$$T_3(x) = -x^3 + x^2 - x + 1$$

S-4: We saw this kind of problem in Question **3**. The fourth order Taylor polynomial for $f(x)$ about $x = 1$ is

$$T_4(x) = f(1) + f'(1)(x-1) + \frac{1}{2}f''(1)(x-1)^2 + \frac{1}{3!}f'''(1)(x-1)^3 + \frac{1}{4!}f^{(4)}(1)(x-1)^4$$

while the third order Taylor polynomial for $f(x)$ about $x = 1$ is

$$T_3(x) = f(1) + f'(1)(x-1) + \frac{1}{2}f''(1)(x-1)^2 + \frac{1}{3!}f'''(1)(x-1)^3$$

In Question **3** we “chopped off” the term of degree 4 to get $T_3(x)$. However, *our polynomial is not in this form*. It's not clear, right away, what the term $f^{(4)}(x-1)^4$ is in our given $T_4(x)$. So, we will use a different method from Question **3**.

One option is to do some fancy algebra to get $T_4(x)$ into the standard form of a Taylor polynomial. Another option (which we will use) is to recover $f(1)$, $f'(1)$, $f''(1)$, and $f'''(1)$ from $T_4(x)$.

Recall that $T_4(x)$ and $f(x)$ have the same values at $x = 1$ (although maybe not anywhere else!), and they also have the same first, second, third, and fourth derivatives at $x = 1$ (but again, maybe not anywhere else, and maybe their fifth derivatives don't agree). This tells us the following:

$$\begin{array}{lll} T_4(x) = x^4 + x^3 - 9 & \Rightarrow & f(1) = T_4(1) = -7 \\ T_4'(x) = 4x^3 + 3x^2 & \Rightarrow & f'(1) = T_4'(1) = 7 \\ T_4''(x) = 12x^2 + 6x & \Rightarrow & f''(1) = T_4''(1) = 18 \\ T_4'''(x) = 24x + 6 & \Rightarrow & f'''(1) = T_4'''(1) = 30 \end{array}$$

Now, we can write the third order Taylor polynomial for $f(x)$ about $x = 1$:

$$\begin{aligned} T_3(x) &= -7 + 7(x-1) + \frac{1}{2}(18)(x-1)^2 + \frac{1}{3!}(30)(x-1)^3 \\ &= -7 + 7(x-1) + 9(x-1)^2 + 5(x-1)^3 \end{aligned}$$

Remark: expanding the expression above, we get the equivalent polynomial $T_3(x) = 5x^3 - 6x^2 + 4x - 10$. From this, it is clear that we can't just “chop off” the term with x^4 to change $T_4(x)$ into $T_3(x)$ when the Taylor polynomial is not centred about $x = 0$.

S-5: The n th order Taylor polynomial for $f(x)$ about $x = 5$ is

$$T_n(x) = \sum_{k=0}^n \frac{1}{k!} f^{(k)}(5)(x-5)^k$$

We expand this somewhat:

$$T_n(x) = f(5) + f'(x-5) + \cdots + \boxed{\frac{1}{10!} f^{(10)}(5)(x-5)^{10}} + \cdots + \frac{1}{n!} f^{(n)}(5)(x-5)^n$$

So, the coefficient of $(x-5)^{10}$ is $\frac{1}{10!} f^{(10)}(5)$. Expanding the given form of the Taylor polynomial:

$$\begin{aligned} T_n(x) &= \sum_{k=0}^{n/2} \frac{2k+1}{3k-9} (x-5)^{2k} \\ &= \underbrace{\frac{1}{-9}}_{k=0} + \underbrace{\frac{3}{-6}(x-5)^2}_{k=1} + \cdots + \boxed{\frac{11}{6}(x-5)^{10}}_{k=5} + \cdots + \underbrace{\frac{n+1}{(3/2)n-9}(x-5)^n}_{k=n/2} \end{aligned}$$

Equating the coefficients of $(x-5)^{10}$ in the two expressions:

$$\begin{aligned} \frac{1}{10!} f^{(10)}(5) &= \frac{11}{6} \\ f^{(10)}(5) &= \frac{11 \cdot 10!}{6} \end{aligned}$$

S-6: Since $T_3(x)$ is the third order Taylor polynomial for $f(x)$ about $x = a$, we know the following things to be true:

- $f(a) = T_3(a)$
- $f'(a) = T'_3(a)$
- $f''(a) = T''_3(a)$
- $f'''(a) = T'''_3(a)$

But, some of these don't look super useful. For instance, if we try to use the first bullet, we get this equation:

$$a^3 \left[2 \log a - \frac{11}{3} \right] = -\frac{2}{3} \sqrt{e^3} + 3ea - 6\sqrt{e}a^2 + a^3$$

Solving this would be terrible. Instead, let's think about how the equations look when we move further down the list. Since $T_3(x)$ is a cubic equation, $T'''_3(x)$ is a constant (and so $T'''_3(a)$ does not depend on a). That sounds like it's probably the simplest option. Let's start differentiating. We'll

need to know both $f'''(a)$ and $T_3'''(a)$.

$$\begin{aligned} f(x) &= x^3 \left[2 \log x - \frac{11}{3} \right] \\ f'(x) &= x^3 \left[\frac{2}{x} \right] + 3x^2 \left[2 \log x - \frac{11}{3} \right] = 6x^2 \log x - 9x^2 \\ f''(x) &= 6x^2 \cdot \frac{1}{x} + 12x \log x - 18x = 12x \log x - 12x \\ f'''(x) &= 12x \cdot \frac{1}{x} + 12 \log x - 12 = 12 \log x \\ f'''(a) &= 12 \log a \end{aligned}$$

Now, let's move to the Taylor polynomial. Remember that e is a constant.

$$\begin{aligned} T_3(x) &= -\frac{2}{3}\sqrt{e^3} + 3ex - 6\sqrt{e}x^2 + x^3 \\ T_3'(x) &= 3e - 12\sqrt{e}x + 3x^2 \\ T_3''(x) &= -12\sqrt{e} + 6x \\ T_3'''(x) &= 6 \\ T_3'''(a) &= 6 \end{aligned}$$

The final bullet point gives us the equation:

$$\begin{aligned} f'''(a) &= T_3'''(a) \\ 12 \log a &= 6 \\ \log a &= \frac{1}{2} \\ a &= e^{\frac{1}{2}} \end{aligned}$$

So, $a = \sqrt{e}$.

Solutions to Exercises [3.4.5](#) — Jump to [TABLE OF CONTENTS](#)

S-1: If we were to find the 16th order Maclaurin polynomial for a generic function, we might expect to have to differentiate 16 times (ugh). But, we know that the derivatives of sines and cosines repeat themselves. So, it's enough to figure out the pattern:

$$\begin{array}{ll} f(x) = \sin x + \cos x & f(0) = 1 \\ f'(x) = \cos x - \sin x & f'(0) = 1 \\ f''(x) = -\sin x - \cos x & f''(0) = -1 \\ f'''(x) = -\cos x + \sin x & f'''(0) = -1 \\ f^{(4)}(x) = \sin x + \cos x & f^{(4)}(0) = 1 \end{array}$$

Since $f^{(4)}(x) = f(x)$, our derivatives repeat from here. They follow the pattern:
 $+1, +1, -1, -1$.

$$\begin{aligned} T_{16}(x) &= 1 + x - \frac{1}{2}x^2 - \frac{1}{3!}x^3 + \frac{1}{4!}x^4 + \frac{1}{5!}x^5 - \frac{1}{6!}x^6 - \frac{1}{7!}x^7 + \frac{1}{8!}x^8 + \frac{1}{9!}x^9 - \frac{1}{10!}x^{10} - \frac{1}{11!}x^{11} \\ &\quad + \frac{1}{12!}x^{12} + \frac{1}{13!}x^{13} - \frac{1}{14!}x^{14} - \frac{1}{15!}x^{15} + \frac{1}{16!}x^{16} \end{aligned}$$

S-2: A Taylor polynomial gives a polynomial approximation for a function $s(t)$. Since $s(t)$ is itself a polynomial, any n th order Taylor polynomial, with n greater than or equal to the degree of $s(t)$, will simply give $s(t)$. So, in our case, $T_{100}(t) = s(t) = 4.9t^2 - t + 10$.

If that isn't satisfying, you can go through the normal method of calculating $T_{100}(t)$.

$$\begin{array}{ll} s(t) = 4.9t^2 - t + 10 & s(5) = 4.9(25) - 5 + 10 = 127.5 \\ s'(t) = 9.8t - 1 & s'(5) = 9.8(5) - 1 = 48 \\ s''(t) = 9.8 & s''(5) = 9.8 \end{array}$$

The rest of the derivatives of $s(t)$ are identically zero, so they are (in particular) zero when $t = 5$. Therefore,

$$\begin{aligned} T_{100}(t) &= 127.5 + 48(t - 5) + \frac{1}{2}9.8(t - 5)^2 \\ &= 127.5 + 48(t - 5) + 4.9(t - 5)^2 \end{aligned}$$

We can now check that $T_{100}(t)$ really is the same as $s(t)$.

$$\begin{aligned} T_{100}(t) &= 127.5 + 48(t - 5) + 4.9(t - 5)^2 \\ &= 127.5 + 48(t - 5) + 4.9(t^2 - 10t + 25) \\ &= [127.5 + 48(-5) + 4.9(25)] + [48 - 4.9(10)]t + 4.9t^2 \\ &= 10 - t + 4.9t^2 = s(t) \end{aligned}$$

as expected.

S-3: Let's start by differentiating $f(x)$ and looking for a pattern. Remember that $\log 2 = \log_e 2$ is a constant number.

$$\begin{aligned} f(x) &= 2^x \\ f'(x) &= 2^x \log 2 \\ f''(x) &= 2^x (\log 2)^2 \\ f^{(3)}(x) &= 2^x (\log 2)^3 \\ f^{(4)}(x) &= 2^x (\log 2)^4 \\ f^{(5)}(x) &= 2^x (\log 2)^5 \end{aligned}$$

So, in general,

$$f^{(k)}(x) = 2^x (\log 2)^k$$

We notice that this formula works even when $k = 0$ and $k = 1$. When $x = 1$,

$$f^{(k)}(1) = 2 (\log 2)^k$$

The n th order Taylor polynomial of $f(x)$ about $x = 1$ is

$$\begin{aligned} T_n(x) &= \sum_{k=0}^n \frac{f^{(k)}(1)}{k!} (x - 1)^k \\ &= \sum_{k=0}^n \frac{2(\log 2)^k}{k!} (x - 1)^k \end{aligned}$$

S-4: We need to know the first six derivatives of $f(x)$ at $x = 1$. Let's get started.

$$\begin{aligned}
 f(x) &= x^2 \log x + 2x^2 + 5 & f(1) &= 7 \\
 f'(x) &= (x^2) \frac{1}{x} + (2x) \log x + 4x & & \\
 &= 2x \log x + 5x & f'(1) &= 5 \\
 f''(x) &= (2x) \frac{1}{x} + (2) \log x + 5 & & \\
 &= 2 \log x + 7 & f''(1) &= 7 \\
 f'''(x) &= 2x^{-1} & f'''(1) &= 2 \\
 f^{(4)} &= -2x^{-2} & f^{(4)}(1) &= -2 \\
 f^{(5)} &= 4x^{-3} & f^{(5)}(1) &= 4 \\
 f^{(6)} &= -12x^{-4} & f^{(6)}(1) &= -12
 \end{aligned}$$

Now, we can plug in.

$$\begin{aligned}
 T_6(x) &= f(1) + f'(1)(x-1) + \frac{1}{2}f''(1)(x-1)^2 + \frac{1}{3!}f'''(1)(x-1)^3 \\
 &\quad + \frac{1}{4!}f^{(4)}(1)(x-1)^4 + \frac{1}{5!}f^{(5)}(1)(x-1)^5 + \frac{1}{6!}f^{(6)}(1)(x-1)^6 \\
 &= 7 + 5(x-1) + \frac{1}{2}(7)(x-1)^2 + \frac{1}{3!}(2)(x-1)^3 \\
 &\quad + \frac{1}{4!}(-2)(x-1)^4 + \frac{1}{5!}(4)(x-1)^5 + \frac{1}{6!}(-12)(x-1)^6 \\
 &= 7 + 5(x-1) + \frac{7}{2}(x-1)^2 + \frac{1}{3}(x-1)^3 - \frac{1}{12}(x-1)^4 + \frac{1}{30}(x-1)^5 - \frac{1}{60}(x-1)^6
 \end{aligned}$$

S-5: We'll start by differentiating and looking for a pattern.

$$f(x) = \frac{1}{1-x} = (1-x)^{-1}$$

Using the chain rule,

$$\begin{aligned}
 f'(x) &= (-1)(1-x)^{-2}(-1) = (1-x)^{-2} \\
 f''(x) &= (-2)(1-x)^{-3}(-1) = 2(1-x)^{-3} \\
 f^{(3)}(x) &= (-3)(2)(1-x)^{-4}(-1) = 2(3)(1-x)^{-4} \\
 f^{(4)}(x) &= (-4)(2)(3)(1-x)^{-5}(-1) = 2(3)(4)(1-x)^{-5} \\
 f^{(5)}(x) &= (-5)(2)(3)(4)(1-x)^{-6}(-1) = 2(3)(4)(5)(1-x)^{-6}
 \end{aligned}$$

Recognizing the pattern,

$$\begin{aligned}
 f^{(k)}(x) &= k!(1-x)^{-(k+1)} \\
 f^{(k)}(0) &= k!(1)^{-(k+1)} = k!
 \end{aligned}$$

The n th order Maclaurin polynomial for $f(x)$ is

$$\begin{aligned} T_n(x) &= \sum_{k=0}^n \frac{f^{(k)}(0)}{k!} x^k \\ &= \sum_{k=0}^n \frac{k!}{k!} x^k \\ &= \sum_{k=0}^n x^k \end{aligned}$$

S-6: We'll need to know the first three derivatives of x^x at $x = 1$. This is a good review of logarithmic differentiation, covered in Section 2.10.

$$\begin{aligned} f(x) &= x^x & f(1) &= 1 \\ \log(f(x)) &= \log(x^x) = x \log x \\ \frac{d}{dx} \{\log(f(x))\} &= \frac{d}{dx} \{x \log x\} \\ \frac{f'(x)}{f(x)} &= x \cdot \frac{1}{x} + \log x = 1 + \log x \\ f'(x) &= x^x [1 + \log x] & f'(1) &= 1 \\ f''(x) &= \frac{d}{dx} \{x^x\} [1 + \log x] + x^x \frac{d}{dx} \{1 + \log x\} \\ &= (x^x [1 + \log x]) [1 + \log x] + x^x \cdot \frac{1}{x} \\ &= x^x \left((1 + \log x)^2 + \frac{1}{x} \right) & f''(1) &= 2 \\ f'''(x) &= \frac{d}{dx} \{x^x\} \left((1 + \log x)^2 + \frac{1}{x} \right) + x^x \frac{d}{dx} \left\{ (1 + \log x)^2 + \frac{1}{x} \right\} \\ &= x^x [1 + \log x] \left((1 + \log x)^2 + \frac{1}{x} \right) + x^x \left[\frac{2}{x} (1 + \log x) - \frac{1}{x^2} \right] \\ &= x^x \left((1 + \log x)^3 + \frac{3}{x} (1 + \log x) - \frac{1}{x^2} \right) & f'''(1) &= 3 \end{aligned}$$

Now, we can plug in:

$$\begin{aligned} T_3(x) &= f(1) + f'(1)(x-1) + \frac{1}{2}f''(1)(x-1)^2 + \frac{1}{3!}f'''(1)(x-1)^3 \\ &= 1 + 1(x-1) + \frac{1}{2}(2)(x-1)^2 + \frac{1}{6}(3)(x-1)^3 \\ &= 1 + (x-1) + (x-1)^2 + \frac{1}{2}(x-1)^3 \end{aligned}$$

S-7: We note that $6 \arctan\left(\frac{1}{\sqrt{3}}\right) = 6\left(\frac{\pi}{6}\right) = \pi$. We will find the 5th order Maclaurin polynomial $T_5(x)$ for $f(x) = 6 \arctan x$. Then $\pi = f\left(\frac{1}{\sqrt{3}}\right) \approx T_5\left(\frac{1}{\sqrt{3}}\right)$. Let's begin by finding the first five

derivatives of $f(x) = 6 \arctan x$.

$$\begin{aligned}
f(x) &= 6 \arctan x & f(0) &= 0 \\
f'(x) &= 6 \left(\frac{1}{1+x^2} \right) & f'(0) &= 6 \\
f''(x) &= 6 \left(\frac{0-2x}{(1+x^2)^2} \right) = -12 \left(\frac{x}{(1+x^2)^2} \right) & f''(0) &= 0 \\
f'''(x) &= -12 \left(\frac{(1+x^2)^2 - x \cdot 2(1+x^2)(2x)}{(1+x^2)^4} \right) \\
&= -12 \left(\frac{(1+x^2) - 4x^2}{(1+x^2)^3} \right) & f'''(0) &= -12 \\
&= -12 \left(\frac{1-3x^2}{(1+x^2)^3} \right) \\
f^{(4)}(x) &= -12 \left(\frac{(1+x^2)^3(-6x) - (1-3x^2) \cdot 3(1+x^2)^2(2x)}{(1+x^2)^6} \right) \\
&= -12 \left(\frac{-6x(1+x^2) - 6x(1-3x^2)}{(1+x^2)^4} \right) \\
&= 144 \left(\frac{x-x^3}{(1+x^2)^4} \right) & f^{(4)}(0) &= 0 \\
f^{(5)}(x) &= 144 \left(\frac{(1+x^2)^4(1-3x^2) - (x-x^3) \cdot 4(1+x^2)^3(2x)}{(1+x^2)^8} \right) \\
&= 144 \left(\frac{(1+x^2)(1-3x^2) - 8x(x-x^3)}{(1+x^2)^5} \right) \\
&= 144 \frac{5x^4 - 10x^2 + 1}{(1+x^2)^5} & f^{(5)}(0) &= 144
\end{aligned}$$

We now use these values to compute the 5th order Maclaurin polynomial for $f(x)$.

$$\begin{aligned}
T_5(x) &= f(0) + f'(0)x + \frac{1}{2}f''(0)x^2 + \frac{1}{3!}f'''(0)x^3 + \frac{1}{4!}f^{(4)}(0)x^4 + \frac{1}{5!}f^{(5)}(0)x^5 \\
&= 6x - \frac{12}{6}x^3 + \frac{144}{120}x^5 \\
&= 6x - 2x^3 + \frac{6}{5}x^5
\end{aligned}$$

Now, if we want to approximate $f\left(\frac{1}{\sqrt{3}}\right) = 6 \arctan\left(\frac{1}{\sqrt{3}}\right) = \pi$:

$$\begin{aligned}
\pi &= f\left(\frac{1}{\sqrt{3}}\right) \approx T_5\left(\frac{1}{\sqrt{3}}\right) = \frac{6}{\sqrt{3}} - \frac{2}{\sqrt{3}^3} + \frac{6}{5\sqrt{3}^5} \\
&= 2\sqrt{3} \left(1 - \frac{1}{3 \cdot 3} + \frac{1}{5 \cdot 9} \right) \approx 3.156
\end{aligned}$$

Remark: There are numerous methods for computing π to any desired degree of accuracy. Many of them use the Maclaurin expansion of $\arctan x$. In 1706 John Machin computed π to 100 decimal digits by using the Maclaurin expansion together with $\pi = 16 \arctan \frac{1}{5} - 4 \arctan \frac{1}{239}$.

S-8: Let's start by differentiating, and looking for a pattern.

$f(x) = x(\log x - 1)$	$f(1) = -1$
$f'(x) = x\left(\frac{1}{x}\right) + \log x - 1 = \log x$	$f'(1) = 0$
$f''(x) = \frac{1}{x} = x^{-1}$	$f''(1) = 1$
$f^{(3)}(x) = (-1)x^{-2}$	$f^{(3)}(1) = -1$
$f^{(4)}(x) = (-2)(-1)x^{-3} = 2!x^{-3}$	$f^{(4)}(1) = 2!$
$f^{(5)}(x) = (-3)(-2)(-1)x^{-4} = -3!x^{-4}$	$f^{(5)}(1) = -3!$
$f^{(6)}(x) = (-4)(-3)(-2)(-1)x^{-5} = 4!x^{-5}$	$f^{(6)}(1) = 4!$
$f^{(7)}(x) = (-5)(-4)(-3)(-2)(-1)x^{-6} = -5!x^{-6}$	$f^{(7)}(1) = -5!$
$f^{(8)}(x) = (-6)(-5)(-4)(-3)(-2)(-1)x^{-7} = 6!x^{-7}$	$f^{(8)}(1) = 6!$

When $k \geq 2$, making use of the fact that $0! = 1$ and $(-1)^{k-2} = (-1)^k$:

$$f^{(k)}(x) = (-1)^{k-2}(k-2)!x^{-(k-1)} \qquad f^{(k)}(1) = (-1)^k(k-2)!$$

Now we use the standard form of a Taylor polynomial. Since the first two terms don't fit the pattern, we add those outside of the sigma.

$$\begin{aligned}
 T_{100}(x) &= \sum_{k=0}^{100} \frac{f^{(k)}(1)}{k!} (x-1)^k \\
 &= f(1) + f'(1)(x-1) + \sum_{k=2}^{100} \frac{f^{(k)}(1)}{k!} (x-1)^k \\
 &= -1 + 0(x-1) + \sum_{k=2}^{100} \frac{(-1)^k(k-2)!}{k!} (x-1)^k \\
 &= -1 + \sum_{k=2}^{100} \frac{(-1)^k}{k(k-1)} (x-1)^k
 \end{aligned}$$

S-9: Recall that

$$T_{2n}(x) = \sum_{k=0}^{2n} \frac{f^{(k)}\left(\frac{\pi}{4}\right)}{k!} \left(x - \frac{\pi}{4}\right)^k$$

Let's start by taking some derivatives. Of course, since we're differentiating sine, the derivatives will repeat every four iterations.

$f(x) = \sin x$	$f\left(\frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}$
$f'(x) = \cos x$	$f'\left(\frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}$
$f''(x) = -\sin x$	$f''\left(\frac{\pi}{4}\right) = -\frac{1}{\sqrt{2}}$
$f'''(x) = -\cos x$	$f'''\left(\frac{\pi}{4}\right) = -\frac{1}{\sqrt{2}}$

So, the pattern of derivatives is $\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}, \dots$, etc. This is a little tricky to write in sigma notation. We can deal with the “doubles” by separating the even and odd powers. The first few terms of T_{2n} that contain even powers of $(x - \frac{\pi}{4})$ are

$$\underbrace{\frac{1}{\sqrt{2}}}_{k=0} - \underbrace{\frac{1}{2!\sqrt{2}} \left(x - \frac{\pi}{4}\right)^2}_{k=2} + \underbrace{\frac{1}{4!\sqrt{2}} \left(x - \frac{\pi}{4}\right)^4}_{k=4}$$

Observe that the signs alternate between successive terms. So if we rename k to 2ℓ these terms are

$$\underbrace{\frac{1}{\sqrt{2}}}_{\ell=0} - \underbrace{\frac{1}{2!\sqrt{2}} \left(x - \frac{\pi}{4}\right)^2}_{\ell=1} + \underbrace{\frac{1}{4!\sqrt{2}} \left(x - \frac{\pi}{4}\right)^4}_{\ell=2}$$

and the ℓ^{th} term here is $\frac{(-1)^\ell}{(2\ell)!\sqrt{2}} \left(x - \frac{\pi}{4}\right)^{2\ell}$. To verify that this really is the ℓ^{th} term, evaluate this for $\ell = 0, 1, 2$ explicitly. When $k = 2n$, $\ell = n$ so that

$$\sum_{\substack{0 \leq k \leq 2n \\ k \text{ even}}} \frac{f^{(k)}\left(\frac{\pi}{4}\right)}{k!} \left(x - \frac{\pi}{4}\right)^k = \sum_{\ell=0}^n \frac{(-1)^\ell}{(2\ell)!\sqrt{2}} \left(x - \frac{\pi}{4}\right)^{2\ell}$$

Now for the odd powers. The first few terms of T_{2n} that contain odd powers of $(x - \frac{\pi}{4})$ are

$$\underbrace{\frac{1}{\sqrt{2}} \left(x - \frac{\pi}{4}\right)}_{k=1} - \underbrace{\frac{1}{3!\sqrt{2}} \left(x - \frac{\pi}{4}\right)^3}_{k=3} + \underbrace{\frac{1}{5!\sqrt{2}} \left(x - \frac{\pi}{4}\right)^5}_{k=5}$$

Observe that the signs again alternate between successive terms. So if we rename k to $2\ell + 1$ these terms are

$$\underbrace{\frac{1}{\sqrt{2}} \left(x - \frac{\pi}{4}\right)}_{\ell=0} - \underbrace{\frac{1}{3!\sqrt{2}} \left(x - \frac{\pi}{4}\right)^3}_{\ell=1} + \underbrace{\frac{1}{5!\sqrt{2}} \left(x - \frac{\pi}{4}\right)^5}_{\ell=2}$$

and the ℓ^{th} term here is $\frac{(-1)^\ell}{(2\ell+1)!\sqrt{2}} \left(x - \frac{\pi}{4}\right)^{2\ell+1}$. To verify that this really is the ℓ^{th} term, evaluate this for $\ell = 0, 1, 2$ explicitly. The largest odd integer that is smaller than $2n$ is $2n - 1$ and when $k = 2n - 1 = 2\ell + 1$, $\ell = n - 1$ so that

$$\sum_{\substack{0 \leq k \leq 2n \\ k \text{ odd}}} \frac{f^{(k)}\left(\frac{\pi}{4}\right)}{k!} \left(x - \frac{\pi}{4}\right)^k = \sum_{\ell=0}^{n-1} \frac{(-1)^\ell}{(2\ell+1)!\sqrt{2}} \left(x - \frac{\pi}{4}\right)^{2\ell+1}$$

Putting the even and odd powers together

$$T_{2n}(x) = \sum_{\ell=0}^n \frac{(-1)^\ell}{(2\ell)!\sqrt{2}} \left(x - \frac{\pi}{4}\right)^{2\ell} + \sum_{\ell=0}^{n-1} \frac{(-1)^\ell}{(2\ell+1)!\sqrt{2}} \left(x - \frac{\pi}{4}\right)^{2\ell+1}$$

S-10: From Example 3.4.12 in the text, we see that the n th Maclaurin polynomial for $f(x) = e^x$ is

$$T_n(x) = \sum_{k=0}^n \frac{1}{k!} x^k = 1 + x + \frac{x^2}{2} + \frac{x^3}{3!} + \frac{x^4}{4!} + \cdots + \frac{x^n}{n!}$$

If $n = 157$ and $x = 1$,

$$T_{157}(1) = \sum_{k=0}^{157} \frac{1}{k!} = 1 + 1 + \frac{1}{2} + \frac{1}{3!} + \frac{1}{4!} + \cdots + \frac{1}{157!}$$

Although we wouldn't expect $T_{157}(1)$ to be exactly equal to e^1 , it's probably pretty close. So, we estimate

$$1 + \frac{1}{2} + \frac{1}{3!} + \frac{1}{4!} + \cdots + \frac{1}{157!} \approx e - 1$$

S-11: While you're working with sums, it's easy to mistake a constant for a function. The sum given in this question is some *number*: k is an index— if you wrote out all 100 terms of this sum, there would be no letter k . So, the sum given is indeed a number, but we don't want to have to add 100 terms to get a good idea of what number it is.

From Example 3.4.14 in the text, we see that the $(2n)$ th order Maclaurin polynomial for $f(x) = \cos x$ is

$$T_{2n}(x) = \sum_{k=0}^n \frac{(-1)^k}{(2k)!} \cdot x^{2k}$$

If $n = 100$ and $x = \frac{5\pi}{4}$, this equation becomes

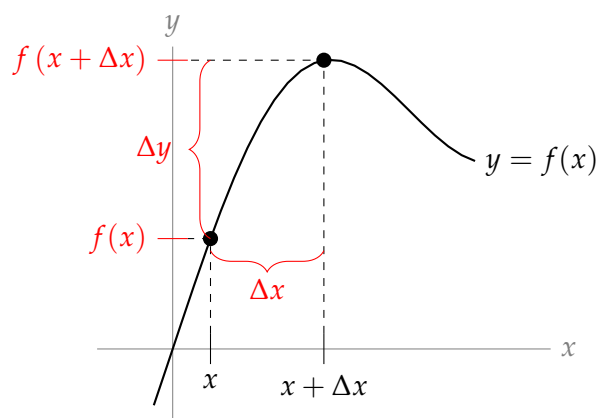
$$T_{200}\left(\frac{5\pi}{4}\right) = \sum_{k=0}^{100} \frac{(-1)^k}{(2k)!} \cdot \left(\frac{5\pi}{4}\right)^{2k}$$

So, the sum corresponds to the 200th Maclaurin polynomial for $f(x) = \cos x$ evaluated at $x = \frac{5\pi}{4}$. We should be careful to understand that $T_{200}(x)$ is *not* equal to $f(x)$, in general. However, when x is reasonably close to 0, these two functions are approximations of one another. So, we estimate

$$\sum_{k=0}^{100} \frac{(-1)^k}{(2k)!} \left(\frac{5\pi}{4}\right)^{2k} = T_{200}\left(\frac{5\pi}{4}\right) \approx \cos\left(\frac{5\pi}{4}\right) = -\frac{1}{\sqrt{2}}$$

Solutions to Exercises 3.4.6 — Jump to [TABLE OF CONTENTS](#)

S-1:



S-2: Let $f(x)$ be the number of problems finished after x minutes of work. The question tells us that $5\Delta y \approx \Delta x$. So, if $\Delta x = 15$, $\Delta y \approx 3$. That is, in 15 minutes more, you will finish about 3 more problems.

Remark: the math behind this problem is intended to be easy! Looking at symbols like Δy and $f(x + \Delta x)$ can be confusing, but the basic idea is pretty simple.

S-3: First, let's find the first and second derivatives of f when $x = 5$.

$$\begin{aligned} f'(x) &= \frac{1}{1+x^2} & f'(5) &= \frac{1}{26} \\ f''(x) &= \frac{-2x}{(1+x^2)^2} & f''(5) &= \frac{-10}{26^2} \end{aligned}$$

The linear approximation for Δy when $\Delta x = \frac{1}{10}$ is

$$\Delta y \approx f'(5)\Delta x = \frac{1}{26} \cdot \frac{1}{10} = \frac{1}{260} \approx 0.003846$$

The quadratic approximation for Δy when $\Delta x = \frac{1}{10}$ is

$$\Delta y \approx f'(5)\Delta x + \frac{1}{2}f''(5)(\Delta x)^2 = \frac{1}{26} \cdot \frac{1}{10} + \frac{1}{2} \cdot \frac{-10}{26^2} \cdot \frac{1}{10^2} = \frac{51}{13520} \approx 0.003772$$

Remark: $\Delta y = \arctan(5.1) - \arctan(5) \approx 0.003774$.

S-4: (a)

When x is near 4, the linear approximation of Δy is

$$\Delta y \approx s'(4)\Delta x$$

From the problem, $\Delta x = 5 - 4 = 1$. Differentiating,

$$\begin{aligned} s'(t) &= \sqrt{19.6} \cdot \frac{1}{2\sqrt{x}} = \sqrt{\frac{4.9}{x}}, \text{ so} \\ s'(4) &= \sqrt{\frac{4.9}{4}} \end{aligned}$$

The linear approximation gives us

$$\Delta y \approx \sqrt{\frac{4.9}{4}}(1) \approx 1.1$$

So moving from 4 metres to 5 metres increases the speed with which you hit the water by about 1.1 metres per second.

(b) Again, we'll use the linear approximation

$$\begin{aligned}\Delta y &\approx s'(a)\Delta x \\ &= \sqrt{\frac{4.9}{a}} \cdot \Delta x\end{aligned}$$

The difference in height between the first two jumps and between the last two jumps is the same, Δx . The initial height of the first jump is smaller than the initial height of the second jump. So, the value corresponding to a is smaller for the first jump than for the second. Therefore, Δy is larger between the first two jumps than it is between the last two jumps. So, the increase in speed from the first jump to the second is larger than the increase in speed from the second jump to the third.

To put that more symbolically, the change in terminal speed between the first two jumps is

$$\Delta y \approx \sqrt{\frac{4.9}{x}} \cdot \Delta x$$

while the change in terminal speed between the next two jumps is

$$\Delta y \approx \sqrt{\frac{4.9}{x + \Delta x}} \cdot \Delta x$$

Since $\sqrt{\frac{4.9}{x}} \cdot \Delta x > \sqrt{\frac{4.9}{x + \Delta x}} \cdot \Delta x$ (when x and Δx are positive), the change in terminal speed is greater between the first two jumps than between the next two jumps.

Solutions to Exercises 3.4.7 — Jump to [TABLE OF CONTENTS](#)

S-1: False. The linear approximation is an *approximation*. It tells us

$$\Delta y \approx f'(x_0)\Delta x = f'(2)(1) = 25$$

However, from our definition of Δy ,

$$\Delta y = f(x_0 + \Delta x) - f(x_0) = f(2 + 1) - f(2) = 58 - 26 = 32$$

Remark: this is to emphasize that the calculations in this subsection are *estimations* of error bounds, rather than actual error bounds. All we can say is that we *estimate* the error will be no more than some number—we don't guarantee it.

In the next subsection, we will introduce an error bound that is guaranteed to be accurate. It is usually harder to calculate than the estimations in this section.

S-2: When an exact value Q_0 is measured as $Q_0 + \Delta Q$, Definition 3.4.25 gives us the absolute error as $|\Delta Q|$, and the percentage error as $100 \frac{|\Delta Q|}{Q_0}$.

In our situation, $Q_0 = 5.83$ and $Q_0 + \Delta Q = 6$, so $\Delta Q = 0.17$. So, the absolute error is 0.17, and the percentage error is

$$100 \frac{0.17}{5.83} \approx 2.92\%$$

S-3: Since $f'(x) = 6x$, when $x = 10$, $f'(10) = 60$. If $\Delta y = f(11) - f(10)$, and $\Delta x = 11 - 10$, then the linear approximation tells us

$$\Delta y \approx 60\Delta x = 60$$

So, the linear approximation estimates the error in $f(x)$ to be about 60.

Since $f''(x) = 6$, the quadratic approximation (using $f'(10) = 60$, $f''(10) = 6$, and $\Delta x = 1$) tells us

$$\Delta y \approx f'(10)\Delta x + \frac{1}{2}f''(10)(\Delta x)^2 = 60 \cdot 1 + \frac{1}{2}(6)(1)^2 = 63$$

So, the quadratic approximation estimates the error in $f(x)$ to be about 63. (Indeed, the exact value of $f(11) - f(10)$ is 63. It is not a fluke that our estimated error, using a quadratic approximation, is exactly the same as our actual error. It is a consequence of the fact that $f(x)$ is a quadratic function.)

S-4: Let A be the area of a pen of radius r . Then

$$A(r) = \pi r^2$$

Differentiating with respect to r ,

$$A'(r) = 2\pi r$$

The exact area desired is A_0 . Let the corresponding exact radius desired be r_0 .

Using the linear approximation formula, where ΔA is the change in A corresponding to a change in r of Δr ,

$$\begin{aligned}\Delta A &\approx A'(r_0)\Delta r = 2\pi r_0\Delta r \\ \Delta r &\approx \frac{\Delta A}{2\pi r_0}\end{aligned}$$

What we're interested in is the percent error r can have. The percent error is:

$$\begin{aligned}100\frac{\Delta r}{r_0} &\approx 100\frac{\Delta A}{2\pi r_0 \cdot r_0} \\ &= 100\frac{\Delta A}{2(\pi r_0^2)} \\ &= 100\frac{\Delta A}{2 \cdot A_0} \\ &= \left(100\frac{\Delta A}{A_0}\right)\frac{1}{2} \\ &\leq (2)\frac{1}{2} = 1\end{aligned}$$

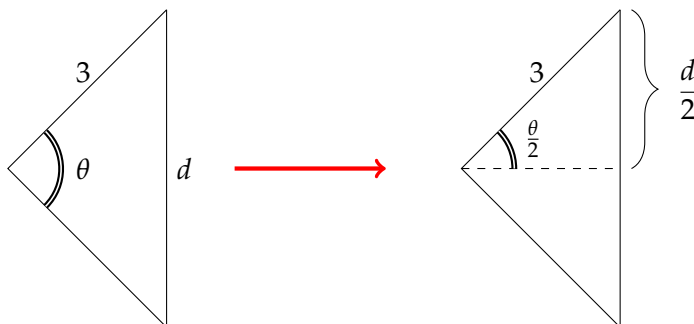
(To get the last line, we used the given information that the percent error in the area, $100\frac{\Delta A}{A_0}$, must be less than 2%.)

We conclude the error in r cannot be more than 1%.

S-5: (a) The area removed represents a proportion of $\frac{\theta}{2\pi}$ of the entire circle, whose area is $\pi(3^2) = 9\pi$. So, the area of the sector removed is

$$\frac{\theta}{2\pi} \cdot 9\pi = \frac{9}{2}\theta$$

(b) To find θ from d , we cut our triangle (with angle θ opposite side of length d) into two equivalent right triangles, as shown below.



Using the information that the radius of the circle (also the hypotenuse of the right triangle) is 3,

$$\sin\left(\frac{\theta}{2}\right) = \frac{\frac{d}{2}}{3} = \frac{d}{6}$$

Since the question tells us the sector is no more than a quarter of the circle, we know $0 \leq \theta \leq \frac{\pi}{2}$, so $0 \leq \frac{\theta}{2} \leq \frac{\pi}{4}$. This puts $\frac{\theta}{2}$ well within the range of arcsine.

$$\theta = 2 \arcsin\left(\frac{d}{6}\right)$$

(c) First, let's get an expression for the area of the sector in terms of d .

$$\begin{aligned} A &= \frac{9}{2}\theta = \frac{9}{2} \left(2 \arcsin\left(\frac{d}{6}\right) \right) \\ &= 9 \arcsin\left(\frac{d}{6}\right) \end{aligned}$$

Differentiating,

$$\begin{aligned} A'(d) &= \frac{9}{\sqrt{1 - \left(\frac{d}{6}\right)^2}} \cdot \frac{1}{6} \\ &= \frac{9}{\sqrt{36 - d^2}} \end{aligned}$$

Let ΔA is the error in A corresponding to an error of Δd in d . Since we measured d to be 0.7 instead of 0.68, in the linear approximation we take $\Delta d = 0.02$ and $d_0 = 0.68$.

$$\begin{aligned} \Delta A &\approx A'(d_0) \cdot \Delta d \\ &= A'(0.68) \cdot 0.02 \\ &= \frac{9}{\sqrt{36 - 0.68^2}} \cdot 0.02 \\ &\approx 0.03 \end{aligned}$$

So, the error in A is about 0.03.

S-6: Suppose we have a function $V(h)$ that gives the volume of water in the tank as a function of its height.

Let $h_0 = 0.5$ metres, $\Delta h = -0.05$, and $\Delta V = V(h_0 + \Delta h) - V(h_0) = V(0.45) - V(0.5)$. Then, by the linear approximation,

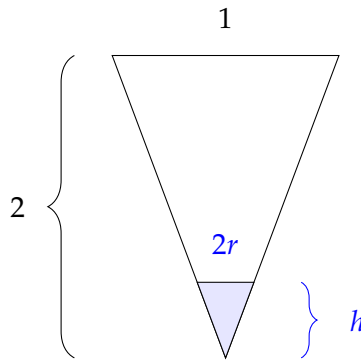
$$\Delta V \approx V'(0.5) \cdot \Delta h = -0.05V'(0.5)$$

In order to solve the problem, we will find a function $V(h)$ giving the volume of water in terms of the height, then find $V'(0.5)$, and finally approximate that the change in the volume of the water is $\Delta V \approx -0.05V'(0.5)$.

The water in the tank forms a cone. The volume of a cone of height h and radius r is

$$V = \frac{1}{3}\pi r^2 h$$

We need to get rid of the variable r . We can do this using similar triangles. The diagram below shows the side view of the tank and the water.



The side view of the tank forms a triangle that is similar to the triangle formed by the side view of the water, so

$$\begin{aligned}\frac{1}{2} &= \frac{2r}{h} \\ r &= \frac{h}{4}\end{aligned}$$

Using this, we find our equation for the volume of the water in terms of h .

$$V(h) = \frac{1}{3}\pi r^2 h = \frac{\pi}{3} \left(\frac{h}{4}\right)^2 h = \frac{\pi h^3}{48}$$

Differentiating,

$$\begin{aligned}V'(h) &= \frac{\pi h^2}{16} \\ V'(0.5) &= \frac{0.25\pi}{16} = \frac{\pi}{64}\end{aligned}$$

Finally, using the approximation $\Delta V \approx -0.05V'(0.5)$,

$$\Delta V \approx \frac{-0.05\pi}{64} = -\frac{\pi}{1280} \approx -0.00245 \text{ m}^3$$

We estimate that the volume decreased by about 0.00245 cubic metres, or about 2450 cubic centimetres.

S-7: Let q be the measured amount of the isotope remaining after 3 years. Let $h(q)$ be the half-life of the isotope that we calculate using q . We measured $q = 0.9$, but we want to know what the change in h is if q moves by 0.05. So, let $\Delta q = \pm 0.05$, and let Δh be the corresponding change in h . The linear approximation tells us

$$\Delta h \approx h'(0.9)\Delta q$$

So,

$$|\Delta h| \approx |h'(0.9)| |\Delta q| = h'(0.9) \cdot 0.05$$

This suggests a plan for solving the problem. We will find the equation $h(q)$ giving the calculated half-life of the isotope based on the measurement q . Then, we will find $h'(0.9)$. Finally, the equation $|\Delta h| = h'(0.9) \cdot 0.05$ will tell us the change in h that corresponds with a change of 0.05 in our measurement.

Let us find the half-life of the isotope, if after three years $q \mu\text{g}$ is remaining. The amount of the isotope present after t years is given by

$$Q(t) = Q(0)e^{-kt}$$

for some constant k . Let's take $t = 0$ to be the time when precisely one μg was present. Then

$$Q(t) = e^{-kt}$$

After three years, q is the amount of the isotope remaining, so

$$\begin{aligned} q &= e^{-k \cdot 3} \\ q^{\frac{1}{3}} &= e^{-k} \\ Q(t) &= \left(e^{-k}\right)^t = q^{\frac{t}{3}} \end{aligned}$$

The half-life is the value of t for which $Q(t) = \frac{1}{2}Q(0) = \frac{1}{2}$.

$$\begin{aligned} \frac{1}{2} &= Q(t) = q^{\frac{t}{3}} \\ \log\left(\frac{1}{2}\right) &= \log\left(q^{\frac{t}{3}}\right) \\ -\log 2 &= \frac{t}{3} \log q \\ t &= \frac{-3 \log 2}{\log q} \end{aligned}$$

So, we calculate the half-life to be $\frac{-3 \log 2}{\log q}$. This gives us our first goal: a function $h(q)$ that tells us the calculated half-life of the element.

$$h(q) = \frac{-3 \log 2}{\log q}$$

Following our plan, we find $h'(0.9)$.

$$\begin{aligned}h'(q) &= \frac{d}{dq} \left\{ \frac{-3 \log 2}{\log q} \right\} \\&= -3 \log 2 \cdot \frac{d}{dq} \left\{ (\log q)^{-1} \right\} \\&= -3 \log 2 \cdot (-1) (\log q)^{-2} \cdot \frac{1}{q} \\&= \frac{3 \log 2}{q \log^2 q} \\h'(0.9) &= \frac{3 \log 2}{0.9 \log^2(0.9)} \approx 208\end{aligned}$$

Finally, as outlined in our plan,

$$\begin{aligned}|\Delta h| &= h'(0.9) \cdot 0.05 \\&= \frac{3 \log 2}{18 \log^2(0.9)} \approx 10.4\end{aligned}$$

If our measurement changes by $\pm 0.05 \mu\text{g}$, then we estimate our calculated half-life changes by about ± 10.4 years. Since our measurement is accurate to within $0.05 \mu\text{g}$, that means we estimate our calculated half-life to be accurate to within about 10.4 years.

Remark: since $h(0.9) = \frac{-3 \log 2}{\log 0.9} \approx 19.7$, an absolute error of 10.4 years corresponds to a percentage error of $100 \frac{10.4}{19.7} \approx 53\%$. The question did not specify absolute or percentage error. Since both make sense, you can use either one.

Solutions to Exercises [3.4.8](#) — Jump to [TABLE OF CONTENTS](#)

S-1: From the given information,

$$|R(10)| = |f(10) - F(10)| = |-3 - 5| = |-8| = 8$$

So, (a) is false (since 8 is not less than or equal to 7), while (b), (c), and (d) are true.

Remark: $R(x)$ is the error in our approximation. As mentioned in the text, we almost never know R exactly, but we can give a bound. We don't need the tightest bound—just a reasonable one that is easy to calculate. If we were dealing with real functions and approximations, we might not know that $|R(10)| = 8$, but if we knew it was at most 9, that would be a pretty decent approximation.

Often in this section, we will make simplifying assumptions to get a bound that is easy to calculate. But, don't go overboard! It is a true statement to say that our absolute error is at most 100, but this statement would probably not be very helpful as a bound.

S-2: Equation 3.4.33 tells us that, when $T_n(x)$ is the n th order Taylor polynomial for a function $f(x)$ about $x = a$, then

$$|f(x) - T_n(x)| = \left| \frac{f^{(n+1)}(c)}{(n+1)!} (x-a)^{n+1} \right|$$

for some c strictly between x and a . In our case, $n = 3$, $a = 0$, $x = 2$, and $f^{(4)}(c) = e^c$, so

$$\begin{aligned} |f(2) - T_3(2)| &= \left| \frac{f^{(4)}(c)}{4!} (2-0)^4 \right| \\ &= \frac{2^4}{4!} e^c = \frac{2}{3} e^c \end{aligned}$$

Since c is strictly between 0 and 2, $e^c < e^2$:

$$\leq \frac{2}{3} e^2$$

but this isn't a number we really know. Indeed: e^2 is the very number we're trying to approximate. So, we use the estimation $e < 3$:

$$< \frac{2}{3} \cdot 3^2 = 6$$

We conclude that the error $|f(2) - T_3(2)|$ is *less than* 6.

Now we'll get a more exact idea of the error using a calculator. (Calculators will also only give approximations of numbers like e , but they are generally very good approximations.)

$$\begin{aligned} |f(2) - T_3(2)| &= \left| e^2 - \left(1 + 2 + \frac{1}{2} \cdot 2^2 + \frac{1}{3!} \cdot 2^3 \right) \right| \\ &= \left| e^2 - \left(1 + 2 + 2 + \frac{4}{3} \right) \right| \\ &= \left| e^2 - \frac{19}{3} \right| \approx 1.056 \end{aligned}$$

So, our actual answer was only off by about 1.

Remark: $1 < 6$, so this does not in any way contradict our bound $|f(2) - T_3(2)| < 6$.

S-3: Whenever you approximate a polynomial with a Taylor polynomial of greater or equal degree, your Taylor polynomial is *exactly the same* as the function you are approximating. So, the error is zero.

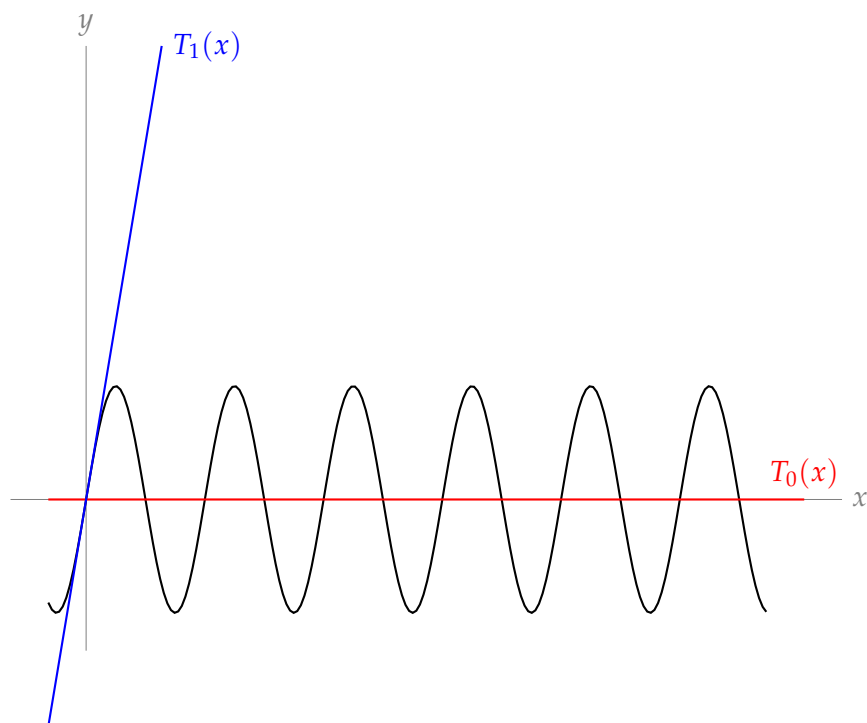
S-4: The constant approximation gives

$$\sin(33) \approx \sin(0) = 0$$

while the linear approximation gives

$$\begin{aligned} f(x) &\approx f(0) + f'(0)x \\ \sin(x) &\approx \sin(0) + \cos(0)x \\ &= x \\ \sin(33) &\approx 33 \end{aligned}$$

Since $-1 \leq \sin(33) \leq 1$, the constant approximation is better. (But both are a little silly.)



S-5: Equation 3.4.33 tells us that, when $T_n(x)$ is the n th order Taylor polynomial for a function $f(x)$ about $x = a$, then

$$|f(x) - T_n(x)| = \left| \frac{f^{(n+1)}(c)}{(n+1)!} (x-a)^{n+1} \right|$$

for some c strictly between x and a . In our case, $n = 5$, $a = 11$, $x = 11.5$, and $f^{(6)}(c) = \frac{6!(2c-5)}{c+3}$.

$$\begin{aligned} |f(11.5) - T_5(11.5)| &= \left| \frac{1}{6!} \left(\frac{6!(2c-5)}{c+3} \right) (11.5-11)^6 \right| \\ &= \left| \frac{2c-5}{c+3} \right| \cdot \frac{1}{2^6} \end{aligned}$$

for some c in $(11, 11.5)$. We don't know exactly which c this is true for, but since we know that c lies in $(11, 11.5)$, we can provide bounds.

- $2c - 5 < 2(11.5) - 5 = 18$
- $c + 3 > 11 + 3 = 14$
- Therefore, $\left| \frac{2c-5}{c+3} \right| = \frac{2c-5}{c+3} < \frac{18}{14} = \frac{9}{7}$ when $c \in (11, 11.5)$.

With this bound, we see

$$\begin{aligned} |f(11.5) - T_5(11.5)| &= \left| \frac{2c-5}{c+3} \right| \cdot \frac{1}{2^6} \\ &< \left(\frac{9}{7} \right) \left(\frac{1}{2^6} \right) \approx 0.0201 \end{aligned}$$

Our error is less than 0.02.

S-6: Equation 3.4.33 tells us that, when $T_n(x)$ is the n th order Taylor polynomial for a function $f(x)$ about $x = a$, then

$$|f(x) - T_n(x)| = \left| \frac{f^{(n+1)}(c)}{(n+1)!} (x-a)^{n+1} \right|$$

for some c strictly between x and a . In our case, $n = 2$, $a = 0$, and $x = 0.1$, so

$$\begin{aligned} |f(0.1) - T_2(0.1)| &= \left| \frac{f^{(3)}(c)}{3!} (0.1 - 0)^3 \right| \\ &= \frac{|f'''(c)|}{6000} \end{aligned}$$

for some c in $(0, 0.1)$.

We will find $f'''(x)$, and use it to give an upper bound for

$$|f(0.1) - T_2(0.1)| = \frac{|f'''(c)|}{6000}$$

when c is in $(0, 0.1)$.

$$\begin{aligned} f(x) &= \tan x \\ f'(x) &= \sec^2 x \\ f''(x) &= 2 \sec x \cdot \sec x \tan x \\ &= 2 \sec^2 x \tan x \\ f'''(x) &= (2 \sec^2 x) \sec^2 x + (4 \sec x \cdot \sec x \tan x) \tan x \\ &= 2 \sec^4 x + 4 \sec^2 x \tan^2 x \end{aligned}$$

When $0 < c < \frac{1}{10}$, also $0 < c < \frac{\pi}{6}$, so:

- $\tan c < \tan\left(\frac{\pi}{6}\right) = \frac{1}{\sqrt{3}}$
- $\cos c > \cos\left(\frac{\pi}{6}\right) = \frac{\sqrt{3}}{2}$
- $\sec c < \frac{2}{\sqrt{3}}$

With these bounds in mind for secant and tangent, we return to the expression we found for our error.

$$\begin{aligned} |f(0.1) - T_2(0.1)| &= \frac{|f'''(c)|}{6000} = \frac{|2 \sec^4 x + 4 \sec^2 x \tan^2 x|}{6000} \\ &< \frac{2 \left(\frac{2}{\sqrt{3}}\right)^4 + 4 \left(\frac{2}{\sqrt{3}}\right)^2 \left(\frac{1}{\sqrt{3}}\right)^2}{6000} \\ &= \frac{1}{1125} \end{aligned}$$

The error is less than $\frac{1}{1125}$.

S-7: Equation 3.4.33 tells us that, when $T_n(x)$ is the n th order Taylor polynomial for a function $f(x)$ about $x = a$, then

$$|f(x) - T_n(x)| = \left| \frac{f^{(n+1)}(c)}{(n+1)!} (x-a)^{n+1} \right|$$

for some c strictly between x and a . In our case, $n = 5$, $a = 0$, and $x = -\frac{1}{4}$, so

$$\begin{aligned} \left| f\left(-\frac{1}{4}\right) - T_5\left(-\frac{1}{4}\right) \right| &= \left| \frac{f^{(6)}(c)}{6!} \left(-\frac{1}{4} - 0\right)^6 \right| \\ &= \frac{|f^{(6)}(c)|}{6! \cdot 4^6} \end{aligned}$$

for some c in $(-\frac{1}{4}, 0)$. We'll need to know the sixth derivative of $f(x)$.

$$\begin{aligned} f(x) &= \log(1-x) \\ f'(x) &= -(1-x)^{-1} \\ f''(x) &= -(1-x)^{-2} \\ f'''(x) &= -2(1-x)^{-3} \\ f^{(4)}(x) &= -3!(1-x)^{-4} \\ f^{(5)}(x) &= -4!(1-x)^{-5} \\ f^{(6)}(x) &= -5!(1-x)^{-6} \end{aligned}$$

Plugging in $|f^{(6)}(c)| = \frac{5!}{(1-c)^6}$:

$$\left| f\left(-\frac{1}{4}\right) - T_5\left(-\frac{1}{4}\right) \right| = \frac{5!}{6! \cdot 4^6 \cdot (1-c)^6} = \frac{1}{6 \cdot 4^6 \cdot (1-c)^6}$$

for some c in $(-\frac{1}{4}, 0)$.

We're interested in an upper bound for the error: we want to know the worst case scenario, so we can say that the error is *no worse* than that. We need to know what the biggest possible value of $\frac{1}{6 \cdot 4^6 \cdot (1-c)^6}$ is, given $-\frac{1}{4} < c < 0$. That means we want to know the biggest possible value of $\frac{1}{(1-c)^6}$. This corresponds to the smallest possible value of $(1-c)^6$, which in turn corresponds to the smallest absolute value of $1-c$.

- Since $-\frac{1}{4} \leq c \leq 0$, the smallest absolute value of $1-c$ occurs when $c = 0$. In other words, $|1-c| \leq 1$.
- That means the smallest possible value of $(1-c)^6$ is $1^6 = 1$.
- Then the largest possible value of $\frac{1}{(1-c)^6}$ is 1.
- Then the largest possible value of $\frac{1}{6 \cdot 4^6} \cdot \frac{1}{(1-c)^6}$ is $\frac{1}{6 \cdot 4^6} \approx 0.0000407$.

Finally, we conclude

$$\left| f\left(-\frac{1}{4}\right) - T_5\left(-\frac{1}{4}\right) \right| = \frac{1}{6 \cdot 4^6 \cdot (1-c)^6} < \frac{1}{6 \cdot 4^6} < 0.00004$$

S-8: Equation 3.4.33 tells us that, when $T_n(x)$ is the n th order Taylor polynomial for a function $f(x)$ about $x = a$, then

$$|f(x) - T_n(x)| = \left| \frac{f^{(n+1)}(c)}{(n+1)!} (x-a)^{n+1} \right|$$

for some c strictly between x and a . In our case, $n = 3$, $a = 30$, and $x = 32$, so

$$\begin{aligned} |f(30) - T_3(30)| &= \left| \frac{f^{(4)}(c)}{4!} (30-32)^4 \right| \\ &= \frac{2}{3} |f^{(4)}(c)| \end{aligned}$$

for some c in $(30, 32)$.

We will now find $f^{(4)}(x)$. Then we can give an upper bound on $|f(30) - T_3(30)| = \frac{2}{3} |f^{(4)}(c)|$ when $c \in (30, 32)$.

$$\begin{aligned} f(x) &= x^{\frac{1}{5}} \\ f'(x) &= \frac{1}{5} x^{-\frac{4}{5}} \\ f''(x) &= -\frac{4}{5^2} x^{-\frac{9}{5}} \\ f'''(x) &= \frac{4 \cdot 9}{5^3} x^{-\frac{14}{5}} \\ f^{(4)}(x) &= -\frac{4 \cdot 9 \cdot 14}{5^4} x^{-\frac{19}{5}} \end{aligned}$$

Using this,

$$\begin{aligned} |f(30) - T_3(30)| &= \frac{2}{3} |f^{(4)}(c)| \\ &= \frac{2}{3} \left| -\frac{4 \cdot 9 \cdot 14}{5^4} c^{-\frac{19}{5}} \right| \\ &= \frac{336}{5^4 \cdot c^{\frac{19}{5}}} \end{aligned}$$

Since $30 < c < 32$,

$$\begin{aligned} &< \frac{336}{5^4 \cdot 30^{\frac{19}{5}}} = \frac{336}{5^4 \cdot 30^3 \cdot 30^{\frac{4}{5}}} \\ &= \frac{14}{5^7 \cdot 9 \cdot 30^{\frac{4}{5}}} \end{aligned}$$

This isn't a number we know. We're trying to find the error in our estimation of $\sqrt[5]{30}$, but $\sqrt[5]{30}$ shows up in our error. From here, we have to be a little creative to get a bound that actually makes sense to us. There are different ways to go about it. You could simply use $30^{\frac{4}{5}} > 1$. We will be a little more careful, and use the following estimation:

$$\begin{aligned} \frac{14}{5^7 \cdot 9 \cdot 30^{\frac{4}{5}}} &= \frac{14 \cdot 30^{\frac{1}{5}}}{5^7 \cdot 9 \cdot 30} \\ &< \frac{14 \cdot 32^{\frac{1}{5}}}{5^7 \cdot 9 \cdot 30} \\ &< \frac{14 \cdot 2}{5^7 \cdot 9 \cdot 30} \\ &< \frac{14}{5^7 \cdot 9 \cdot 15} \\ &< 0.000002 \end{aligned}$$

We conclude $|f(30) - T_3(30)| < 0.000002$.

S-9: Equation 3.4.33 tells us that, when $T_n(x)$ is the n th order Taylor polynomial for a function $f(x)$ about $x = a$, then

$$|f(x) - T_n(x)| = \left| \frac{f^{(n+1)}(c)}{(n+1)!} (x-a)^{n+1} \right|$$

for some c strictly between x and a . In our case, $n = 1$, $a = \frac{1}{\pi}$, and $x = 0.01$, so

$$\begin{aligned} |f(0.01) - T_1(0.01)| &= \left| \frac{f''(c)}{2} \left(0.01 - \frac{1}{\pi}\right)^2 \right| \\ &= \frac{1}{2} \left(\frac{100 - \pi}{100\pi} \right)^2 \cdot |f''(c)| \end{aligned}$$

for some c in $(\frac{1}{100}, \frac{1}{\pi})$.

Let's find $f''(x)$.

$$\begin{aligned} f(x) &= \sin\left(\frac{1}{x}\right) \\ f'(x) &= \cos\left(\frac{1}{x}\right) \cdot \frac{-1}{x^2} = \frac{-\cos\left(\frac{1}{x}\right)}{x^2} \\ f''(x) &= \frac{x^2 \sin\left(\frac{1}{x}\right) (-x^{-2}) + \cos\left(\frac{1}{x}\right) (2x)}{x^4} \\ &= \frac{2x \cos\left(\frac{1}{x}\right) - \sin\left(\frac{1}{x}\right)}{x^4} \end{aligned}$$

Now we can plug in a better expression for $f''(c)$:

$$\begin{aligned} |f(0.01) - T_1(0.01)| &= \frac{1}{2} \left(\frac{100 - \pi}{200\pi} \right)^2 \cdot |f''(c)| \\ &= \frac{1}{2} \left(\frac{100 - \pi}{100\pi} \right)^2 \cdot \frac{|2c \cos\left(\frac{1}{c}\right) - \sin\left(\frac{1}{c}\right)|}{c^4} \end{aligned}$$

for some c in $(\frac{1}{100}, \frac{1}{\pi})$.

What we want to do now is find an upper bound on this expression containing c ,

$$\frac{1}{2} \left(\frac{100 - \pi}{100\pi} \right)^2 \cdot \frac{|2c \cos(\frac{1}{c}) - \sin(\frac{1}{c})|}{c^4}.$$

- Since $c \geq \frac{1}{100}$, it follows that $c^4 \geq \frac{1}{100^4}$, so $\frac{1}{c^4} \leq 100^4$.
- For any value of x , $|\cos x|$ and $|\sin x|$ are at most 1. Since $|c| < 1$, also $|c \cos(\frac{1}{c})| < |\cos(\frac{1}{c})| \leq 1$. So, $|2c \cos(\frac{1}{c}) - \sin(\frac{1}{c})| < 3$
- Therefore,

$$\begin{aligned} |f(0.01) - T_n(0.01)| &= \frac{1}{2} \left(\frac{100 - \pi}{100\pi} \right)^2 \cdot \frac{1}{c^4} \cdot \left| 2c \cos\left(\frac{1}{c}\right) - \sin\left(\frac{1}{c}\right) \right| \\ &< \frac{1}{2} \left(\frac{100 - \pi}{100\pi} \right)^2 \cdot 100^4 \cdot 3 \\ &= \frac{3 \cdot 100^2}{2} \left(\frac{100}{\pi} - 1 \right)^2 \end{aligned}$$

Equation 3.4.33 gives the bound $|f(0.01) - T_1(0.01)| \leq \frac{3 \cdot 100^2}{2} \left(\frac{100}{\pi} - 1 \right)^2$.

The bound above works out to approximately fourteen million. One way to understand why the bound is so high is that $\sin(\frac{1}{x})$ moves about crazily when x is near zero—it moves up and down incredibly fast, so a straight line isn't going to approximate it very well at all.

That being said, because $\sin(\frac{1}{x})$ is still “sine of something,” we know $-1 \leq f(0.01) \leq 1$. To get a better bound on the error, let's find $T_1(x)$.

$$\begin{aligned} f(x) &= \sin\left(\frac{1}{x}\right) & f\left(\frac{1}{\pi}\right) &= \sin(\pi) = 0 \\ f'(x) &= \frac{-\cos(\frac{1}{x})}{x^2} & f'\left(\frac{1}{\pi}\right) &= -\pi^2 \cos(\pi) = \pi^2 \end{aligned}$$

$$\begin{aligned} T_1(x) &= f\left(\frac{1}{\pi}\right) + f'\left(\frac{1}{\pi}\right) \left(x - \frac{1}{\pi}\right) \\ &= 0 + \pi^2 \left(x - \frac{1}{\pi}\right) \\ &= \pi^2 x - \pi \\ T_1(0.01) &= \frac{\pi^2}{100} - \pi \end{aligned}$$

Now that we know $T_1(0.01)$, and we know $-1 \leq f(0.01) \leq 1$, we can give the bound

$$\begin{aligned}
 |f(0.01) - T_1(0.01)| &\leq |f(0.01)| + |T_1(0.01)| \\
 &\leq 1 + \left| \frac{\pi^2}{100} - \pi \right| \\
 &= 1 + \pi \left| 1 - \frac{\pi}{100} \right| \\
 &< 1 + \pi \\
 &< 1 + 4 = 5
 \end{aligned}$$

A more reasonable bound on the error is that it is less than 5.

Still more reasonably, we would not use $T_1(x)$ to evaluate $\sin(100)$ approximately. We would write $\sin(100) = \sin(100 - 32\pi)$ and approximate the right hand side, which is roughly $\sin(-\pi/6)$.

S-10: Equation 3.4.33 tells us that, when $T_n(x)$ is the n th order Taylor polynomial for a function $f(x)$ about $x = a$, then

$$|f(x) - T_n(x)| = \left| \frac{f^{(n+1)}(c)}{(n+1)!} (x-a)^{n+1} \right|$$

for some c strictly between x and a . In our case, $n = 2$, $a = 0$, and $x = \frac{1}{2}$, so

$$\begin{aligned}
 \left| f\left(\frac{1}{2}\right) - T_2\left(\frac{1}{2}\right) \right| &= \left| \frac{f^{(3)}(c)}{3!} \left(\frac{1}{2} - 0\right)^3 \right| \\
 &= \frac{|f^{(3)}(c)|}{3! \cdot 2^3}
 \end{aligned}$$

for some c in $(0, \frac{1}{2})$.

The next task that suggests itself is finding $f^{(3)}(x)$.

$$\begin{aligned}
 f(x) &= \arcsin x \\
 f'(x) &= \frac{1}{\sqrt{1-x^2}} = (1-x^2)^{-\frac{1}{2}} \\
 f''(x) &= -\frac{1}{2}(1-x^2)^{-\frac{3}{2}}(-2x) \\
 &= x(1-x^2)^{-\frac{3}{2}} \\
 f'''(x) &= x\left(-\frac{3}{2}\right)(1-x^2)^{-\frac{5}{2}}(-2x) + (1-x^2)^{-\frac{3}{2}} \\
 &= 3x^2(1-x^2)^{-\frac{5}{2}} + (1-x^2)^{-\frac{5}{2}+1} \\
 &= (1-x^2)^{-\frac{5}{2}}(3x^2 + (1-x^2)) \\
 &= (1-x^2)^{-\frac{5}{2}}(2x^2 + 1)
 \end{aligned}$$

Since $\left|f\left(\frac{1}{2}\right) - T_2\left(\frac{1}{2}\right)\right| = \frac{|f^{(3)}(c)|}{3! \cdot 2^3}$ for some c in $(0, \frac{1}{2})$,

$$\left|f\left(\frac{1}{2}\right) - T_2\left(\frac{1}{2}\right)\right| = \frac{\left|\frac{1+2c^2}{(\sqrt{1-c^2})^5}\right|}{3! \cdot 2^3} = \frac{1+2c^2}{48(\sqrt{1-c^2})^5}$$

for some c in $(0, \frac{1}{2})$.

We want to know what is the worst case scenario-what's the biggest this expression can be. **So, now we find an upper bound on $\frac{1+2c^2}{48(\sqrt{1-c^2})^5}$ when $0 \leq c \leq \frac{1}{2}$.** Remember that our bound doesn't have to be exact, but it should be relatively easy to calculate.

- When $0 \leq c \leq \frac{1}{2}$, the biggest $1+2c^2$ can be is $1+2\left(\frac{1}{2}\right)^2 = \frac{3}{2}$.

So, the **numerator** of $\frac{1+2c^2}{48(\sqrt{1-c^2})^5}$ is at most $\frac{3}{2}$.

- The smallest $1-c^2$ can be is $1-\left(\frac{1}{2}\right)^2 = \frac{3}{4}$.

- So, the smallest $(\sqrt{1-c^2})^5$ can be is $\left(\sqrt{\frac{3}{4}}\right)^5 = \left(\frac{\sqrt{3}}{2}\right)^5$.

- Then smallest possible value for the **denominator** of $\frac{1+2c^2}{48(\sqrt{1-c^2})^5}$ is $48\left(\frac{\sqrt{3}}{2}\right)^5$

- Then

$$\begin{aligned} \frac{1+2c^2}{48(\sqrt{1-c^2})^5} &\leq \frac{\frac{3}{2}}{48\left(\frac{\sqrt{3}}{2}\right)^5} \\ &= \frac{1}{\sqrt{3}^5} = \frac{1}{9\sqrt{3}} \\ &< \frac{1}{10} \end{aligned}$$

Let's put together these pieces. We found that

$$\left|f\left(\frac{1}{2}\right) - T_2\left(\frac{1}{2}\right)\right| = \frac{1+2c^2}{48(\sqrt{1-c^2})^5}$$

for some c in $(0, \frac{1}{2})$. We also found that

$$\frac{1 + 2c^2}{48(\sqrt{1 - c^2})^5} < \frac{1}{10}$$

when c is in $(0, \frac{1}{2})$. We conclude

$$\left| f\left(\frac{1}{2}\right) - T_2\left(\frac{1}{2}\right) \right| < \frac{1}{10}.$$

For the second part of the question, we need to find $f\left(\frac{1}{2}\right)$ and $T_2\left(\frac{1}{2}\right)$.

Finding $f\left(\frac{1}{2}\right)$ is not difficult.

$$\begin{aligned} f(x) &= \arcsin x \\ f\left(\frac{1}{2}\right) &= \arcsin\left(\frac{1}{2}\right) = \frac{\pi}{6} \end{aligned}$$

In order to find $T_2\left(\frac{1}{2}\right)$, we need to find $T_2(x)$.

$$T_2(x) = f(0) + f'(0)x + \frac{1}{2}f''(0)x^2$$

Conveniently, we've already found the first few derivatives of $f(x)$.

$$\begin{aligned} T_2(x) &= \arcsin(0) + \left(\frac{1}{\sqrt{1-0^2}}\right)x + \frac{1}{2}\left(\frac{0}{(\sqrt{1-0^2})^3}\right)x^2 \\ &= 0 + x + 0 \\ &= x \\ T_2\left(\frac{1}{2}\right) &= \frac{1}{2} \end{aligned}$$

So, the actual error is

$$\left| f\left(\frac{1}{2}\right) - T_2\left(\frac{1}{2}\right) \right| = \left| \frac{\pi}{6} - \frac{1}{2} \right| = \frac{\pi}{6} - \frac{1}{2}$$

A calculator tells us that this is about 0.02.

S-11: Our error will have the form $\frac{f^{(n+1)}(c)}{(n+1)!}(x-1)^{n+1}$ for some constant c , so let's find an equation for $f^{(n)}(x)$. This has been done before in the text, but we'll do it again here: we'll take several derivatives, then notice the pattern.

$$\begin{aligned} f(x) &= \log x \\ f'(x) &= x^{-1} \\ f''(x) &= -x^{-2} \\ f'''(x) &= 2! x^{-3} \\ f^{(4)}(x) &= -3! x^{-4} \\ f^{(5)}(x) &= 4! x^{-5} \end{aligned}$$

So, when $n \geq 1$,

$$f^{(n)}(x) = (-1)^{n-1}(n-1)! \cdot x^{-n}$$

Now that we know the derivative of $f(x)$, we have a better idea what the error in our approximation looks like.

$$\begin{aligned} |f(1.1) - T_n(1.1)| &= \left| \frac{f^{(n+1)}(c)}{(n+1)!} (1.1 - 1)^{n+1} \right| \\ &= \left| f^{(n+1)}(c) \right| \frac{0.1^{n+1}}{(n+1)!} \\ &= \left| \frac{n!}{c^{n+1}} \right| \frac{1}{10^{n+1}(n+1)!} \\ &= \frac{1}{|c|^{n+1} \cdot 10^{n+1} \cdot (n+1)} \end{aligned}$$

for some c in $(1, 1.1)$

$$\begin{aligned} &< \frac{1}{(n+1)10^{n+1} \cdot 1^{n+1}} \\ &= \frac{1}{(n+1)10^{n+1}} \end{aligned}$$

What we've shown so far is

$$|f(1.1) - T_n(1.1)| < \frac{1}{(n+1)10^{n+1}}$$

If we can show that $\frac{1}{(n+1)10^{n+1}} \leq 10^{-4}$, then we'll be able to conclude

$$|f(1.1) - T_n(1.1)| < \frac{1}{(n+1)10^{n+1}} \leq 10^{-4}$$

That is, our error is less than 10^{-4} .

So, our goal for the problem is to find a value of n that makes $\frac{1}{(n+1)10^{n+1}} \leq 10^{-4}$. Certainly, $n = 3$ is such a number. Therefore, any n greater than or equal to 3 is an acceptable value.

S-12: We will approximate $f(x) = x^{\frac{1}{7}}$ using a Taylor polynomial. Since $3^7 = 2187$, we will use $x = 2187$ as our centre.

We need to figure out which order Taylor polynomial will result in a small enough error.

If we use the n th Taylor polynomial, our error will be

$$\begin{aligned} |f(2200) - T_n(2200)| &= \left| \frac{f^{(n+1)}(c)}{(n+1)!} (2200 - 2187)^{n+1} \right| \\ &= \left| f^{(n+1)}(c) \right| \cdot \frac{13^{n+1}}{(n+1)!} \end{aligned}$$

for some c in $(2187, 2200)$. In order for this to be less than 0.001, we need

$$\begin{aligned} \left| f^{(n+1)}(c) \right| \cdot \frac{13^{n+1}}{(n+1)!} &< 0.001 \\ \left| f^{(n+1)}(c) \right| &< \frac{(n+1)!}{1000 \cdot 13^{n+1}} \end{aligned}$$

It's a tricky thing to figure out which n makes this true. Let's make a table. We won't show all the work of filling it in, but the work is standard.

n	$\frac{(n+1)!}{1000 \cdot 13^{n+1}}$	$\left f^{(n+1)}(c) \right $	Is $\left f^{(n+1)}(c) \right < \frac{(n+1)!}{1000 \cdot 13^{n+1}}$?
0	$\frac{1}{1000 \cdot 13}$	$ f'(c) = \frac{1}{7c^{6/7}} < \frac{1}{7 \cdot 3^6}$	
1	$\frac{2}{1000 \cdot 13^2}$	$ f''(c) = \frac{6}{7^2 \cdot c^{13/7}} < \frac{6}{7^2 \cdot 3^{13}}$	Yes!

That is: if we use the first order Taylor polynomial, then for some c between 2187 and 2200,

$$\begin{aligned} |f(2200) - T_1(2200)| &= |f''(c)| \cdot \frac{13^2}{2!} \\ &= \frac{6}{7^2 \cdot c^{13/7}} \cdot \frac{13^2}{2} \\ &< \frac{6}{7^2 \cdot 3^{13}} \cdot \frac{13^2}{2} \\ &= \frac{3 \cdot 13^2}{7^2 \cdot 3^{13}} \approx 0.0000065 \end{aligned}$$

So, actually, the linear Taylor polynomial (or any higher order Taylor polynomial) will result in an approximation that is much more accurate than required. (We don't know, however, that the constant approximation will be accurate enough—so we'd better stick with $n \geq 1$.)

Now that we know we can take the first order Taylor polynomial, let's compute $T_1(x)$. Recall we

are taking the Taylor polynomial for $f(x) = x^{\frac{1}{7}}$ about $x = 2187$.

$$\begin{aligned}
 f(2187) &= 2187^{\frac{1}{7}} = 3 \\
 f'(x) &= \frac{1}{7}x^{-\frac{6}{7}} \\
 f'(2187) &= \frac{1}{7\sqrt[7]{2187^6}} = \frac{1}{7 \cdot 3^6} \\
 T_1(x) &= f(2187) + f'(2187)(x - 2187) \\
 &= 3 + \frac{x - 2187}{7 \cdot 3^6} \\
 T_1(2200) &= 3 + \frac{2200 - 2187}{7 \cdot 3^6} \\
 &= 3 + \frac{13}{7 \cdot 3^6} \\
 &\approx 3.00255
 \end{aligned}$$

We conclude $\sqrt[7]{2200} \approx 3.00255$.

S-13: If we're going to use Equation 3.4.33, then we'll probably be taking a Taylor polynomial. Using Example 3.4.16, the 6th order Maclaurin polynomial for $\sin x$ is

$$T_6(x) = T_5(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!}$$

so let's play with this a bit. Equation 3.4.33 tells us that the error will depend on the seventh derivative of $f(x)$, which is $-\cos x$:

$$\begin{aligned}
 f(1) - T_6(1) &= f^{(7)}(c) \frac{1^7}{7!} \\
 \sin(1) - \left(1 - \frac{1}{3!} + \frac{1}{5!}\right) &= \frac{-\cos c}{7!} \\
 \sin(1) - \frac{101}{5!} &= \frac{-\cos c}{7!} \\
 \sin(1) &= \frac{4242 - \cos c}{7!}
 \end{aligned}$$

for some c between 0 and 1. Since $-1 \leq \cos c \leq 1$,

$$\begin{aligned}
 \frac{4242 - 1}{7!} &\leq \sin(1) \leq \frac{4242 + 1}{7!} \\
 \frac{4241}{7!} &\leq \sin(1) \leq \frac{4243}{7!} \\
 \frac{4241}{5040} &\leq \sin(1) \leq \frac{4243}{5040}
 \end{aligned}$$

Remark: there are lots of ways to play with this idea to get better estimates. One way is to take a higher order Maclaurin polynomial. Another is to note that, since $0 < c < 1 < \frac{\pi}{3}$, then

$\frac{1}{2} < \cos c < 1$, so

$$\begin{aligned}\frac{4242-1}{7!} &< \sin(1) < \frac{4242-\frac{1}{2}}{7!} \\ \frac{4241}{5040} &< \sin(1) < \frac{8483}{10080} < \frac{4243}{5040}\end{aligned}$$

If you got tighter bounds than asked for in the problem, congratulations!

S-14: (a) For every whole number n , the n th derivative of e^x is e^x . So:

$$T_4(x) = \sum_{n=0}^4 \frac{e^0}{n!} x^n = \sum_{n=0}^4 \frac{x^n}{n!}$$

(b)

$$\begin{aligned}T_4(1) &= \sum_{n=0}^4 \frac{1^n}{n!} = \sum_{n=0}^4 \frac{1}{n!} \\ &= \frac{1}{0!} + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \frac{1}{4!} \\ &= \frac{1}{1} + \frac{1}{1} + \frac{1}{2} + \frac{1}{6} + \frac{1}{24} \\ &= \frac{65}{24}\end{aligned}$$

(c) Using Equation 3.4.33,

$$\begin{aligned}e^1 - T_4(1) &= \frac{1}{5!} e^c \quad \text{for some } c \text{ strictly between 0 and 1. So,} \\ e - \frac{65}{24} &= \frac{e^c}{120} \\ e &= \frac{65}{24} + \frac{e^c}{120}\end{aligned}$$

Since e^x is a strictly increasing function, and $0 < c < 1$, we conclude $e^0 < e^c < e^1$:

$$\frac{65}{24} + \frac{1}{120} < e < \frac{65}{24} + \frac{e}{120}$$

Simplifying the left inequality, we see

$$\frac{326}{120} < e$$

From the right inequality, we see

$$\begin{aligned}e &< \frac{65}{24} + \frac{e}{120} \\ e - \frac{e}{120} &< \frac{65}{24} \\ e \cdot \frac{119}{120} &< \frac{65}{24} \\ e &< \frac{65}{24} \cdot \frac{120}{119} = \frac{325}{119}\end{aligned}$$

So, we conclude

$$\frac{326}{120} < e < \frac{325}{119},$$

as desired.

Remark: $\frac{326}{120} \approx 2.717$, and $\frac{325}{119} \approx 2.731$.

Solutions to Exercises 3.4.9 — Jump to [TABLE OF CONTENTS](#)

S-1: The third Maclaurin polynomial for $f(x)$ is

$$f(0) + f'(0)x + \frac{f''(0)}{2} \cdot x^2 + \frac{f'''(0)}{6} \cdot x^3 = 4 + 3x^2 + \frac{1}{2}x^3.$$

The coefficient of x is $f'(0)$ on one side, and 0 on the other, so $f'(0) = 0$.

The coefficient of x^2 is $\frac{1}{2}f''(0)$ on one side, and 3 on the other, so $f''(0) = 6$.

S-2: The third Maclaurin polynomial for $h(x)$ is

$$h(0) + h'(0)x + \frac{h''(0)}{2} \cdot x^2 + \frac{h^{(3)}(0)}{3!} \cdot x^3 = 1 + 4x - \frac{1}{3}x^2 + \frac{2}{3}x^3$$

The coefficient of x^3 is $\frac{1}{3!}h^{(3)}(0)$ on one side, and $\frac{2}{3}$ on the other.

Thus $\frac{h^{(3)}(0)}{6} = \frac{2}{3}$, so $h^{(3)}(0) = 6 \cdot \frac{2}{3} = 4$.

S-3: The third order Taylor polynomial for $h(x)$ about $x = 2$ is

$$h(2) + h'(2)(x-2) + \frac{h''(2)}{2} \cdot (x-2)^2 + \frac{h'''(2)}{6} \cdot (x-2)^3$$

The coefficient of $(x-2)$ is $h'(2)$ in the definition, and $\frac{1}{2}$ in the given function, so $h'(2) = \frac{1}{2}$.

The coefficient of $(x-2)^2$ is $\frac{1}{2}h''(2)$ in the definition, and 0 in the given function, so $h''(2) = 0$.

S-4:

(a) For x near 3,

$$f(x) \approx f(3) + f'(3)(x-3) = 2 + 4(x-3)$$

In particular

$$f(2.98) \approx 2 + 4(2.98 - 3) = 2 - 0.08 = 1.92$$

(b) For x near 3,

$$\begin{aligned} f(x) &\approx f(3) + f'(3)(x-3) + \frac{1}{2}f''(3)(x-3)^2 \\ &= 2 + 4(x-3) - \frac{1}{2}10(x-3)^2 \end{aligned}$$

In particular

$$\begin{aligned}f(2.98) &\approx 2 + 4(2.98 - 3) - 5(2.98 - 3)^2 \\&= 2 - 0.08 - 0.002 = 1.918\end{aligned}$$

S-5: Let's name $g(x) = x^{1/3}$. Then $g'(x) = \frac{1}{3}x^{-2/3}$ and $g''(x) = -\frac{2}{9}x^{-5/3}$.

In particular, $g(8) = 2$, $g'(8) = \frac{1}{12}$, and $g''(x) < 0$ for all $x > 0$.

The tangent line approximation to $10^{1/3}$ is

$$\begin{aligned}g(10) &\approx g(8) + g'(8)(10 - 8) \\&= 2 + \frac{1}{12}(2) = \frac{13}{6}\end{aligned}$$

Using the error formula:

$$g(10) = g(8) + g'(8)(10 - 8) + \frac{1}{2}g''(c)(10 - 8)^2$$

for some $8 < c < 10$. Since $g''(c) = -\frac{2}{9c^{5/3}}$, $g''(c)$ is negative, so $g(10)$ is $\frac{13}{6}$ plus some *negative* quantity. So, the tangent line approximation is too big.

S-6: We use the function $f(x) = \sqrt{x}$ and point $a = 1$ as the centre of our approximation, since we can easily calculate

$$f(a) = f(1) = \sqrt{1} = 1.$$

We compute $f'(x) = \frac{1}{2\sqrt{x}}$, so

$$f'(1) = \frac{1}{2\sqrt{1}} = \frac{1}{2}.$$

So, a linear approximation of $\sqrt{2} = f(2)$ is

$$\begin{aligned}\sqrt{2} &\approx T_1(2) = f(1) + f'(1) \cdot (2 - 1) \\&= 1 + \frac{1}{2} = \frac{3}{2}.\end{aligned}$$

S-7: We use the function $f(x) = x^{1/3}$ and point $a = 27$ as the centre of our approximation since we can easily compute

$$f(27) = 3$$

We compute $f'(x) = \frac{1}{3}x^{-2/3}$, so

$$f'(27) = \frac{1}{3} \cdot (27)^{-2/3} = \frac{1}{27}$$

So, the linear approximation of $26^{1/3} = f(26)$ is

$$\begin{aligned} 26^{1/3} &\approx T_1(26) = f(27) + f'(27) \cdot (26 - 27) \\ &= 3 - \frac{1}{27} = \frac{80}{27} \end{aligned}$$

S-8: We use the function $f(x) = x^5$ and point $a = 10$ as the centre of our approximation since we know that $f(a) = f(10) = 10^5$.

Since $f'(x) = 5x^4$ we have $f'(10) = 50,000$.

So, a linear approximation of 10.1^5 is

$$\begin{aligned} T_1(10.1) &= f(10) + f'(10) \cdot (10.1 - 10) \\ &= 100,000 + 50,000 \cdot 0.1 = 105,000. \end{aligned}$$

S-9: We use the function $f(x) = \sin(x)$ and point $a = \pi$ as the centre of our approximation since we know that

$$\sin(a) = f(\pi) = \sin \pi = 0.$$

and π is reasonably close to $\frac{101\pi}{100}$.

We compute $f'(x) = \cos(x)$, so

$$f'(\pi) = \cos(\pi) = -1.$$

So, the linear approximation of $\sin\left(\frac{101\pi}{100}\right)$ is

$$\begin{aligned} f\left(\frac{101\pi}{100}\right) &\approx T_1\left(\frac{101\pi}{100}\right) \\ &= f(\pi) + f'(\pi) \cdot \left(\frac{101\pi}{100} - \pi\right) \\ &= 0 + (-1) \cdot \frac{\pi}{100} = -\frac{\pi}{100}. \end{aligned}$$

S-10: Set $f(x) = \arctan(x)$. Then $f'(x) = \frac{1}{1+x^2}$, so $f'(1) = \frac{1}{2}$ and

$$f(1.1) \approx f(1) + f'(1)(1.1 - 1) = \frac{\pi}{4} + \frac{1}{20}$$

S-11: Set $f(x) = (2+x)^3$, so we are approximating $f(0.001)$.

The obvious choice of a is $a = 0$.

Then $f'(x) = 3(2+x)^2$, so

$$(2.001)^3 = f(0.001) \approx f(0) + f'(0)(0.001 - 0) = 8 + \frac{12}{1000} = \frac{8012}{1000}$$

Remark: if we had chosen $f(x) = x^3$ and $a = 2$, the result would have been exactly the same.

S-12: We set $f(x) = (8 + x)^{2/3}$, and choose $a = 0$ as our centre.

Then $f'(x) = \frac{2}{3}(8 + x)^{-1/3}$, so that

$$\begin{aligned}(8.06)^{2/3} = f(0.06) &\approx f(0) + f'(0) \cdot 0.06 \\&= 8^{2/3} + \frac{2}{3}8^{-1/3} \cdot 0.06 \\&= \sqrt[3]{8^2} + \frac{2}{3\sqrt[3]{8}} \cdot 0.06 \\&= 2^2 + \frac{2}{3 \cdot 2} \cdot 0.06 \\&= 4 + \frac{1}{3} \cdot 0.06 \\&= 4.02 = \frac{402}{100}\end{aligned}$$

S-13: We begin by finding the derivatives of f at $x = 0$.

$$\begin{aligned}f(x) &= (1 - 3x)^{-1/3} & f(0) &= 1 \\f'(x) &= (-3)\frac{-1}{3}(1 - 3x)^{-4/3} = (1 - 3x)^{-4/3} & f'(0) &= 1 \\f''(x) &= (-3)\frac{-4}{3}(1 - 3x)^{-7/3} = 4(1 - 3x)^{-7/3} & f''(0) &= 4 \\f^{(3)}(x) &= (-3)(4)\frac{-7}{3}(1 - 3x)^{-10/3} = 28(1 - 3x)^{-10/3} & f^{(3)}(0) &= 28\end{aligned}$$

Plugging these into the definition of a Taylor Polynomial, we find that the third-order Taylor polynomial for f around $x = 0$ is

$$\begin{aligned}T_3(x) &= 1 + x + \frac{4}{2!}x^2 + \frac{28}{3!}x^3 \\&= 1 + x + 2x^2 + \frac{14}{3}x^3\end{aligned}$$

S-14:

- By Equation 3.4.33, the absolute value of the error is

$$\left| \frac{f'''(c)}{3!} \cdot (2 - 1)^3 \right| = \left| \frac{c}{6(22 - c^2)} \right|$$

for some $c \in (1, 2)$.

- When $1 \leq c \leq 2$, we know that $18 \leq 22 - c^2 \leq 21$, and that numerator and denominator are non-negative, so

$$\begin{aligned} \left| \frac{c}{6(22 - c^2)} \right| &= \frac{c}{6(22 - c^2)} \leq \frac{2}{6(22 - c^2)} \leq \frac{2}{6 \cdot 18} \\ &= \frac{1}{54} \leq \frac{1}{50} \end{aligned}$$

as required.

- Alternatively, notice that c is an increasing function of c , while $22 - c^2$ is a decreasing function of c . Hence the fraction is an increasing function of c and takes its largest value at $c = 2$. Hence

$$\left| \frac{c}{6(22 - c^2)} \right| \leq \frac{2}{6 \times 18} = \frac{1}{54} \leq \frac{1}{50}.$$

S-15:

- By Equation 3.4.33, there is $c \in (0, 0.5)$ such that the error is

$$\begin{aligned} R_4 &= \frac{f^{(4)}(c)}{4!} (0.5 - 0)^4 \\ &= \frac{1}{24 \cdot 16} \cdot \frac{\cos(c^2)}{3 - c} \end{aligned}$$

- For any c we have $|\cos(c^2)| \leq 1$, and for $c < 0.5$ we have $3 - c > 2.5$, so that

$$\left| \frac{\cos(c^2)}{3 - c} \right| \leq \frac{1}{2.5}.$$

- We conclude that

$$|R_4| \leq \frac{1}{2.5 \cdot 24 \cdot 16} = \frac{1}{60 \cdot 16} < \frac{1}{60 \cdot 10} = \frac{1}{600} < \frac{1}{500}$$

S-16:

- By Equation 3.4.33, there is $c \in (0, 1)$ such that the error is

$$\left| \frac{f'''(c)}{3!} \cdot (1 - 0)^3 \right| = \left| \frac{e^{-c}}{6(8 + c^2)} \right|.$$

- When $0 < c < 1$, we know that $1 > e^{-c} > e^{-1}$ and $8 \leq 8 + c^2 < 9$, so

$$\begin{aligned} \left| \frac{e^{-c}}{6(8 + c^2)} \right| &= \frac{e^{-c}}{6(8 + c^2)} \\ &< \frac{1}{6|8 + c^2|} \\ &< \frac{1}{6 \times 8} = \frac{1}{48} < \frac{1}{40} \end{aligned}$$

as required.

- Alternatively, notice that e^{-c} is a decreasing function of c , while for $0 < c$ $8 + c^2$ is an increasing function of c . Hence the fraction is a decreasing function of c and takes its largest value at $c = 0$. Hence

$$\left| \frac{e^c}{6(8 + c^2)} \right| \leq \frac{1}{6 \times 8} = \frac{1}{48} < \frac{1}{40}.$$

S-17: (a), (b):

Let $f(x) = x^{1/3}$ and $x_0 = 27$. Then

$$\begin{aligned} f(x) &= x^{1/3} & f'(x) &= \frac{1}{3}x^{-2/3} & f''(x) &= -\frac{2}{9}x^{-5/3} \\ f(27) &= 27^{1/3} = 3 & f'(27) &= \frac{1}{3} \cdot \frac{1}{3^2} = \frac{1}{27} & f''(27) &= -\frac{2}{9} \cdot \frac{1}{3^5} = -\frac{2}{2187} \end{aligned}$$

so that, with $x = 25$,

$$5^{2/3} = f(25) \approx f(27) + f'(27)(25 - 27) = 3 - \frac{2}{27} \approx 2.9259 \text{ (linear approx)}$$

$$\begin{aligned} 5^{2/3} = f(25) &\approx f(27) + f'(27)(25 - 27) + \frac{1}{2}f''(27)(25 - 27)^2 \\ &= 3 - \frac{2}{27} - \frac{1}{2} \cdot \frac{2 \cdot 4}{2187} \approx 2.9241 \text{ (quadratic app)} \end{aligned}$$

(c) To obtain an error estimate for the linear approximation, we use that

$$5^{2/3} = f(25) = f(27) + f'(27)(25 - 27) + \frac{1}{2}f''(z)(25 - 27)^2$$

for some z between 25 and 27. The error is exactly

$$\left| \frac{1}{2}f''(z)(25 - 27)^2 \right| = \left| \frac{1}{2} \left(-\frac{2}{9}z^{-5/3} \right) (-2)^2 \right| = \frac{4}{9}z^{-5/3}.$$

For z between 25 and 27, $z^{-5/3}$ is between $25^{-5/3}$ and $27^{-5/3}$. The biggest this can be is $25^{-5/3}$, so the maximum possible error is $\frac{4}{9}25^{-5/3}$.

To get a better idea of what this number is, we note $2.9^3 < 25$, so $\frac{4}{9}25^{-5/3} < \frac{4}{9}2.9^{-5} = 0.0022$.

S-18: The fourth order Maclaurin polynomial for $f(x)$ is

$$T_4(x) = f(0) + f'(0)x + \frac{1}{2}f''(0)x^2 + \frac{1}{3!}f'''(0)x^3 + \frac{1}{4!}f^{(4)}(0)x^4$$

while the third order Maclaurin polynomial for $f(x)$ is

$$T_3(x) = f(0) + f'(0)x + \frac{1}{2}f''(0)x^2 + \frac{1}{3!}f'''(0)x^3$$

So, we simply “chop off” the part of $T_4(x)$ that includes x^4 . Since that’s already 0, in this case $T_3(x) = T_4(x)$.

$$T_3(x) = 5x^2 - 9$$

S-19: y is a function of x that obeys

$$y(x)^4 + xy(x) = x^2 - 1$$

By implicit differentiation (and then subbing in $x = 2, y(2) = 1$)

$$4y(x)^3 y'(x) + y(x) + xy'(x) = 2x$$

$$4y'(2) + 1 + 2y'(2) = 4$$

$$y'(2) = \frac{1}{2}$$

Differentiating with respect to x a second time and then subbing in $x = 2, y(2) = 1, y'(2) = \frac{1}{2}$:

$$12y(x)^2 y'(x)^2 + 4y(x)^3 y''(x) + y'(x) + y'(x) + xy''(x) = 2$$

$$12 \times 1 \times \frac{1}{4} + 4y''(2) + \frac{1}{2} + \frac{1}{2} + 2y''(2) = 2$$

$$6y''(2) = -2$$

$$y''(2) = -\frac{1}{3}$$

The tangent line approximation to $y(x)$ at $x = 2$ is

$$y(x) \approx y(2) + y'(2)(x - 2) = 1 + \frac{1}{2}(x - 2)$$

In particular,

$$y(2.1) \approx y(2) + y'(2)(2.1 - 2) = 1 + \frac{1}{2}(.1) = \boxed{1.05}$$

The quadratic approximation to $y(x)$ at $x = 2$ is

$$y(x) \approx y(2) + y'(2)(x - 2) + \frac{1}{2}y''(2)(x - 2)^2$$

$$= 1 + \frac{1}{2}(x - 2) - \frac{1}{6}(x - 2)^2$$

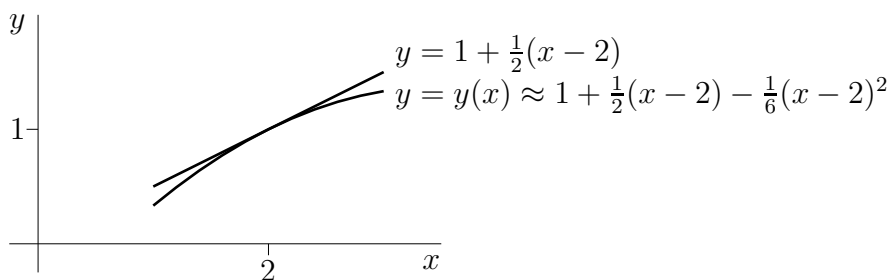
In particular,

$$y(2.1) \approx y(2) + y'(2)(2.1 - 2) + \frac{1}{2}y''(2)(2.1 - 2)^2$$

$$= 1 + \frac{1}{2}(.1) - \frac{1}{6}(.1)^2 = \boxed{1.0483}$$

At $x = 2, y = 1$ and $y' = \frac{1}{2}$. So the tangent line passes through $(2, 1)$ and has slope $\frac{1}{2}$. At $x = 2, y'' = -\frac{1}{3}$, so the graph $y = f(x)$ (locally!) looks like a parabola pointing down near $x = 2$. This gives the graph fragment below.

Alternatively, we could observe that, near $x = 2, y(x)$ will be quite close to its quadratic approximation, $1 + \frac{1}{2}(x - 2) - \frac{1}{6}(x - 2)^2$.



S-20: (a) y is a function of x that obeys

$$1 = x^4 + y(x) + xy(x)^4$$

By implicit differentiation (and then subbing in $x = -1$, $y(-1) = 1$)

$$0 = 4x^3 + y'(x) + y(x)^4 + 4xy(x)^3y'(x)$$

$$0 = -4 + y'(-1) + 1 - 4y'(-1)$$

$$-1 = y'(-1)$$

Differentiating with respect to x a second time and then subbing in $x = -1$, $y(-1) = 1$, and $y'(-1) = -1$:

$$0 = 12x^2 + y''(x) + 4y(x)^3y'(x) + 4y(x)^3y'(x) + 12xy(x)^2y'(x)^2 + 4xy(x)^3y''(x)$$

$$0 = 12 + y''(-1) - 4 - 4 - 12 - 4y''(-1)$$

$$-8 = 3y''(-1)$$

$$y''(-1) = -\frac{8}{3}$$

The tangent line approximation to $y(x)$ at $x = -1$ is

$$y(x) \approx y(-1) + y'(-1)(x + 1) = 1 - (x + 1) = -x$$

In particular,

$$y(-0.9) \approx 0.9$$

(b) The quadratic approximation to $y(x)$ at $x = -1$ is

$$\begin{aligned} y(x) &\approx y(-1) + y'(-1)(x + 1) + \frac{1}{2}y''(-1)(x + 1)^2 \\ &= 1 - (x + 1) - \frac{4}{3}(x + 1)^2 \end{aligned}$$

In particular,

$$y(-0.9) \approx 1 - (.1) - \frac{4}{3}(.1)^2 \approx 0.8867$$

(c) At $x = -1$, the slope of the curve is $y'(-1) = -1$. Its tangent line is falling at 45° . At $x = -1$, $y''(-1) = -\frac{8}{3}$, so the slope of the curve is decreasing as x passes through -1 . Zoomed in very close, the curve looks like a parabola opening downwards. This gives the figure

